

Universal Work Station Installation Operation, and Service

UW02-500

LCN Service - 2

***Universal Work Station
Installation, Operation,
and Service***

UW02-500
Release 520
CE Compliant
6/97

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About This Publication

This publication supports the Universal Work Station (UWS) in an existing LCN-based TPS system. The need to refer to several publications in the standard LCN bookset is minimized by providing, in this publication, the information needed by experienced users to install, checkout, implement, operate, and service a UWS.

The publication has been restructured to improve the presentation of information and include the CE Compliant equipment which is now being produced. New information is indicated by change bars at the side of the page.

This publication supports **TotalPlant** Solution (TPS) Network Release 520. TPS is the evolution of TDC 3000^X.

This publication supports CE Compliant equipment. Any equipment designated as “CE Compliant” complies with the European Union EMC and its health and safety directives. All equipment entering the European countries after January 1, 1996 requires this type of compliance, denoted by the “CE Mark.”

Operation of the UWS with Universal personality software requires a minimum of 4 megawords of memory and the Enhanced keyboard assembly, which includes both the Operator’s keyboard and the Engineer’s keyboard. Refer to the *Five/Ten-Slot Module Service* manual for the compatibility of software release and processor board types that are required. Refer to the *Keyboards* manual for detailed keyboard information.

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Acronyms

CRT	Cathode Ray Tube
ESD	Electrostatic Discharge
HM	History Module
LCN	Local Control Network
Mw	Megaword
NCF	Network Configuration File
ORU	Optimum Replaceable Unit
SP	Setpoint
US	Universal Station
UWS	Universal Work Station

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>TPS System Site Planning</i>	SW02-550	System Site Planning - 1	TPS 3020-1
<i>Local Control Network (LCN) Planning</i>	SW02-501	System Site Planning - 1	TPS 3020-1
<i>System Maintenance Guide</i>	SW13-500	LCN Service - 1	TPS 3060-1
<i>Maintenance Test Operations</i>	SW11-502	LCN Service - 1	TPS 3060-1
<i>Universal Station Service</i>	US13-500	LCN Service - 1	TPS 3060-1
<i>Five/Ten-Slot Module Service</i>	LC13-500	LCN Service - 2	TPS 3060-2
<i>Test System Executive</i>	SW13-210	LCN Service - 3	TPS 3060-3
<i>Hardware Verification Test System</i>	SW13-211	LCN Service - 3	TPS 3060-3
<i>Core Module test System</i>	SW12-212	LCN Service - 3	TPS 3060-3
<i>LCNI Network Communications Test</i>	SW13-208	LCN Service - 3	TPS 3060-3
<i>Keyboards</i>	SW09-508	Implementation/Engineering Operations - 1	TPS 3032-1
<i>Button Configuration Form Instructions</i>	SW12-570	Implementation/Engineering Operations - 1	TPS 3032-1
<i>Button Configuration Data Entry</i>	SW11-570	Implementation/Engineering Operations - 1	TPS 3032-1
<i>Network Forms</i>	SW88-505	Implementation/Configuration Forms	TPS 3031
<i>Network Form Instructions</i>	SW12-505	Implementation/Startup and Reconfiguration - 1	TPS 3030-1
<i>Network Data Entry</i>	SW11-505	Implementation/Startup & Reconfiguration - 1	TPS 3030-1
<i>Engineer's Reference manual</i>	SW09-505	Implementation/Startup & Reconfiguration - 2	TPS 3030-2
<i>Process Operations Manual</i>	SW11-501	Operations/Process Operations	TPS 3050

Section 1 – Introduction

1.1 Overview

Section contents

These are the topics covered in this section:

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Introduction

The Universal Work Station (UWS) is a lower-cost alternative to the Universal Station (US). The UWS is intended for use by Engineers and Supervisors, and can be located away from the control room in an office, laboratory, or other location.

The UWS consists of an electronics cabinet, a 14-inch or 21-inch color monitor, one or two keyboards, and a mouse. An Integrated Keyboard (IKB) can also be included with the UWS. If the UWS is current production hardware, the 21-inch monitor is CE Compliant. The 21-inch monitor can also include a touchscreen as a replacement for the mouse.

The UWS may also include one of three types of removable-media drives and a printer. The removable-media drives are either two floppy disk drives, two high-capacity Bernoulli cartridge drives, or two high-capacity Zip disk drives.

UWS Setup

As illustrated in Figure 1-1, the monitor, keyboard, and optional printer are placed on a desk or table, and the electronics cabinet rests on the floor, adjacent to the desk. When present, the removable-media drives are installed in the top of the electronics cabinet.

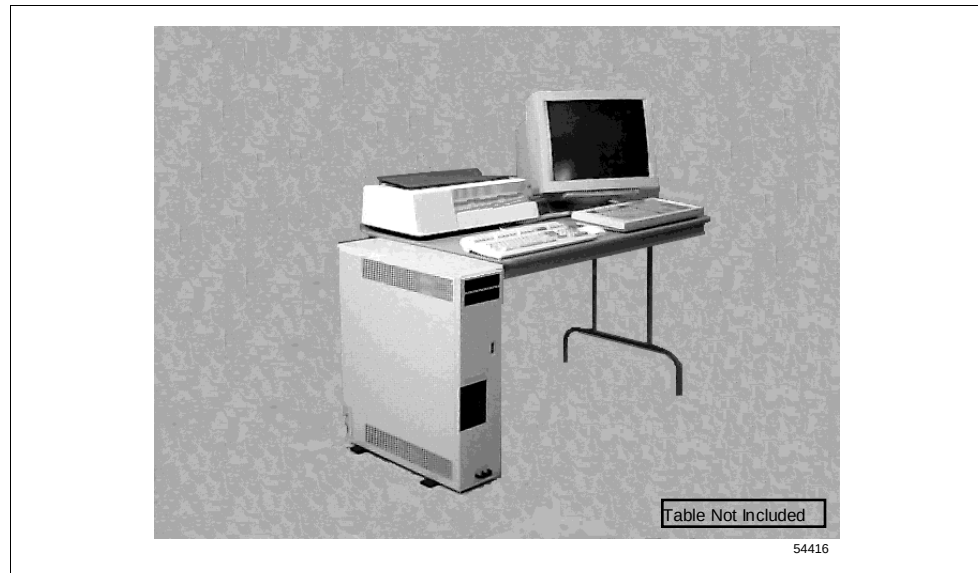
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1.1 Overview, Continued

UWS layout

Figure 1-1 illustrates the equipment layout of a Universal Work Station.

Figure 1-1 Universal Work Station



1.2 General Description

Current production CE Compliant UWS

The current production UWS is CE Compliant and has the following features.

- An improved 21-inch multisync FST color monitor
- A mouse
- A standard Supervisor's keyboard. The optional Engineer's keyboard is cabled to it.
- A K2LCN or K4LCN processor board with a minimum of 4 megawords of memory
- Dual high-capacity (100 megabytes) Zip disk drives
- An optional Engineer's keyboard
- An optional Integrated Keyboard (IKB)
- An optional Remote User LCN Access (RULA) with 8 megawords of memory
- An optional touchscreen that replaces the mouse

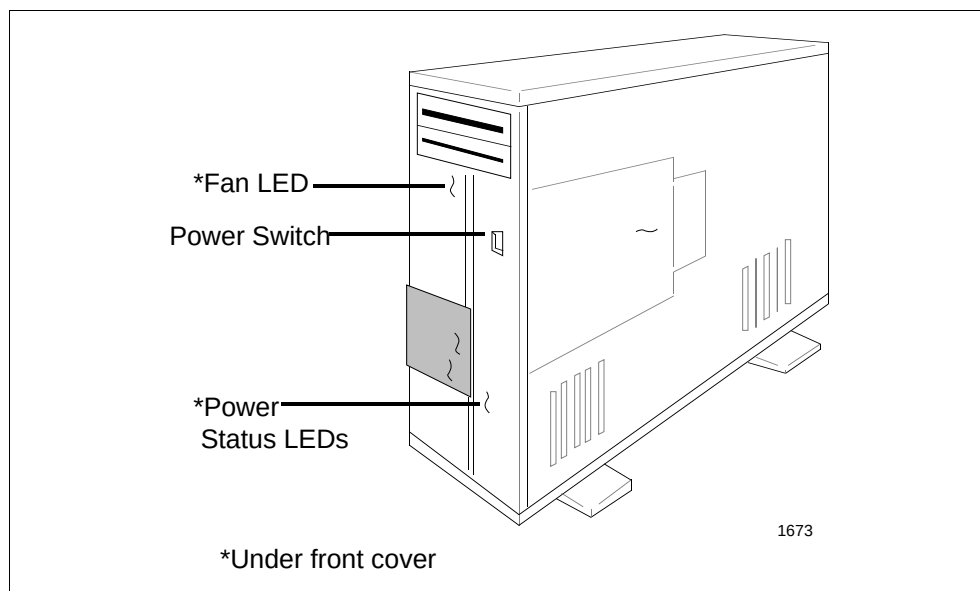
ATTENTION

ATTENTION—CE Compliant hardware requires unique spare parts to maintain CE Compliancy. Some CE Compliant parts cannot be used as a replacement for a non-CE Compliant component.

UWS front profile

Figure 1-2 is a front profile of the UWS electronics cabinet.

Figure 1-2 UWS Cabinet Front Profile



Continued on next page

1.2 General Description, Continued

Three UWS versions	There are three versions of the Universal Work Station. • Supervisor Station—uses the Supervisor's keyboard (see Figures 1-3 and 1-4) and can have removable-media drives and a printer as options. • Engineer Station—uses the Engineer's keyboard (see Figure 1-5) and can also include optional removable-media drives and a printer. • Universal Personality Station—uses the Enhanced keyboard assembly, which includes the Enhanced Operator's keyboard (see Figure 1-3) and the Engineer's keyboard (see Figure 1-5), and can also include the optional removable-media drives and a printer. The Integrated Keyboard can also be used in place of the Operator's and Engineer's keyboards. Refer to Section 6 for detailed information.
Early production UWS	In an early production UWS, the Supervisor's keyboard and the Engineer's keyboard could not be connected to the UWS at the same time. However, it is possible to change keyboards, but to do so, jumper pins on the EPDGP I/O interface board must be changed.
Later production UWS	In a later production “dual personality” UWS, both the Enhanced Operator's keyboard and the Engineer's keyboard can be connected to the UWS at the same time. The Enhanced Operator's keyboard is connected to the non-CE Compliant EPDGP I/O board or CE Compliant EPDGC I/O board, and the Engineer's keyboard connects to the Enhanced Operator's keyboard.
UWS operation	Operation of the UWS is very similar to that of the US. The UWS can operate in either the Operator personality or the Engineer personality. The Supervisor Station is intended to operate in only the Operator personality. The Engineer Station can operate in either personality. The Universal Personality Station operates in the Universal personality which includes both the Operator personality and the Engineer personality.
Standard US displays are used	In either personality, all of the standard displays are available at the UWS. Because the schematic displays (custom or graphic displays) in the Operator personality are related to the area database, they appear only if their pathnames are configured in the area database that is loaded in the UWS. If a unique area database is configured and loaded into a UWS, which may exist if it is separated from the control room, the UWS cannot view the schematics in other areas unless their pathnames are also configured in its area database. You can use the area change function on the Console Status display to acquire the database for other areas.

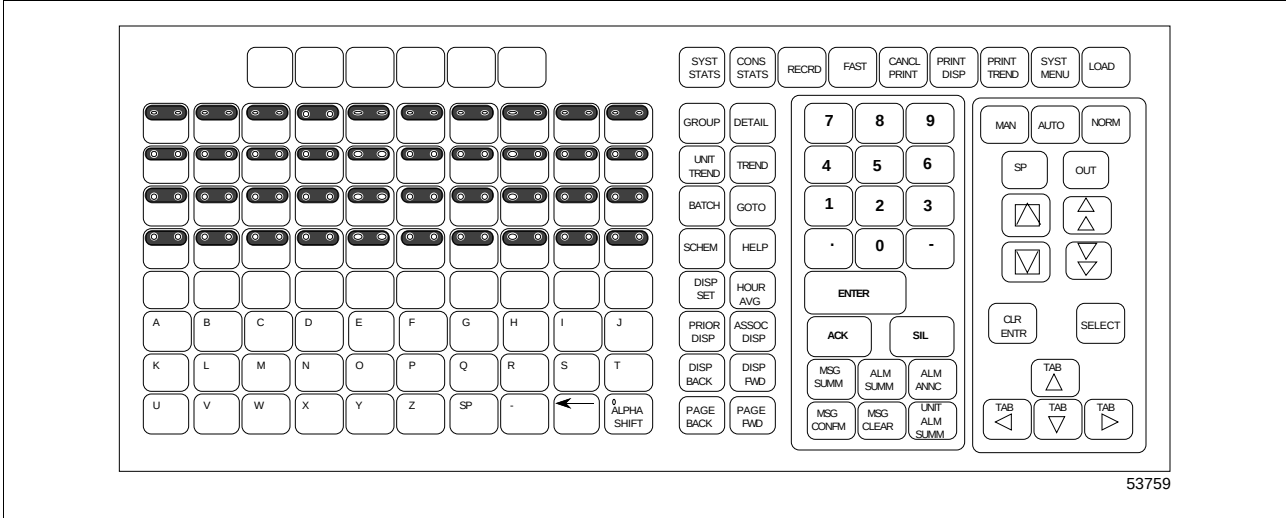
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1.2 General Description, Continued

Supervisor's and Enhanced Operator's keyboard layout

Figure 1-3 illustrates the layout of the Supervisor's and Enhanced Operator's keyboard.

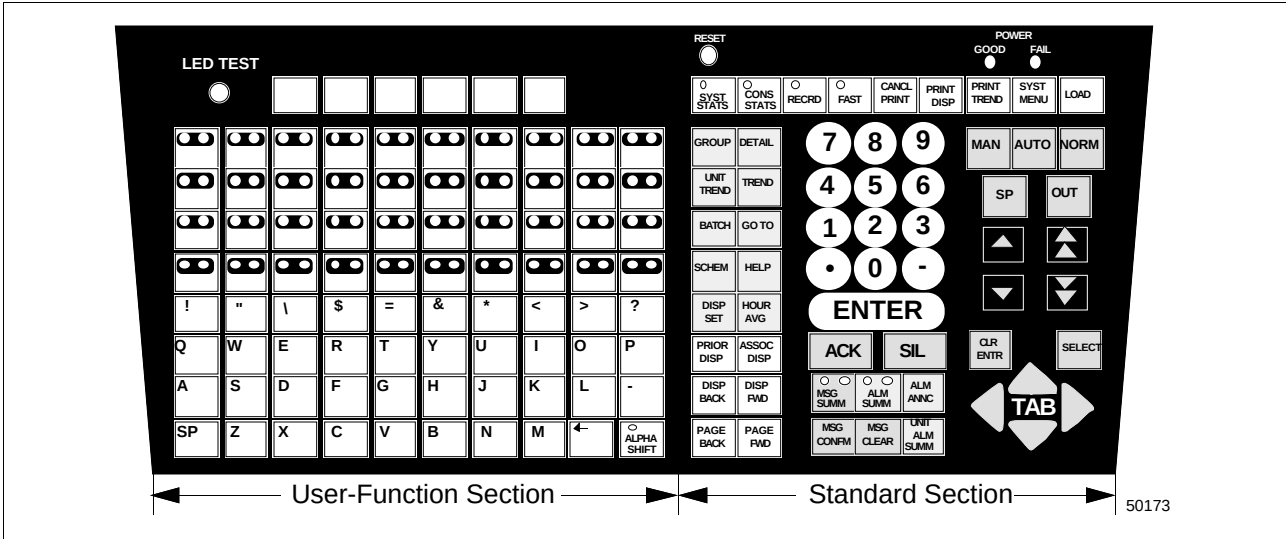
Figure 1-3 Supervisor's Keyboard and Enhanced Operator's Keyboard



Supervisor's QWERTY keyboard layout

Figure 1-4 illustrates the layout of the Supervisor's QWERTY keyboard.

Figure 1-4 Supervisor's QWERTY Keyboard



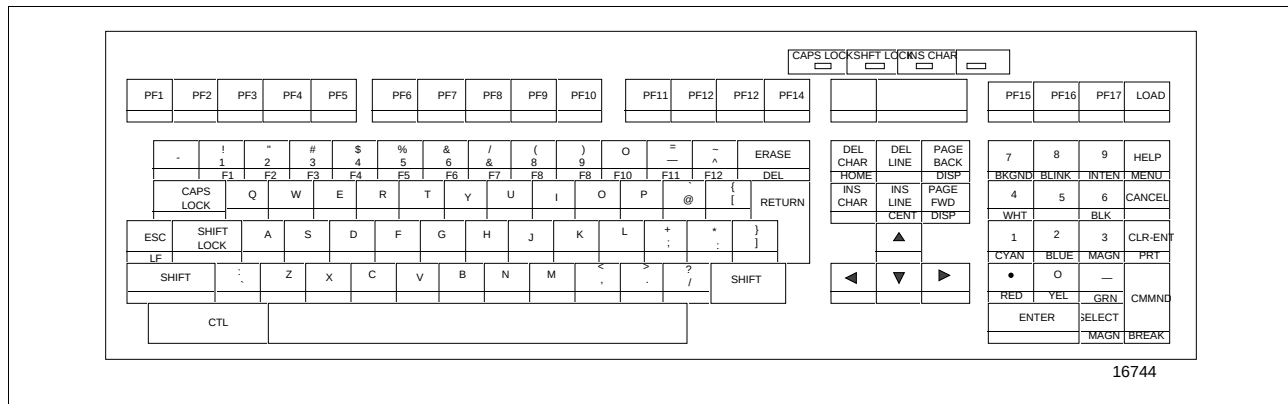
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1.2 General Description, Continued

Original Engineer's
keyboard layout

Figure 1-5 illustrates the Engineer's keyboard layout.

Figure 1-5 Original Engineer's Keyboard



Section 2 – Site Preparation and Installation

2.1 Overview

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Introduction

This section provides Universal Work Station information such as storage requirements, site requirements, electrical requirements, unpacking the UWS equipment, cabling the UWS, and LCN node address configuration.

2.2 Storage Conditions

Shipping packaging

The Universal Work Station (UWS) is protectively packaged for shipment in a cardboard box. The color monitor, keyboard or keyboards, mouse, and printer are in separate packages.

Storage specifications

If the Universal Work Station is to be placed in storage, the “general industrial” environmental constraints listed in Table 2-1 must be followed.

Table 2-1 Storage Specifications

Condition	Specification
Temperature	-35° +70°C
Humidity	10 to 80%, maximum, wet bulb 20 to 30°C
Vibration	1.0 g*
Shock	3.0 g*

* When enclosed in the original shipping container.

ATTENTION

ATTENTION—The temperature/relative humidity must not be cycled such that moisture condensation occurs in the equipment. The rate of change must be less than 6% per hour.

2.3 Site Operating Requirements

Introduction

The Universal Work Station (UWS) is intended for office use, that is, a relatively dust-free environment (dust particles filtered to ≤ 10 micron size), and an operating temperature range of 0°C to 35°C (32°F to 95°F).

Power requirements

In UWS models that are configured for 120 Vac operation, the Five-Slot Module, monitor, and printer are each equipped with 1.8-meter (6-foot), standard 3-prong grounding power cords (NEMA 5-15P). There must be two standard duplex 120 Vac grounded electrical outlets (NEMA 5-15R) that are accessible to these items.

In UWS models configured for 220/240 Vac operation, CEE 7 Standard VII Dual Earthing power cables are supplied. Two duplex outputs of this type must be accessible to the UWS for power.

LCN coax and fiber optic cable limits

When selecting a location for the UWS, consider the LCN coax or fiber-optic cable limits specified in Appendix A of the *Local Control Network Planning* or the *TPS System Site Planning* manual. The monitor, keyboard(s), printer, and mouse are equipped with 1.8-meter (6-foot) signal cables that connect to the EPDGP I/O or EPDGC I/O board.

2.4 Dimensions and Weight

Weight and dimension specifications

Table 2-2 lists the approximate weight and dimensions of the electronics cabinet and 21-inch monitor.

Table 2-2 Dimensions and Weight Specifications

	Electronics Cabinet	21-Inch Color Monitor
Height	72.24 cm (28.44")	455 mm (17.91")
Width	18.4 cm (7.23")	484 mm (19.05")
Depth	57.15 cm (22.50")	525 mm (20.66")
Weight	34.02 kg (75 lb)	33 kg (72.8 lb)

2.5 Electrical Requirements

Introduction

The customer selected ac power option is installed in the UWS before shipment.

Inrush specifications

Table 2-3 lists the inrush current specifications.

Table 2-3 Inrush Current (Maximum Load)

Power Selected	Current Draw	Frequency
120 Vac +10%, -15%	10 A	47 - 63 Hz
220/240 Vac +10%, -15%	5 A	47 - 63 Hz

Table 2-4 UWS Component AC Loading

	Average RMS	True RMS	Peak	Watts	BTU/hr
Five-Slot Module	*	*	*	*	*
Typical	1.10/0.88	1.75/1.30	8.60/1.75	210/156	716/532
Maximum	1.63/1.35	2.65/2.02	10.8/9.06	271/206	924/702
14-Inch Monitor					
Typical	0.53	0.85	2.0	64	218
Maximum	0.75	1.25	3.0	150	512
21-Inch Monitor					
Typical	0.8	1.5	4.1	83	283
Maximum	1.2	2.4	6.6	160	545
Cartridge Drive	*	*	*	*	*
Typical	0.57/0.37	0.89/0.57	5.50/1.90	107/68	365/232
Maximum	1.30/0.82	2.05/1.20	9.10/3.24	210/122	716/416
Floppy Drive	*	*	*	*	*
Typical	0.17/0.37	0.34/0.58	2.90/1.92	41/69	140/235
Maximum	0.38/0.74	0.98/1.08	5.30/2.95	100/110	341/376

* The first value is for a UWS with the early production Five-Slot Module power supply (51400712). The second value is for a later production enhanced power supply (51195066).

2.6 Unpacking

Inspection

When the equipment arrives at the system site, open each shipping box, remove the protective wrapping, and carefully inspect each piece of equipment for any physical damage. If damaged, immediately notify the carrier and your Honeywell sales representative as to extent and type of damage. Also, check each piece of equipment against the invoice list for missing items.

If the equipment is to be stored, repackage the equipment and place it back in the original shipping container.

2.7 Assembly and Cabling

ATTENTION

ATTENTION—For convenience, do all LCN cabling, LCN node address configuration, keyboard configuration, and component cabling at the same time because all the UWS cabling enters the cabinet through the rear panel which must be removed.

2.7.1 LCN Cabling and Node Address Configuration

LCN node number	The LCN node address is unique for each UWS. The node number is configured on the CLCN A/B (I/O) board by setting slide switches and on the LCN I/O board by inserting jumpers.
------------------------	---

Accessing the LCN I/O or CLCN A/B I/O board	To access the rear of the Five-Slot Module and configure the LCN I/O or CLCN A/B (I/O) board, the rear cover must be removed from the UWS cabinet. Use the procedure in Table 2-5 and reference Figure 2-1 as a guide.
--	--

Accessing the card cage	Use the procedure in Table 2-5 to access the Five-Slot Module's card cage and configure the LCN I/O or CLCN A/B board for the appropriate LCN node address.
--------------------------------	---

Table 2-5 Accessing the Five-Slot Module I/O Card Cage

Step	Action
1	If applicable, place the power switch at the front of the cabinet in the OFF (0) position. Disconnect the ac power cord from from its connector at the rear of the cabinet.
2	Remove the six screws and star washers that secure the rear panel as illustrated in Figure 2-1 for a non-CE Compliant UWS cabinet or Figure 2-2 for a CE Compliant cabinet.
3	Carefully set the rear panel close to, and beside, the cabinet's rear opening. Do not strain any cable connections.
4	Remove the LCN I/O or CLCN A/B (I/O) board from slot 1 of the card cage at the rear of the Five-Slot Module and place it on an ESD-safe work surface. The CLCN A/B board's two faceplate screws must be loosened.
5	Configure the LCN I/O or CLCN A/B board for the appropriate LCN node address as illustrated in Figure 2-3 (LCN I/O) or Figure 2-4 (CLCN A/B).
6	Reinstall the LCN I/O or CLCN A/B board in slot 1. Be sure the two screws at each side of the CLCN A/B board's faceplate are tight to ensure proper grounding of the faceplate and the cable shielding.

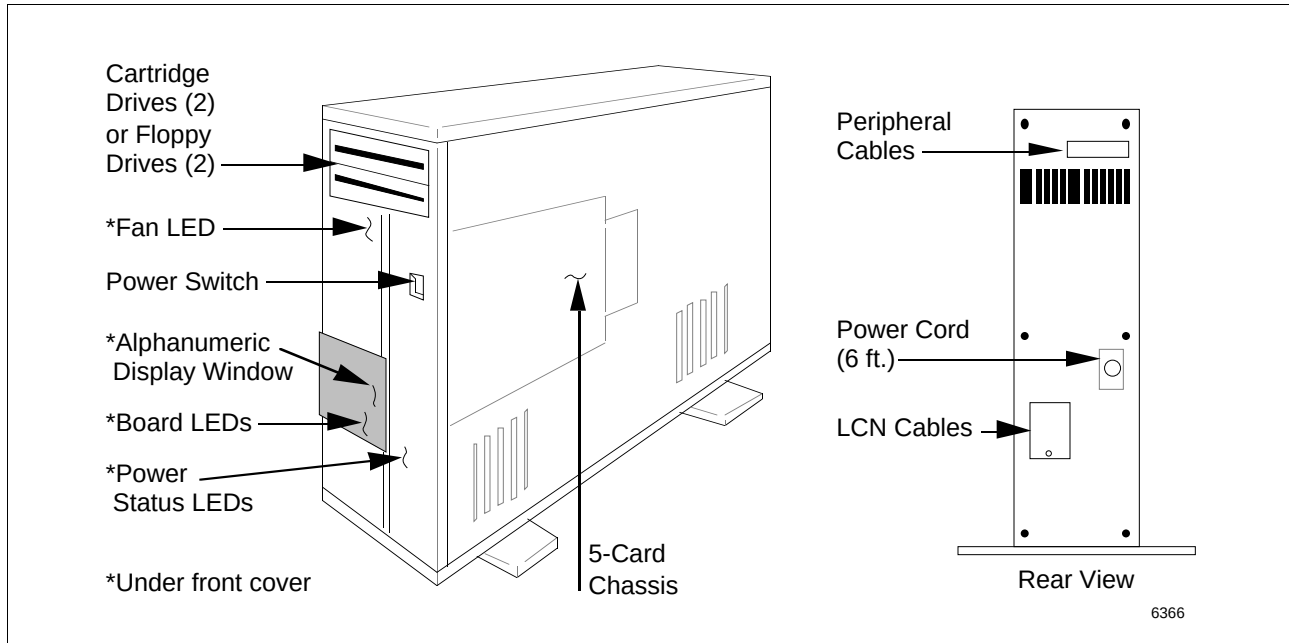
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2.7.1 LCN Cabling and Node Address Configuration, Continued

Non-CE Compliant UWS cabinet

Figure 2-1 provides front and rear views of the non-CE Compliant UWS cabinet.

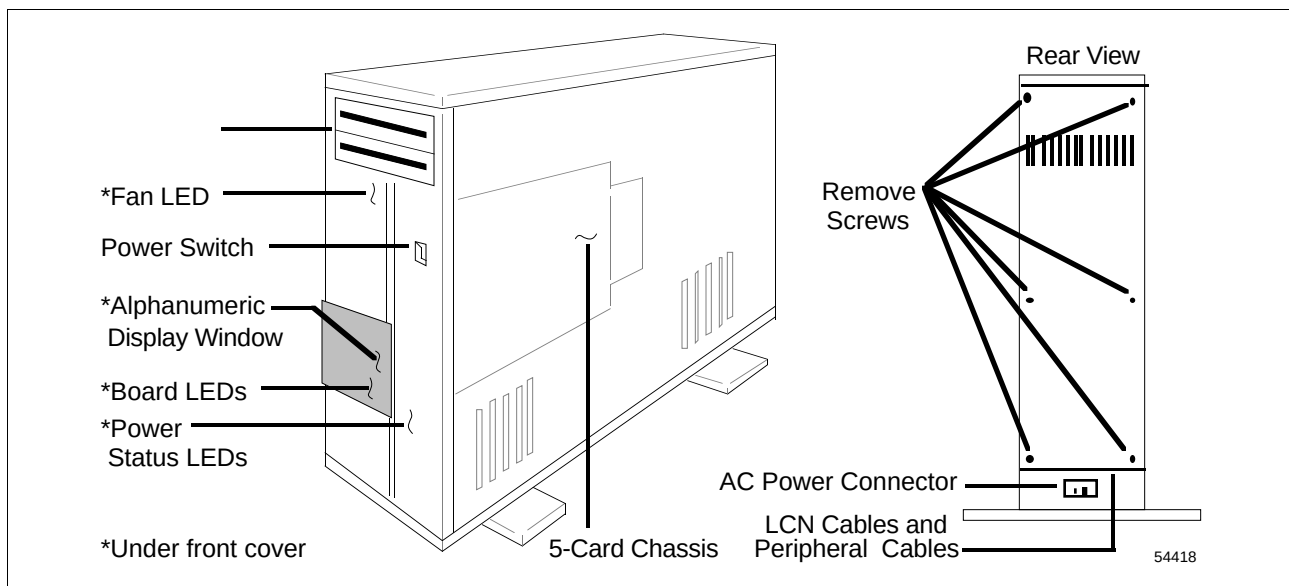
Figure 2-1 Front and Rear Views of Non-CE Compliant UWS Cabinet



CE Compliant UWS cabinet

Figure 2-2 provides front and rear views of the CE Compliant UWS cabinet.

Figure 2-2 Front and Rear Views of CE Compliant UWS Cabinet



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2.7.1 LCN Cabling and Node Address Configuration, Continued

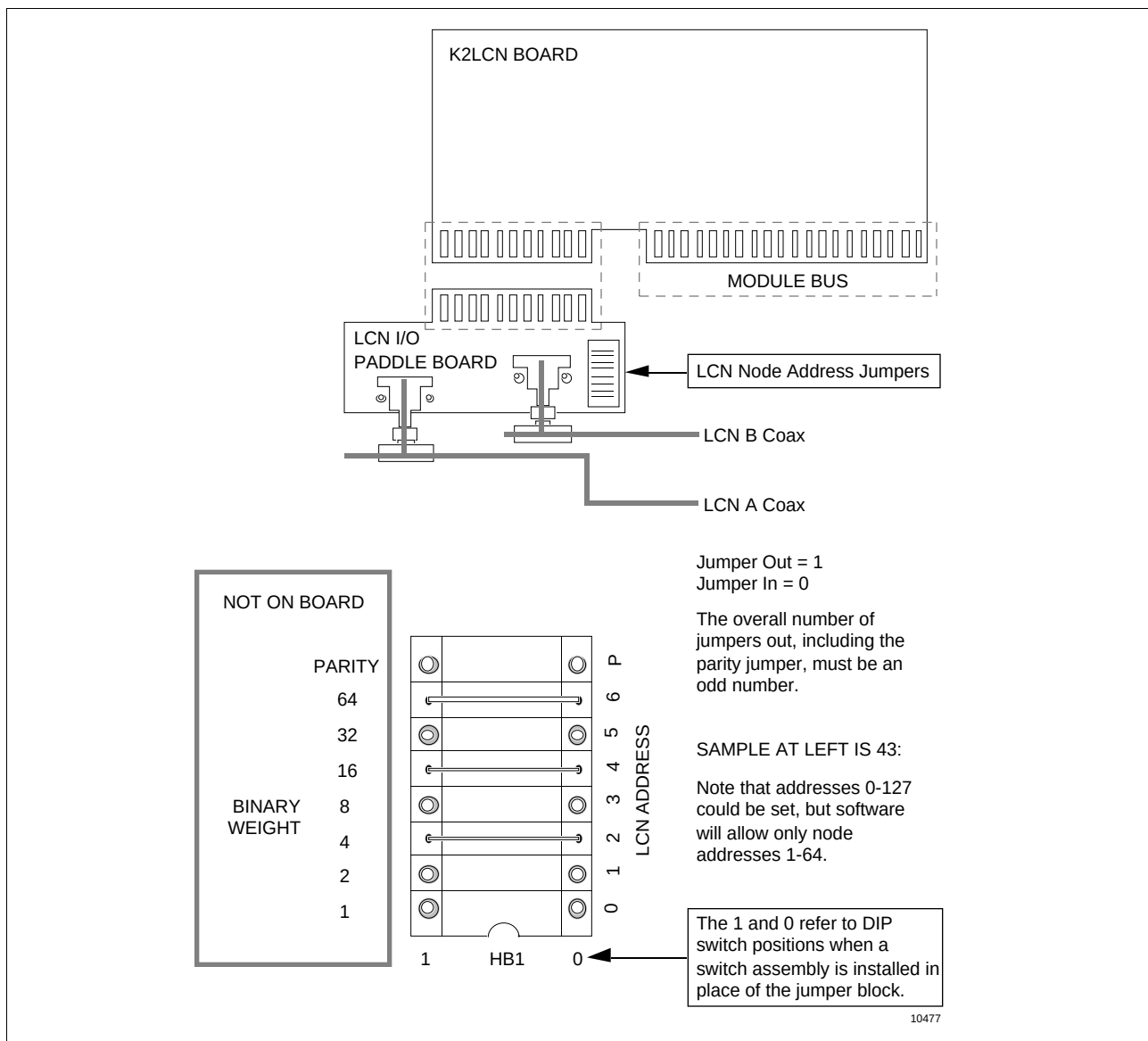
LCN I/O board configuration example

Figure 2-3 illustrates an LCN node address configuration example of 43. The binary number equivalent is “101011.” A missing jumper represents a binary one (1).

An odd number of jumpers, which includes the parity position “P,” must be missing. In this example of an address of 43, there is no jumper installed in position “P.”

Switches are used in place of jumpers on later production boards. A switch setting of zero (0) represents a binary one (1).

Figure 2-3 LCN I/O Board LCN Node Address Configuration



Continued on next page

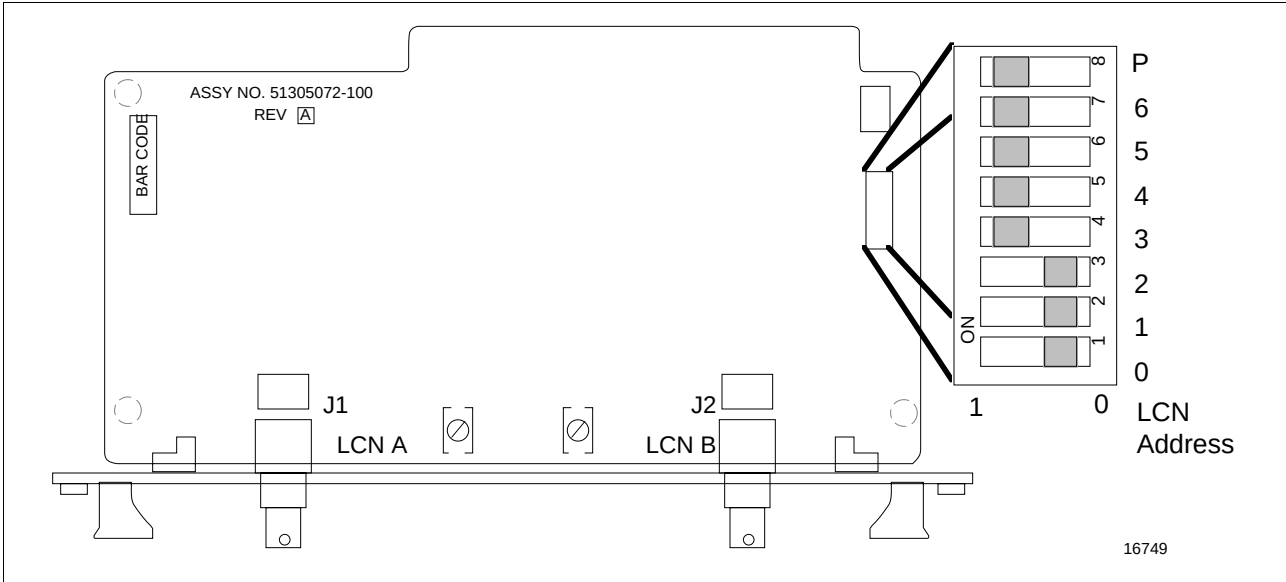
2.7.1 LCN Cabling and Node Address Configuration, Continued

CLCN A/B board configuration example

Figure 2-4 illustrates a configuration example of an LCN node address of 7. The binary number equivalent is “111.” A switch setting of zero (0) represents a binary one (1).

An odd number of switches, which includes the parity switch “P,” must be set in the “0” position. In this example of an LCN address of 7, the “P” switch is set to the “1” position.

Figure 2-4 CLCN A/B Board LCN Node Address Configuration



Continued on next page

2.7.1 LCN Cabling and Node Address Configuration, Continued

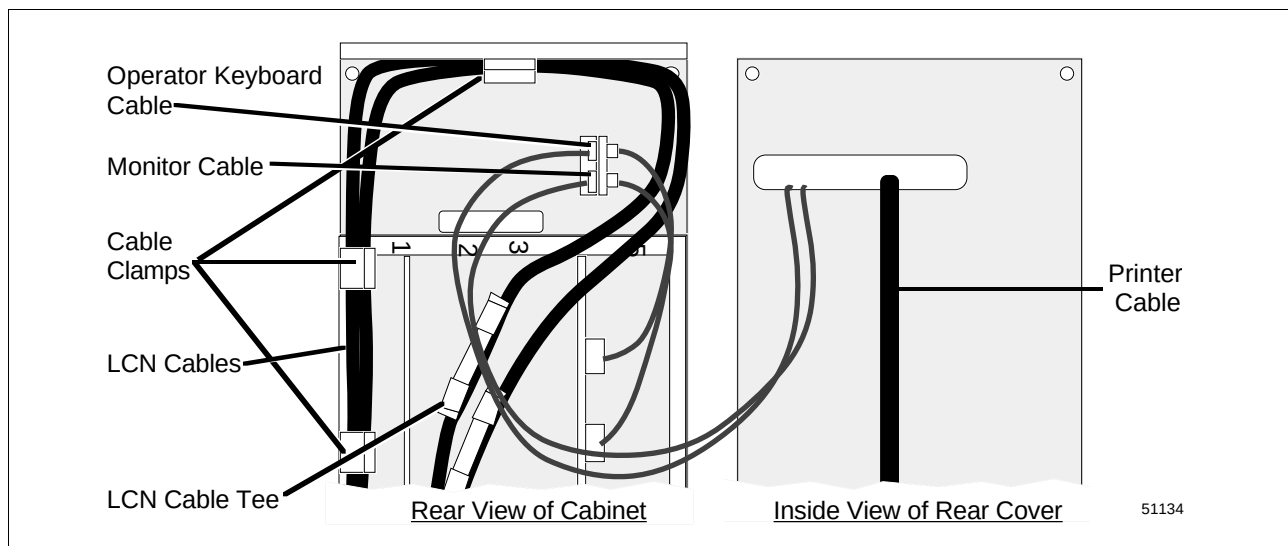
LCN cabling procedure After configuring the LCN I/O or CLCN A/B board for the appropriate LCN node address, use the procedure in Table 2-6 to install the LCN cables.

Table 2-6 LCN Cable Installation Procedure

Step	Action
1	Route the LCN cables through the cable cutout in the cabinet's rear panel. See Figures 2-1 or 2-2.
2	Connect LCN Cable A (color-coded green) to the connector that is labeled LCN A on the LCN I/O or CLCN A/B board. Connect LCN Cable B (color-coded yellow) to the connector that is labeled LCN B. The LCN cables must be connected to the LCN I/O or CLCN A/B board with two (2) BNC T-connectors (51190728-105). If the T-connectors are at the end of the LCN, they must be terminated with a terminator (30732052-001) connected to the unused connection of the T-connector. It is essential that Cable A is Cable A and Cable B is Cable B for all LCN nodes. Be certain that the LCN cables are not crossed (Cable A to Cable B). Also, do not substitute unauthorized cables for Honeywell-approved LCN cables.
3	Loop the LCN cables around the top inside of the cabinet as illustrated in Figure 2-5. Secure the cables with cable clamps on the side and top of the cabinet as shown. Do not have a bend radius that is less than the width of the cabinet.

LCN cable UWS cabinet routing Figure 2-5 illustrates the LCN cable routing inside the UWS cabinet.

Figure 2-5 UWS Internal Cabinet LCN Cable Routing



2.7.2 EPDGP I/O or EPDGC I/O Board Cabling and Configuration

CAUTION

CAUTION—Some cables may already be attached to the EPDGP I/O or EPDGC I/O board. Be careful not to disturb the connection during the following procedures.

ATTENTION

ATTENTION—The adapter cables that are used in Steps 2 and 3 may have been installed in the factory with their connectors mounted and secured on top of the I/O card cage as illustrated in Figure 2-5.

Continued on next page

2.7.2 EPDGP I/O or EPDGC I/O Board Cabling and Configuration, Continued

EPDGP I/O cabling and configuration

If necessary, use the procedure in Table 2-7 to cable and configure the EPDGP I/O board.

Table 2-7 EPDGP I/O Board Cabling and Configuration Procedure

Step	Action
1	If applicable, place the power switch at the front of the cabinet in the OFF (0) position. Disconnect the ac power cord from from its connector at the rear of the cabinet.
2	Remove the six screws and star washers that secure the rear panel as illustrated in Figure 2-1 for a non-CE Compliant UWS cabinet or Figure 2-2 for a CE Compliant cabinet.
3	Carefully set the rear panel close to, and beside, the cabinet's rear opening. Do not strain any cable connections.
4	Remove the EPDGP I/O board from slot 4 at the rear of the Five-Slot Module and place it on an ESD-safe work surface.
5	Determine if you have a Supervisor's or Engineer's keyboard. Configure TS1 on the EPDGP I/O board for either the Supervisor's or Engineer's keyboard as illustrated in Figure 2-6. If the UWS has the Enhanced keyboard assembly (both the Operator's and Engineer's keyboard), configure the EPDGP I/O board as a Supervisor's keyboard.
6	If appropriate, connect one end of the mouse adapter cable (51304033-045) to the mouse connector, J1 on the EPDGP I/O board. See Figures 2-7 and 2-8. Route the mouse cable through the rear panel cutout and connect the mouse cable to the adapter cable. Fasten the mating ends of the adapter and mouse cables to the top of the I/O card cage with two cable ties. Carefully reinstall the EPDGP I/O board.
7	If appropriate, connect one end of the monitor adapter cable (51304399-100) to the monitor connector, J7, on the EPDGP I/O board. See Figures 2-7 and 2-8. Route the monitor cable through the rear panel cutout and connect the monitor cable to the adapter cable. Fasten the mating ends of the adapter and monitor cables to the top of the I/O card cage with two cable ties. Connect the BNC ends of the monitor cable to the connectors on the monitor. Match the red, green, and blue color-coded labels. Connect the grey color-coded connector to H/HV and the black color-coded connector to V.
8	If appropriate, route the keyboard cable through the rear panel cutout and connect it to either the Supervisor connector, J3, or the Engineer connector, J6.
9	If appropriate, route the printer cable through the rear panel cutout and connect it to the printer connector, J2.

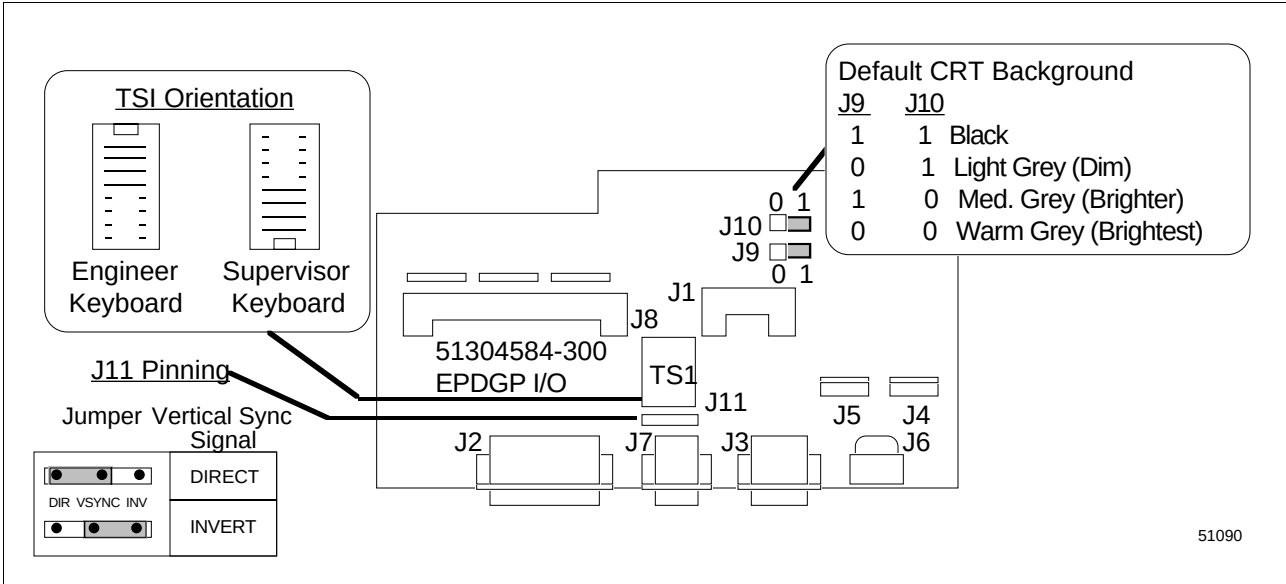
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2.7.2 EPDGP I/O or EPDGC I/O Board Cabling and Configuration, Continued

EPDGP I/O board configuration

Figure 2-6 illustrates the configuration options on the EPDGP I/O board.

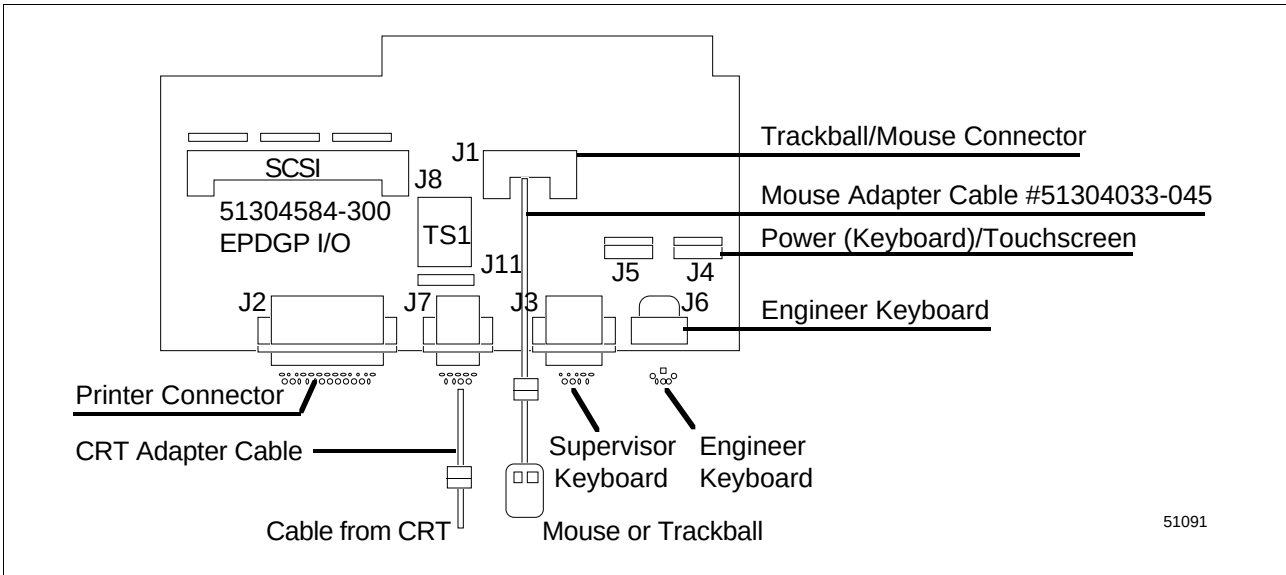
Figure 2-6 EPDGP I/O Board Configuration



EPDGP I/O board cabling

Figure 2-7 illustrates the cabling to the EPDGP I/O board.

Figure 2-7 EPDGP I/O Board Cabling



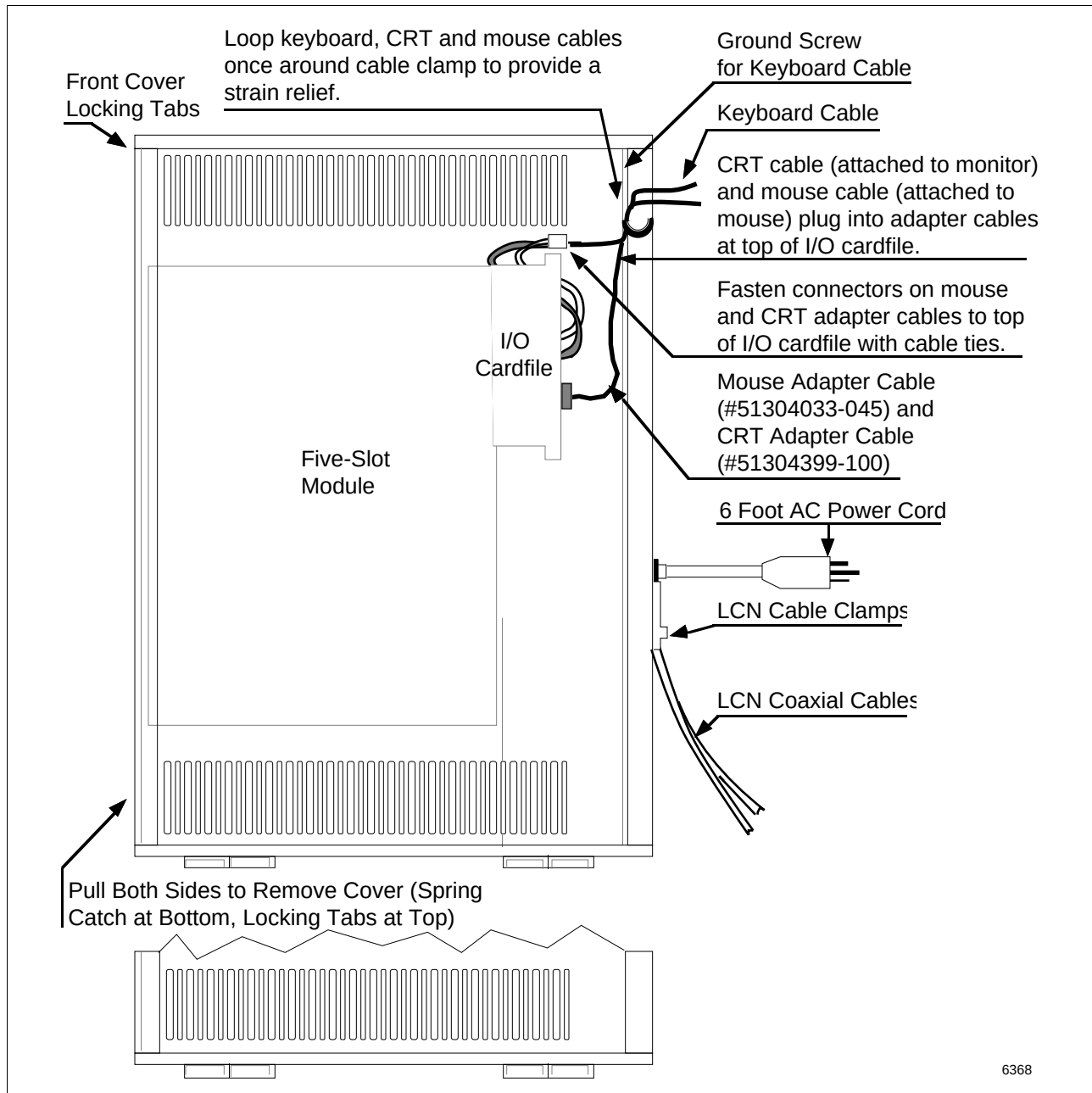
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2.7.2 EPDGP I/O or EPDGC I/O Board Cabling and Configuration, Continued

UWS cable routing

Figure 2-8 illustrates the UWS cable routing.

Figure 2-8 UWS Cable Routing



Continued on next page

2.7.2 EPDGP I/O or EPDGC I/O Board Cabling and Configuration, Continued

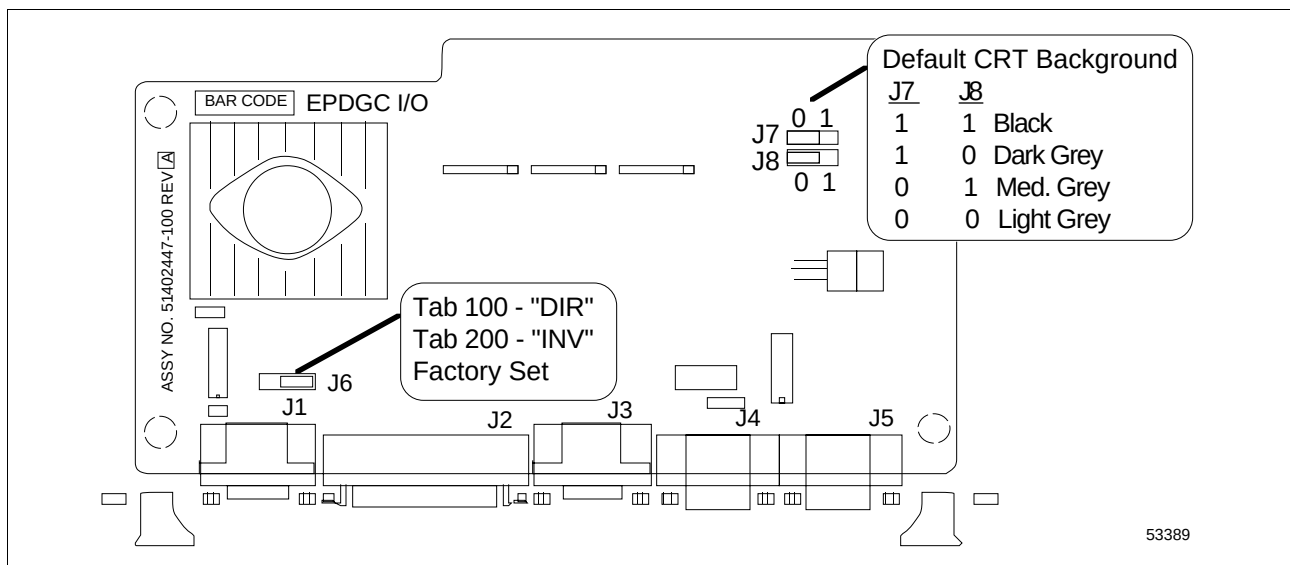
EPDGC I/O board monitor configuration

The EPDGC I/O board has two monitor functions that are normally configured at the factory.

- The background color that is displayed on the monitor for the period of time that the station is being loaded and the color palate is active. The default configuration is a black background.
- Synchronization of the monitor is normally configured at the factory for **DIR**. If an Hitachi monitor is installed, the board can be configured for **DIR** or **INV**.

Figure 2-9 illustrates the slide switch choices at J6, J7, and J8 on the EPDGC I/O board.

Figure 2-9 EPDGC I/O Board Monitor Configuration



2.8 UWS Component Assembly

UWS component installation procedure

The equipment components that constitute the UWS are designed to be positioned adjacent to and on top of a single table or desk in close proximity of each other. Refer to Figure 1-1 for a typical installation. The procedure in Table 2-8 assumes that the equipment being installed is the current production CE Compliant hardware.

Table 2-8 UWS Component Installation Procedure

Step	Action
1	Place the UWS electronics cabinet on the floor beside a table or desk with the front of the cabinet facing the same direction that the monitor will face.
2	Place the ac power switch at the front of the electronics cabinet in the OFF (0) position.
3	Place the monitor on the table top or desktop.
4	Place the printer on the table top or desktop within cabling distance of the electronics cabinet.
5	Place the keyboard(s) and optional mouse on top of the table or desktop in front of the monitor.
6	Remove the electronics cabinet's rear panel by removing the six screws and star washers that secure it in place.
7	Remove the bundle of cables from the rear of the tower by unrolling the cables. One end of the cables are connected to the EPDGC I/O board and are fastened to the cable bulkhead with ties.
8	Locate the keyboard cable, 51308105-200, that is connected to the KEYBD connector on the faceplate of the EPDGC I/O board in slot 4 of the I/O card cage. Connect the loose end of the keyboard cable, 51308105-200, to the Supervisor's keyboard.
9	If you are installing an Engineer's keyboard (second keyboard), connect it to the connector that is located at the right-hand rear edge of the Supervisor's keyboard.
10	Locate the monitor interface cable, 51192126-400, that is connected to the VIDEO connector on the faceplate of the EPDGC I/O board. Connect the five BNC type connectors to the connectors at the back of the monitor by matching the color codes of the connectors. Refer to Figure 2-10.
11	Locate the printer cable, 51308103-0xx, that is connected to the PRINTER connector on the EPDGC I/O board in slot 4. Connect the loose end of the cable to the printer.

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2.8 UWS Component Assembly, Continued

UWS component assembly procedure, continued

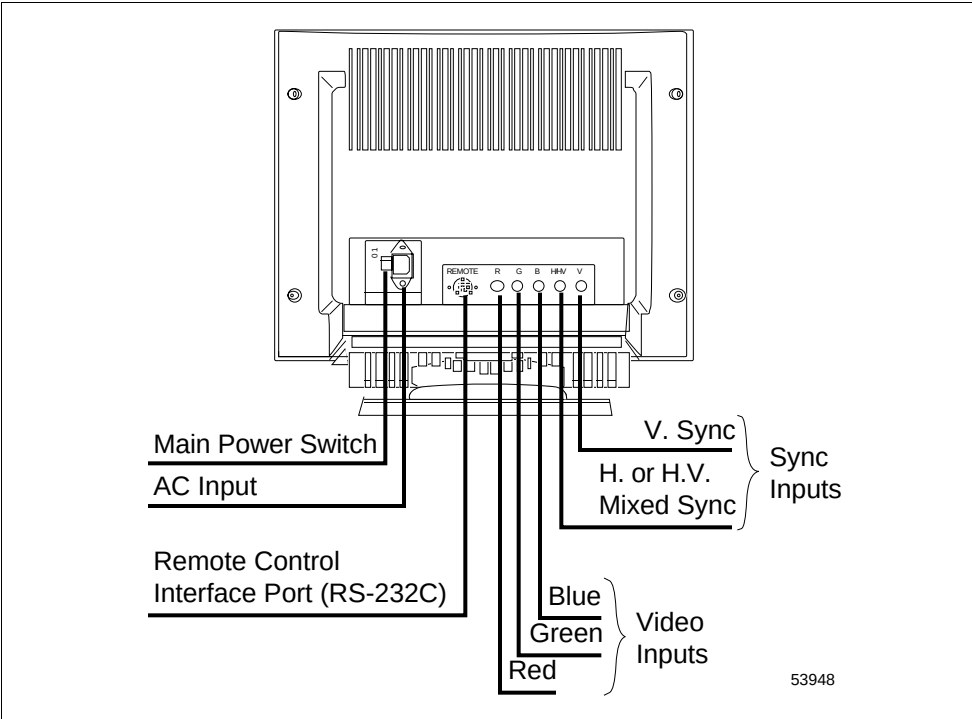
Table 2-5 UWS Component Assembly procedure, Continued

Step	Action
12	If you are installing a mouse, model MP-UWSMS1, connect the mouse cable to the MOUSE connector on the cable bulkhead. A short adapter cable is connected between the CURSOR connector on the EPDGC I/O board and the MOUSE connector on the cable bulkhead.
13	If you are installing a touchscreen, model MP-UWSTS1, that is mounted on the front of the monitor, connect the interface cable, 51308114-100, to the touchscreen control assembly that is mounted at the rear bottom of the Five-Slot Module.
14	If there is a RULA option in the cabinet (a small assembly that is located at the bottom of the cabinet) the transceiver cable, 51308138-100, must be connected between the transceiver connector on the bulkhead and the AUI unit that is connected to the Ethernet. Refer to Figure 2-11

Hitachi 4821-D monitor cable connectors

Figure 2-10 illustrates the location of the cable connectors and switches at the rear of the Hitachi model 4821-D monitor.

Figure 2-10 Hitachi Model 4821-D Monitor Cable Connections

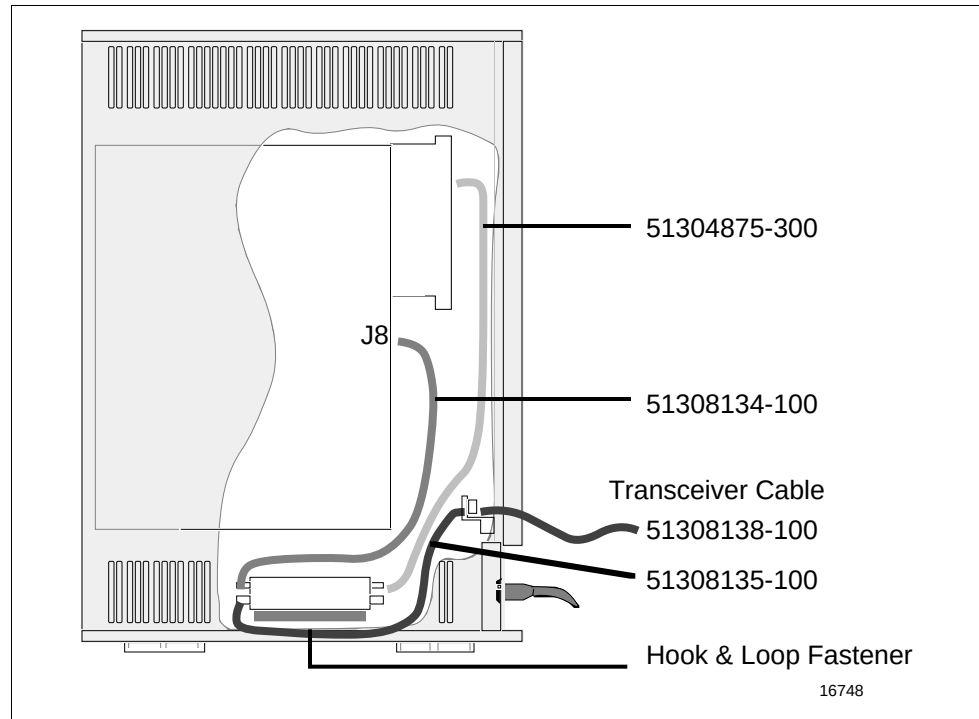


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2.8 Assembly of UWS Components, Continued

RULA cable installation Figure 2-11 illustrates the location of the RULA unit and its cabling.

Figure 2-11 RULA Cable Installation



2.9 LCN Cabling Installation

Introduction

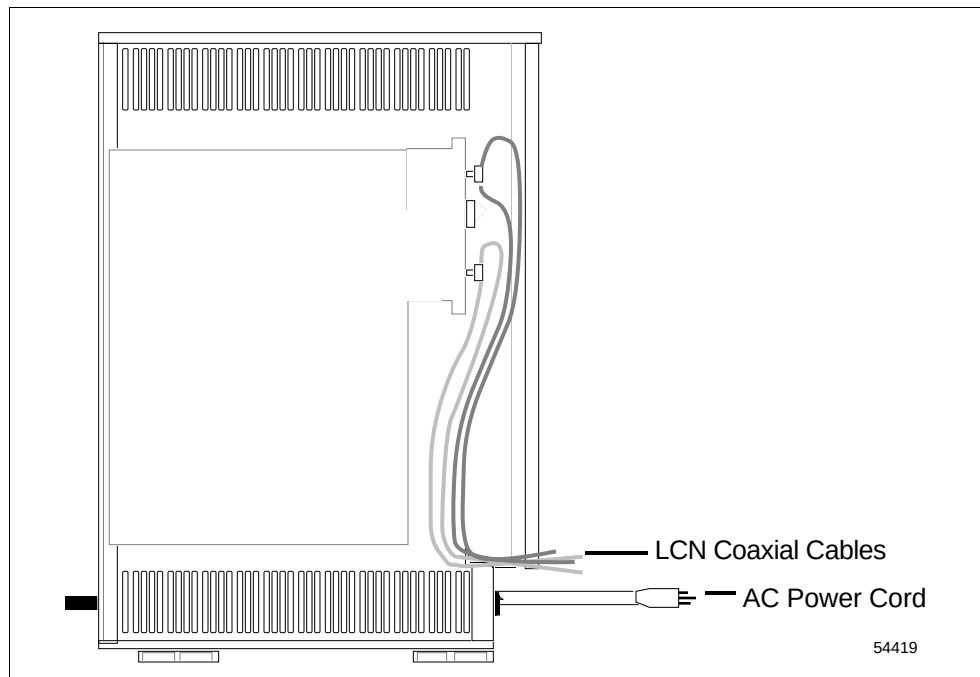
The LCN Cable A and Cable B coaxial cables connect to the CLCN A/B (I/O) board through a 51190728-105 T-connector as illustrated in Figure 2-12. Consider the following.

- If the cable is continuing on to another LCN node, connect a coax cable, 51109181-xxx, to both ends of the T-connector.
- If the series of LCN cables end at the T-connector that is connected to the UWS, terminate the network cables by connecting a 75-ohm terminator, 30732052-105, to the unused side of the T-connector.

UWS coaxial cable routing

Figure 2-12 illustrates the routing of the LCN coaxial cable in the UWS cabinet.

Figure 2-12 UWS LCN Coaxial Cable Routing



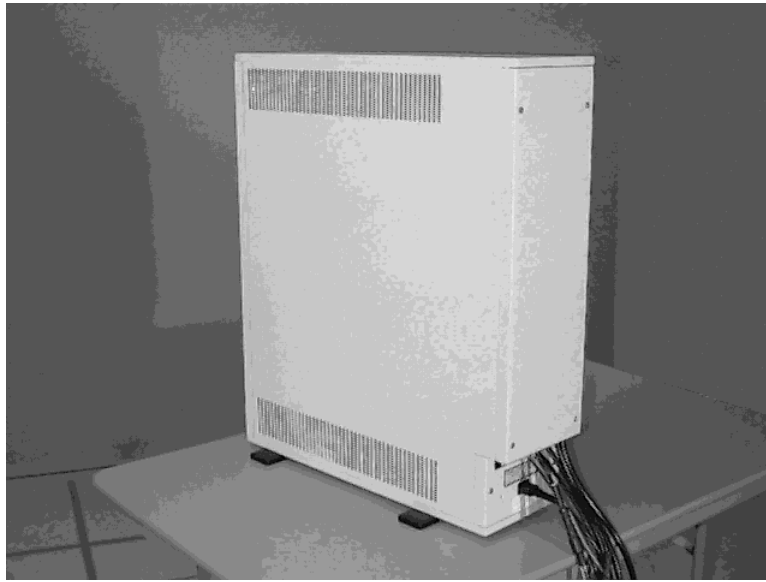
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2.9 LCN Cabling Installation, Continued

CE Compliant UWS entry cabling

Figure 2-13 illustrates the CE Compliant UWS entry cabling.

Figure 2-13 CE Compliant UWS Cable Routing



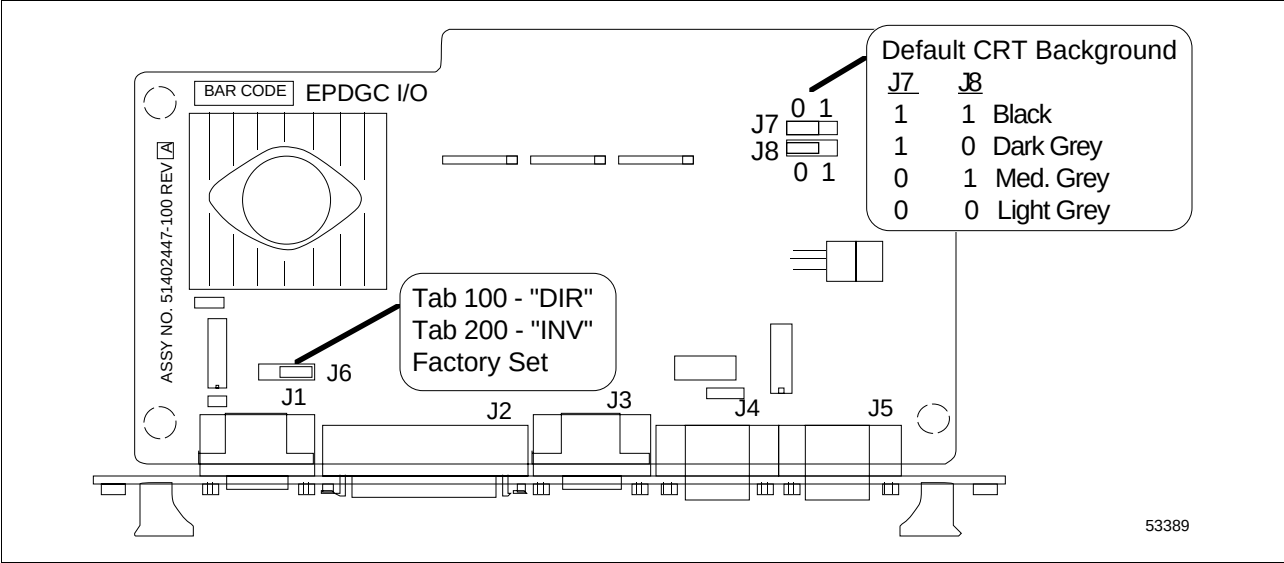
54421

2.10 EPDGC Pinning and Cabling

EPDGC I/O raster background color pinning

Figure 2-14 illustrates the pinning on the EPDGC I/O board for raster background color pinning.

Figure 2-14 EPDGC I/O Board Raster Background Color Pinning



Cable restraints

Cable restraint is provided by tying the cables to the restraint brackets with cable ties.

2.11 Power Connections

The current production CE Compliant UWS cabinet has an ac power connector on the outside of the cabinet, below the rear panel. Use the procedure in Table 2-9 to install the ac power cables.

Table 2-9 AC Power Connections

Step	Action
1	Attach the ac power cords to the monitor and optional printer.
2	Connect their power cords to a designated utility socket.
3	If the UWS has a 120 Vac power cord, 51190921-110, or 240 Vac power cord, 51190921-210, connect it to the rear of the UWS cabinet.
4	Connect the ac power cord to a designated utility socket.

2.12 Cabinet Closure

Cabinet closure procedure

Use the procedure in Table 2-10 to close the cabinet after cable installation is complete.

Table 2-10 Replacing Rear Cover

Step	Action
1	Replace the rear cover panel being careful not pinch any cables.
2	Replace the six screws and star washers. Tighten the screws firmly to provide good EMI protection.

Section 3 – Installation Checkout

3.1 Overview

Section contents

These are the topics covered in this section:

	Topic	See Page
3.1	Overview	29
3.2	Applying Power	30
3.3	Tests.....	32

Introduction

This section details equipment checkout procedures that should be followed after power has been applied to the UWS components.

Preliminary checks

After the power is applied to the electronics cabinet, monitor, and optional printer, there are a few preliminary checks that should be performed before loading the station with the personality software.

- Node number display
 - LED indicators
 - Monitor cursor
-

3.2 Applying Power

Applying power

Use the procedure in Table 3-1 to apply power to the UWS components in an orderly manner.

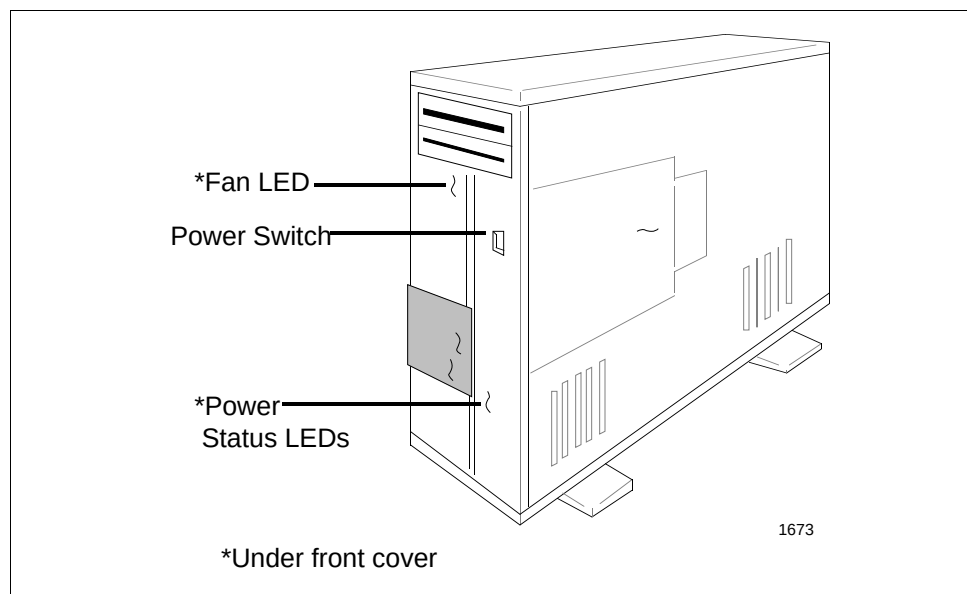
Table 3-1 Power-On Procedure

Step	Action
1	Place the ac power switch at the front of the UWS electronics cabinet in the ON (1) position. The location of the switch is illustrated in Figure 3-1.
2	The 21-inch Hitachi monitor has a power switch at the rear of the monitor. Place the ac power switch at the back of the monitor in the ON (1) position. The location of the switch is illustrated in Figure 3-2.
3	Place the ac power switch at the front of the monitor in the ON (1) position. The location of the switch is illustrated in Figure 3-3.
4	Place the optional printer's power switch in the ON (1) position. The location of the power switch is dependent upon the printer model that is in use.

UWS cabinet ac power switch and indicators

Figure 3-1 illustrates the location of the UWS electronics cabinet's ac power switch and the power indicators.

Figure 3-1 UWS Cabinet AC Power Switch and Indicators Locations



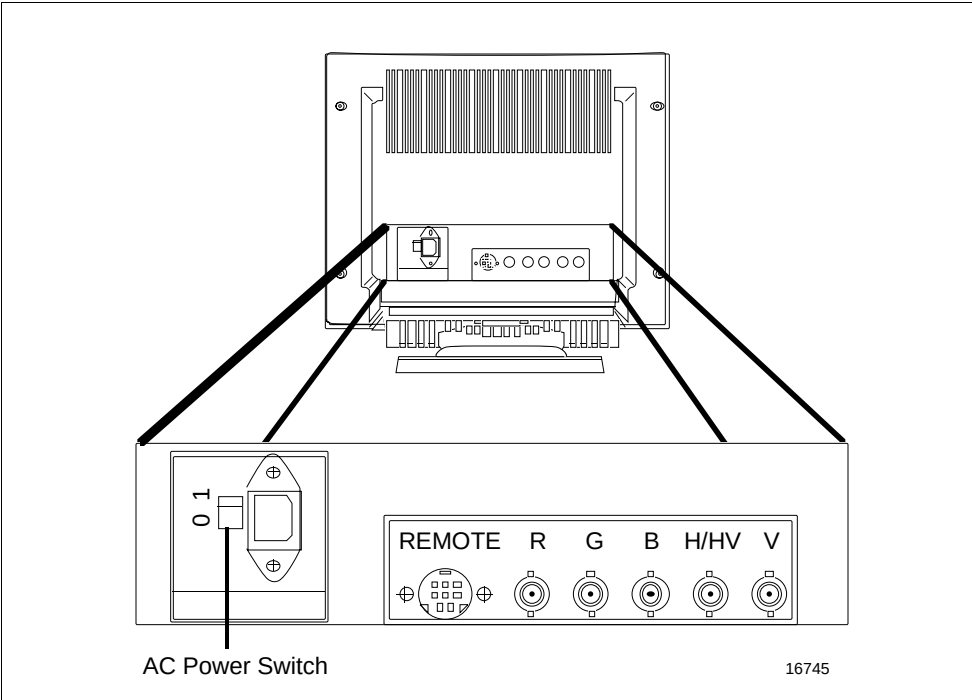
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3.2 Applying Power, Continued

Rear ac power switch

Figure 3-2 illustrates the location of the ac power switch at the rear of the Hitachi monitor.

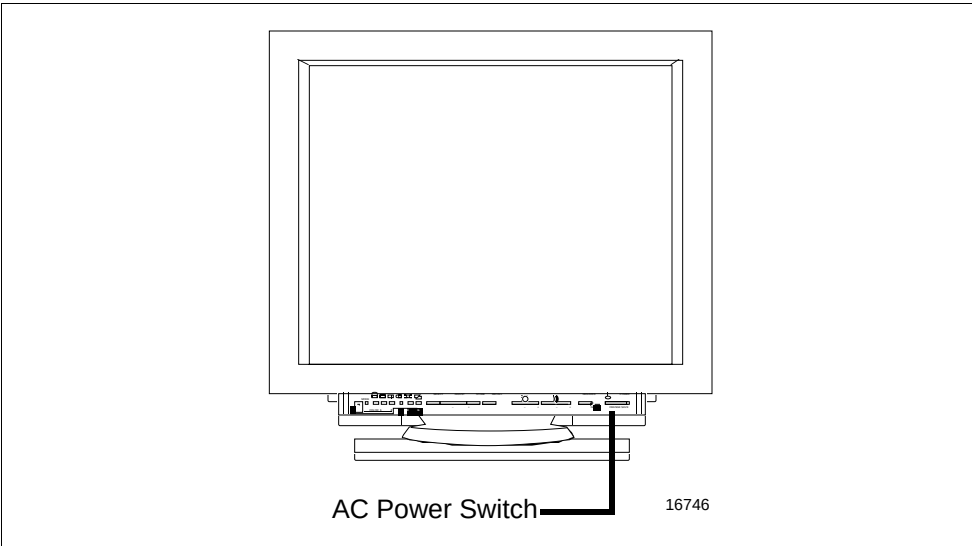
Figure 3-2 Hitachi Monitor Rear Power Switch



Front ac power switch

Figure 3-3 illustrates the location of the ac power switch at the front of the Hitachi monitor.

Figure 3-3 Hitachi Monitor Front Power Switch



3.3 Tests and Adjustments

Power-on self-tests and indicators

Use Table 3-2 to verify that the UWS successfully completed its power-on self-tests and is operational.

Table 3-2 Power-On Tests and Indicators

Step	Action
1	After applying power to electronics cabinet, check the indicators on the Five-Slot Module's power supply. • If the green LED is lit, go to Step 2. • If neither the red or green LEDs are lit, check to be sure that the ac power cord is connected to the cabinet and an ac power source. • If the red LED is lit, troubleshoot the power supply. Refer to the <i>Five-Slot Module Service</i> manual for guidance.
2	Check the monitor for a ">" symbol above "square" symbol. This is the cursor. When the cursor is present, the appropriate LCN node number will be displayed on the processor board (HMPU, HPK2, K2LCN, or K4LCN). This indicates that the board's self-tests have been successfully completed. If a number other than the correct LCN node number is displayed, refer to the <i>Five/Ten-Slot Module Service</i> manual for troubleshooting guidance.
3	Check the indicators on the EPDGC board. If the red indicator is lit, refer to the <i>Five/Ten-Slot Module Service</i> manual for troubleshooting guidance.

Bernoulli cartridge drive testing

Use Table 3-3 to confirm that the Bernoulli cartridge drives are functioning properly.

Table 3-3 Bernoulli Cartridge Drive Functionality Testing

Step	Action
1	If the UWS has Bernoulli cartridge drives, insert a cartridge into the drive.
2	The LED at the front of the drive will blink for 5 to 10 seconds and then remain on. The yellow LED will blink once. After the initial cartridge startup, the yellow LED will flash only when the cartridge is accessed.
3	Remove the cartridge from the drive by pressing the stop button at the front of the drive. The green LED should blink for several seconds while the drive spins down, and then go out.
4	Remove the cartridge from the drive. DO NOT ATTEMPT TO REMOVE THE CARTRIDGE FROM THE DRIVE WHILE THE GREEN LED IS BLINKING.

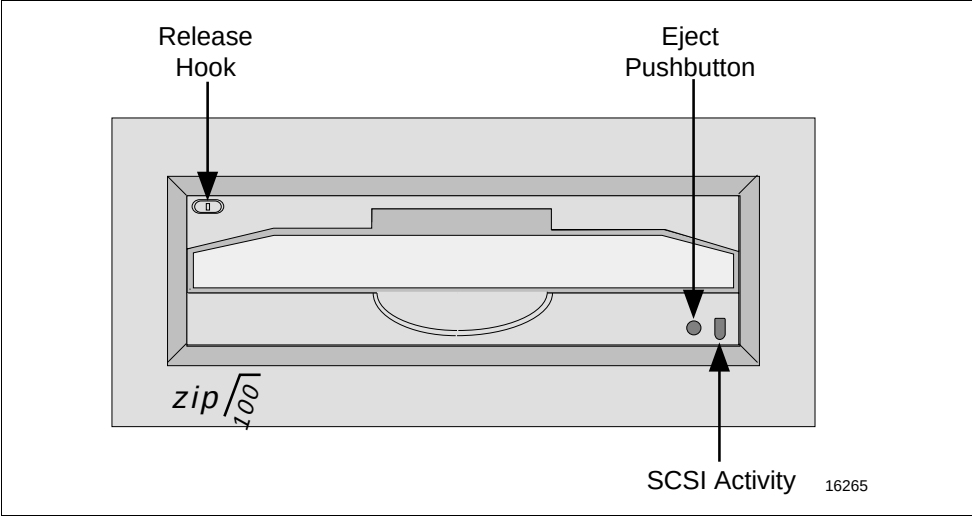
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3.3 Tests and Adjustments, Continued

Front panel of Zip drive

Figure 3-4 illustrates the front panel of the Zip disk drive and identifies its features. Depressing the eject pushbutton normally ejects the disk. If the disk cannot be ejected by this method because of an electronic failure, the release hook that is located in the upper left corner of the front panel can be used to mechanically eject the disk. Use a small paper clip or similar tool to release the disk by pulling forward on the hook.

Figure 3-4 Zip Disk Drive Front Panel



Bernoulli cartridge drive testing

Use Table 3-4 to confirm that the Zip disk drives are functioning properly.

Table 3-4 Zip Disk Drive Functionality Testing

Step	Action
1	Insert a disk into the drive with a smooth motion until a click is felt. The disk should start to rotate and the activity LED indicator should flash and then extinguish. The edge of the disk should be flush with the front surface of the drive.
2	The LED indicator is essentially a disk activity monitor. The intensity of the indicator represents the rate of data transfer on the SCSI bus.

Continued on next page

3.3 Tests and Adjustments, Continued

Hitachi monitor startup	Adjust the brightness and contrast of the 21-inch Hitachi monitor screen for the best resolution of the blinking character that appears on the upper-left corner of the screen. The brightness and contrast controls are the two left-most controls under the front of the monitor. Position (tilt) the screen display for comfortable viewing. Refer to Section 4 for any monitor adjustments that may be required.
Printer testing	If you have an optional printer, run the printer's self-tests. Refer to the appropriate section in the <i>Universal Station Service</i> manual for a detailed description of the self-tests and the procedure.

Section 4 – Implementation

4.1 Overview

Section contents

These are the topics covered in this section:

	Topic	See Page
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4.2	Network Configuration.....	35
4.3	Area Names.....	36
4.4	Console Names	36
4.5	LCN Nodes.....	37
4.6	Button Configuration.....	38
4.6.1	Engineer's Keyboard PF1-PF17 Key Configuration	39
4.6.2	Supervisor's Keyboard Keyswitch Buttons Configuration	41
4.7	Area Configuration.....	42

Introduction

This section guides you through the configuration and data entry process for the Universal Work Station (UWS). If you have been through a network, area, and button configuration, you may be able to implement your UWS with little or no reference to the standard LCN bookset, but in case you need them, references are provided to the publications that contain the detailed implementation references and instructions.

4.2 Network Configuration

Introduction

Each of the network configuration activities that you use can be made effective through the use of an on-line reconfiguration installation unit by allowing the LCN and most of the nodes to remain operational and the process to remain “on control.” If you are adding the UWS to an LCN that is inoperative or on which all nodes need to be shut down and restarted, you can use off-line network configuration to add the UWS to the network.

- Off-line network configuration procedures are provided in subsection 7.3 in the *Network Data Entry* manual.
 - On-line reconfiguration procedures are provided in subsection 7.4 in the *Network Data Entry* manual.
 - Instructions for adding a node to an operating LCN are provided in the *LCN Guidelines - Implementation, Troubleshooting, and Service* manual.
-

4.3 Area Names

Using an existing area database

If you are going to use an existing area database in your UWS, or if you are going to use only the Engineer personality, no configuration data needs to be entered.

If you are using an existing area database, you should record the exact area name for use in subsection 4.5, *LCN Nodes*. The area name should be recorded on *Network Configuration Form SW88-507* in the *Network Forms* manual. It may also be available in an NCF printout. This procedure is in Table 7-16 in the *Network Data Entry* manual.

Configuring a unique or new area database

If you are configuring a unique area database for the UWS, or if it is to be a member of a new area, you should add the new area name and description on *Area Names Configuration Form SW88-507* in the *Network Forms* manual, and you need to enter it through on-line reconfiguration or off-line network configuration.

For form instructions, see subsection 7.3 in the *Network Form Instructions* manual.

For data entry instructions, see subsection 7.3 or 7.4 in the *Data Entry Instructions* manual.

4.4 Console Names

Existing console

If you are configuring the UWS as a member of an existing console, you need to note the console number for use in subsection 4.5, *LCN Nodes*. The console numbers and names should be recorded on *Network Form SW88-508* in the *Network Forms* manual. They may also be available in an NCF printout. This procedure is available in Table 7-17 in the *Network Data Entry* manual.

New console

If you are configuring the UWS as a new one-station console, or as a member of a new console, you should add the new console description on *Network Form SW88-508*, in the *Network Forms* manual, and you need to enter it through on-line reconfiguration or off-line network configuration.

For form instructions, see subsection 7.3 in the *Network Form Instructions* manual.

For data entry instructions, see subsection 7.3 or 7.4 in the *Data Entry Instructions* manual.

4.5 LCN Nodes

Required additions

You should add the UWS as a new node on the *Network Form SW88-509* in the *Network Forms* manual. For the node number assigned to the UWS, enter US in the NODE TYPE column. The node number is selected on the CLCN A/B board. In the REDUNDANT NODE column, enter N.

On the *Network Form SW88-510* in the *Network Forms* manual, enter data for the UWS as specified in Table 4-1.

Table 4-1 Network Form SW88-510 Information

NODE	The UWS node number
CONSOLE #	Enter the console number recorded in subsection 4.4.
STATION #	Enter any unique number for the UWS in the console.
DEFAULT AREA	Enter the area name recorded in subsection 4.3.
FLOPPY DISC NUMBERS	If the UWS has media storage capability, enter any two unique numbers for the drives in this console.
TREND PEN NUMBER	No entry
STATION DEFAULT ACCESS LEVEL	Select either VIEW or OPER.
TOUCHSCREEN	Select YES.
OPERATOR'S KEYBOARD	If this UWS has a Supervisor's keyboard, select YES. If the UWS has an Engineer's keyboard, select NO. If the UWS has the Enhanced keyboard assembly or IKB (both keyboards), select YES.
ENGINEER'S KEYBOARD	If the UWS has an Engineer's keyboard, select YES. If the UWS has a Supervisor's keyboard, select NO. If the UWS has the Enhanced keyboard assembly (both keyboards), select YES.

SELECT DESIRED NODE display

Select LCN NODES on the Engineering Main Menu and enter the data for the UWS in the SELECT DESIRED NODE display. For the appropriate NODE NUMBER:

- Enter US in the TYPE column.
- Enter N (none) in the REDUNDANT NODE column.

4.6 Button Configuration

Introduction

Like the Operator's keyboard on the US, you can configure any 85 configurable buttons on the Supervisor's keyboard to invoke displays or to initiate other actions or functions. There is virtually no difference in button configuration between the UWS and US, but for the UWS, you can also configure the PF1 through PF17 keys on the Engineer's keyboard, and you can configure buttons on either keyboard to emulate a keyswitch in the Operator personality.

No button LEDs

There are no LEDs in the configurable buttons on the Supervisor's keyboard. Therefore, any entry in the LED UNIT field of the *Button Configuration Form SW88-470* is meaningless. If you have the Enhanced keyboard assembly, the LEDs on the Enhanced Operator's keyboard are functional and entries in the form would be meaningful.

Button configuration pathname

The pathname for your button-configuration object file must be entered in the pathname catalog in the area database that the UWS uses when operating in the Operator personality. See subsection 4.3, *Area Names*.

4.6.1 Engineer's Keyboard PF1-PF17 Key Configuration

Introduction

To allow the Engineer's keyboard to be used in the Operator personality, each of the PF1 through PF17 programmable keys can be configured to emulate a configurable button or a function key (SYST STATS, CONS STATS, MAN, AUTO, NORM, SP, OUT, etc.) on the Supervisor's or Operator's keyboard in the US.

Function description

PF1 through PF17 operate as if they are 51 programmable keys. There are three possible functions for each key.

- when the key alone is pressed.
- when the SHIFT key is held and a PF key is pressed.
- when the CTL key is held and a PF key is pressed.

Configuring the keys

The PF1 through PF17 keys are configured through the BUTTON CONFIGURATION activity on the Engineering Main Menu. Figure 4-1 illustrates the Button Configuration display that is used to select one of the keys for configuration. To invoke the display, select the BUTTON CONFIGURATION target on the Engineering Main Menu, and when the first page of configurable buttons appears, press the PAGE FWD key.

Figure 4-1 Button Configuration Display for PF1-PF17

The display shows the Button Configuration activity for PF1-PF17. It is divided into three sections: NORMAL, SHIFT, and CONTROL. Each section contains a grid of PF keys and their corresponding numeric values.

DD MMM YY 09:47:01 1									
NORMAL									
PF1 86	PF2 87	PF3 88	PF4 89	PF5 90	PF6 91	PF7 92	PF8 93	PF9 94	PF10 95
PF11 96	PF12 97	PF13 98	PF14 99	PF15 100	PF16 101	PF17 102			
SHIFT									
PF1 103	PF2 104	PF3 105	PF4 106	PF5 107	PF6 108	PF7 109	PF8 110	PF9 111	PF10 112
PF11 113	PF12 114	PF13 115	PF14 116	PF15 117	PF16 118	PF17 119			
CONTROL									
PF1 120	PF2 121	PF3 122	PF4 123	PF5 124	PF6 125	PF7 126	PF8 127	PF9 128	PF10 129
PF11 130	PF12 131	PF13 132	PF14 133	PF15 134	PF16 135	PF17 136			

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Continued on next page

4.6.1 Engineer's Keyboard PF1-PF17 Key Configuration, Continued

Key switch configuration

There is no OPER/SUPVR/ENG keyswitch on the Engineer's keyboard. In the Operator personality, if you need to change the keyswitch access level for the UWS, you must configure one, two, or three of these keys to do so. See subsection 4.6.2.

4.6.2 Supervisor’s Keyboard Keyswitch Buttons Configuration

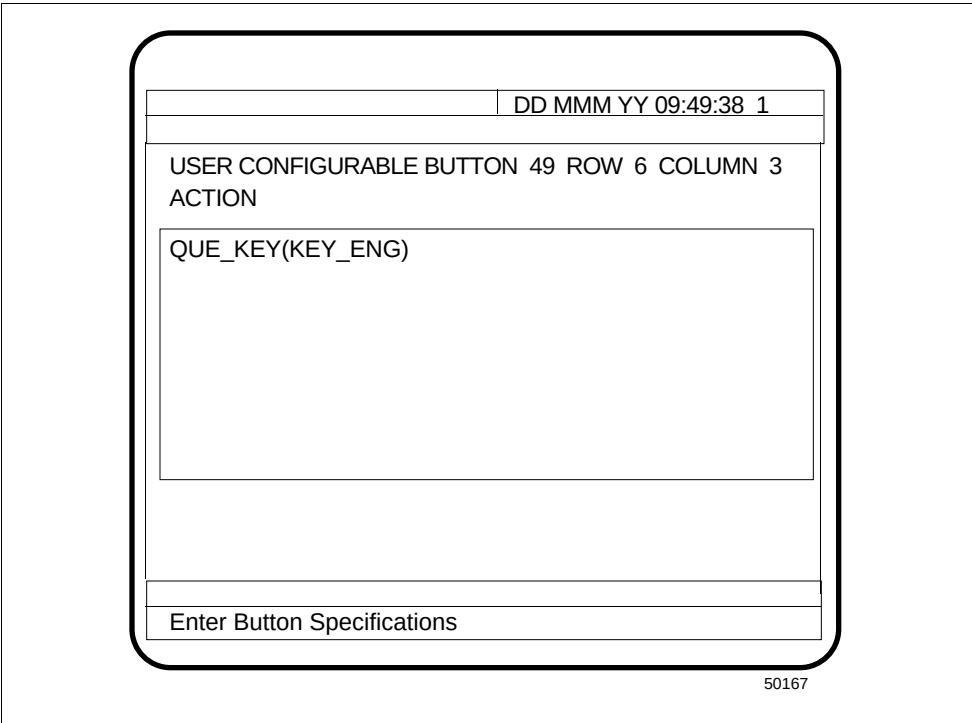
Introduction

There is no OPER/SUPVR/ENG keyswitch on the Supervisor’s keyboard, but you can configure one, two, or three of the configurable buttons for the function. When starting in the Operator personality, the UWS is at the Operator access level. A button must be configured for each keyswitch position that you intend to use. If you need to simulate all three positions, you must configure three buttons. Configure a button to select the Supervisor access level, one to select the Engineer access level, and one to go back to the Operator access level.

Button Configuration display

Figure 4-2 illustrates the Button Configuration display that is used to specify button actions. To invoke the display, select BUTTON CONFIGURATION on the Engineering Main Menu, then select any button on the first two button selection pages.

Figure 4-2 Button Configuration Display for Keyswitch Actions



4.7 Area Configuration

Introduction	When loading and starting a UWS (or US) with the Operator personality, an area database is loaded. You can use an existing area database (in volume &Dnn, where "nn" is a number from 1 to 10 that indicates the specific area database), or you can configure a completely new area database for the UWS, or for the console of which it is a member.
Area Change	Once the UWS is operating in the Operator personality, you can use the AREA CHANGE activity on the Console Status display to change to any other available area database on the LCN, in a History Module, or on a media storage cartridge or disk.
Addition to Area Pathname Catalog	If you configure the PF1-PF17 keys or one or more keyswitch buttons, the configuration must be in the Button Configuration object file that the area's Pathname catalog points to. This means that when you are using more than one area database in the UWS, you must add your PF1-PF17 configuration, or your keyswitch button configuration to the Button Configuration object file in each area. You can do this through the BUTTON CONFIGURATION activity on the Engineering Main Menu. During activity, you read in the appropriate button configuration source file, make the changes, and recompile the source file, which results in a new object file with the changes in it that are ready for use by each area whose Pathname catalog points to it.
No unique area configuration	There is no unique area configuration activity for the UWS. It is the same as a US.

Section 5 – Operation

5.1 Overview

Section contents

These are the topics covered in this section:

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5.2	Supervisor Station Operation.....	43
5.3	Engineer Station Operation.....	44
5.3.1	Engineer Station with Operator Personality.....	44
5.3.2	Engineer Station with Engineer Personality.....	45
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5.2 Supervisor Station Operation

Functional exceptions

A UWS with a Supervisor's keyboard is intended to operate in only the Operator personality. The operation of the UWS with a Supervisor's keyboard is the same as a US, which is described in the *Process Operations Manual* with the following exceptions.

- **No touchscreen**—When there is no touchscreen, either use the TAB keys to move the cursor or use the mouse to move the cursor. Once the cursor is in a target (pick) that you wish to select, press the SELECT button.
- **No keyswitch**—The UWS starts at the Operator keyswitch access level. If configurable buttons have been configured to change the access level, those buttons can be used to change to the Supervisor or Engineer level, or back to Operator level.
- **No audible annunciators**—There is no terminal board in the UWS to accept connections to customer-provided devices such as audible alarm annunciators.
- **No LEDs in configurable buttons**—There are no light emitting diodes (LEDs) in the user-configurable buttons on the Supervisor's keyboard.
- **RESET button and POWER GOOD/FAIL LEDs not present**—To reset the UWS, place the power switch on the Five-Slot Module in the OFF (0) position, wait a few seconds, and then place it in the ON (1) position.

5.3 Engineer Station Operation

Functional exceptions	The UWS with an Engineer's keyboard is intended to operate in either the Operator personality or the Engineer personality. The operation is the same as a US with an Engineer's keyboard with the following exceptions. • No touchscreen—When there is no touchscreen, either use the TAB keys to move the cursor or use the mouse to move the cursor. Once the cursor is in a target (pick) that you wish to select, press the SELECT button. • RESET button and POWER GOOD/FAIL LEDs not present—To reset the UWS, place the power switch on the Five-Slot Module in the OFF (0) position, wait a few seconds, and then place it in the ON (1) position. • LOAD key on Engineer's keyboard—On a US, the Engineer's keyboard and the Operator's keyboard share the same LOAD button. On a UWS, the Supervisor's keyboard has a LOAD button and the Engineer's keyboard has a LOAD key.
------------------------------	---

5.3.1 Engineer Station with Operator Personality

Additional functional exceptions	The following additional functional exceptions to US operation apply for a UWS with an Engineer's keyboard that is operating in the Operator personality. • No keyswitch—The UWS starts at the Operator keyswitch access level. If one, two, or three of the PF1-PF17 keys have been configured to change the access level, the keys can be used to change to the Supervisor or Engineer level, or back to Operator personality. • No audible annunciators—There is no terminal board in the UWS to accept connections to customer-provided devices such as audible alarm annunciators. • Configurable PF1-PF17 keys—The PF1 through PF17 keys at the top of the Engineer's keyboard can be configured to emulate buttons and function keys on the Supervisor's (or Operator's) keyboard. One, two, or three of these keys can be configured to change the keyswitch access level.
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5.3.2 Engineer Station with Engineer Personality

Additional functional exceptions

The following additional functional exceptions to US operation apply for a UWS with an Engineer's keyboard and operating in the Engineer personality.

- **No keyswitch**—The UWS starts at the Operator keyswitch access level. In the Engineer personality, the PF1 key will change the keyswitch access level as follows.
 - Change to Engineer level: Press the PF1 key.
 - Change to Supervisor level: Hold the SHIFT and press the PF1 key.
 - Change to Operator level: Hold the CTL key and press the PF1 key.
- **CMND key**—On the US's Engineer's keyboard, the Command key is labeled COMND. On the UWS's Engineer's keyboard it is labeled CMMD.
- **DEL key label on the front side of ERASE key**—On the US's Engineer's keyboard, the DEL (delete) key is a separate key. On the UWS's Engineer's keyboard it is labeled on the front side of the ERASE key. For the delete function, hold the CTL key and press the ERASE key.
- **LF key label on the front side of ESC key**—On the US's Engineer's keyboard, the LF (line feed) key is a separate key. On the UWS's Engineer's keyboard it is labeled on the front side of the ESC key. For the line feed function, hold the CTL key and press the LF key.
- **All Caps, Shift, and Insert Character modes shown by LEDs**—The CAPS LOCK, SHIFT, and INS CHAR keys are alternate action keys. The CAPS LOCK, SHIFT LOCK, and INS CHAR indicators at the top of the keyboard indicate the current mode.

For example, the first time you press the INS CHAR key, the INS CHAR indicator lights, which indicates that the keyboard is in the insert-character mode (characters are inserted at the cursor's position). The next time you press the INS CHAR key, the indicator extinguishes, which indicates the return to the normal character-overwrite mode.

5.4 Universal Personality Operation

Introduction

A Universal personality station has two keyboards.

- Enhanced Operator's keyboard
- Engineer's keyboard

Operation with the Enhanced Operator's keyboard is the same as operation with the Supervisor's keyboard with the exception that the Enhanced Operator's keyboard has LEDs in the configurable buttons. The forty configurable keys with LEDs are in a cluster of four rows of ten keys each in the upper-left section of the keyboard. See Figure 1-3. The keys are configured by selecting the BUTTON CONFIGURATION target on the Engineering Main Menu. For further information, refer to the *Button Configuration Data Entry* and *Button Configuration Form Instructions* manuals.

Operation of the Engineer's keyboard is the same as operation of the Engineer's Station with Engineer personality software loaded.

Enhanced keyboard operation

The Enhanced keyboard assembly is intended to operate with the Universal personality software, which includes all of the functions of both the Operator personality and the Engineer personality. When first loaded with the Universal personality, the Console Status display appears. To access the Engineer personality functions from the Console Status display or any other Operator personality display, use the Engineer's keyboard to invoke the Engineering Personality Main Menu by holding the CTL key and pressing the HELP key. To access Operator personality functions from the Engineer personality, return to the Engineering Personality Main Menu and use the Operator's keyboard to invoke any Operator personality display.

5.5 Monitor Operation

Introduction

There are two models of IDEK 21-inch monitor that may be in use at the Universal Work Station.

- The older model MF-5221 IDEK 21-inch multiscan color monitor. Subsection 5.5.1 discusses the use of its analog, rotary-type screen control knobs.
 - The newer model MF-8221 IDEK 21-inch Vision Master color monitor. Subsection 5.5.2 discusses its digital, pushbutton screen controls.
-

Optional touchscreen

Both monitors except an optional touchscreen (infrared touchscreen frame with plastic bezel assembly) that are physically different.

The touchscreen option for the newer IDEK monitor has four mounting “buttonholes” (two on each side), and is compatible only with the newer monitor. The older IDEK monitor touchscreen has two “buttonholes” (one on each side).

The touchscreen controller (electronics) and cable for the newer monitor is identical to the older monitor and can be easily transferred.

Vision Master monitor

The newer IDEK monitor is essentially identical to the older monitor, including the touchscreen option. There are small physical differences between them, primarily in the layout and location of the screen controls. The newer monitor has its screen controls mounted inside a recessed, sliding tray (drawer) that is located below the screen. The screen controls on the new monitor cannot be accessed after the drawer is closed (pushed in). The overall dimensions of the newer monitor are slightly larger than the older monitor. Electrically, the video cable plug and pinning are identical for both monitors.

Operation of both monitors

Subsection 5.5.1 and 5.5.2 describes the operation, differences in layout, and location of the screen controls on the two IDEK monitor models, models MF-5221 and MF-8221.

5.5.1 IDEK Model MF-5221 Monitor

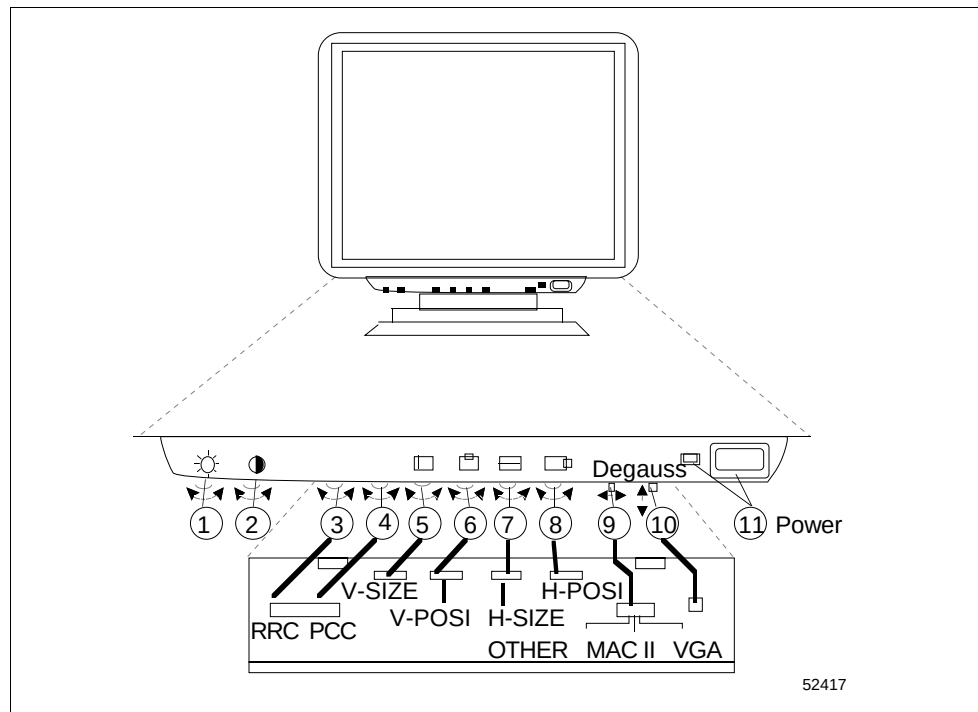
Introduction

This subsection describes the operational controls of the older IDEK model MF-5221 21-inch multiscan color monitor. The numbered description in Table 5-1 for the analog, rotary-type screen controls, switches, and connectors are a reference to the identification numbers that are illustrated in Figures 5-1 and 5-2.

Monitor front controls

Figure 5-1 illustrates the controls and switches at the front of the IDEK model MF-5221 monitor.

Figure 5-1 Model MF-5221 Monitor Front Controls



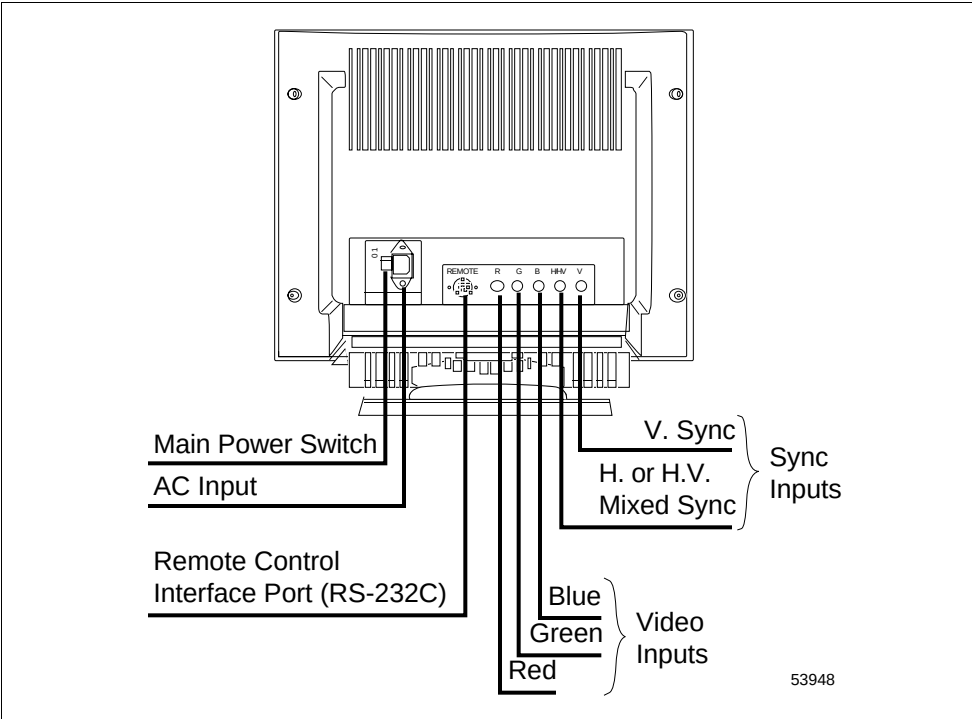
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5.5.1 IDEK Model MF-5221 Monitor, Continued

Monitor rear controls

Figure 5-2 illustrates the switches and connectors at the rear of the IDEK model MF-5221 monitor.

Figure 5-2 Model MF-5221 Monitor Rear Switches and Connectors



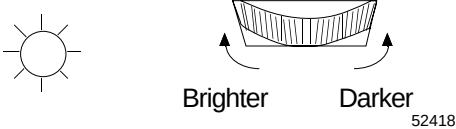
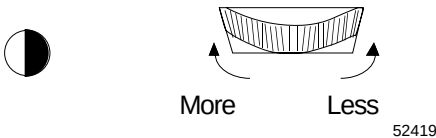
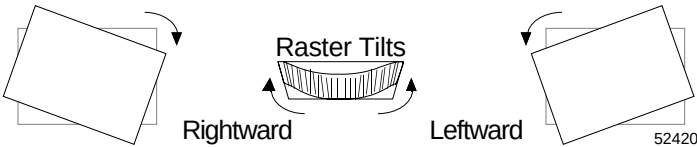
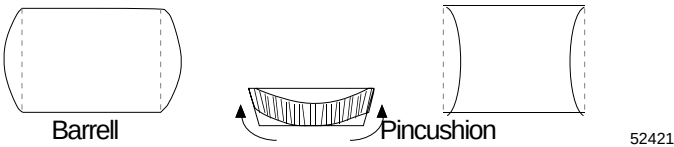
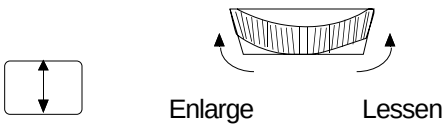
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5.5.1 IDEK Model MF-5221 Monitor, Continued

Monitor controls and connectors

Table 5-1 provides the function of the IDEK model MF-5221 monitor's controls, switches, and connectors.

Table 5-1 IDEK Model MF-5221 Monitor Control and Connector Descriptions

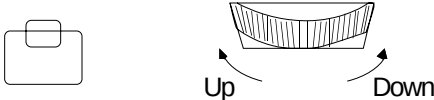
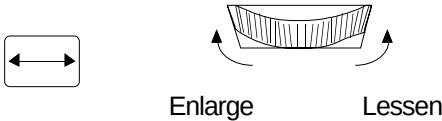
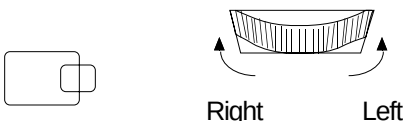
Reference Number	Control	Function
1	Brightness Control	 Adjusts the brightness of the display.
2	Contrast Control	 Adjusts the contrast of the display.
3	RRC Control (Raster Rotation Circuit)	 Corrects the raster tilt. The control also removes poor purity from around the CRT, which is generated when swiveling the monitor.
4	PCC Control (Pincushion Correction Circuit)	 Corrects both right and left side distortion of the display.
5	V-SIZE Control (Vertical Size)	 Adjusts the vertical size of the display.

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5.5.1 IDEK Model MF-5221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-1 IDEK Model MF-5221 Monitor Control and Connector Descriptions, Continued

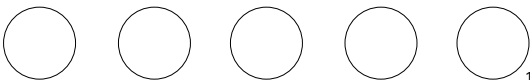
6	V-POSI Control (Vertical Position)	 52423 Adjusts the center of the display vertically.
7	H-SIZE Control (Horizontal Size)	 52424 Adjusts the horizontal size of the display.
8	H-POSI Control (Horizontal Position)	 52425 Adjusts the center of the display horizontally.
9	Signal Select Switch	Select the VGA position when using a VGA graphics card. Select the MACII position when using an Apple Macintosh II (with the standard Apple video card) or PGC graphics card. Select the OTHER position when using any other graphics card or generator with a horizontal frequency between 30 and 80 kHz.
10	DEGAUSS Button	Degausses the CRT when pressed.
11	POWER Switch and Indicator	Pressing the switch applies power to or removes power from the monitor. The indicator lights while power is applied.
12	AC Input Voltage Switch	Selects the local ac power source voltage, 115 V or 220 V.
13	AC Input Connector	AC power cord connector.
14	BNC INPUT VOLTAGE Switch	BNC INPUT  1.0 V VOLTAGE selector  0.7 V 52427 Selects the appropriate BNC video input voltage, 0.7 V (0.7 V) or 1.0 V (1.0 V).

Continued on next page

5.5.1 IDEK Model MF-5221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-1 IDEK Model MF-5221 Monitor Control and Connector Descriptions, Continued

15	Analog RGB Input BNC Connectors (total of five)	R G-SYNC B H/HV V  Connectors for RGB analog output from EPDGP I/O or EPDGC I/O board. BNC analog signals <table><tr><th>Connector</th><th>Sync on Green</th><th>Composite sync</th><th>Separate sync</th></tr><tr><td>R</td><td>Red</td><td>Red</td><td>Red</td></tr><tr><td>G/SYNC</td><td>Sync on Green</td><td>Green</td><td>Green</td></tr><tr><td>B</td><td>Blue</td><td>Blue</td><td>Blue</td></tr><tr><td>H.HV</td><td>GND/unused</td><td>HV sync</td><td>H-sync</td></tr><tr><td>V</td><td>GND/unused</td><td>GND /unused</td><td>V-sync</td></tr></table>	Connector	Sync on Green	Composite sync	Separate sync	R	Red	Red	Red	G/SYNC	Sync on Green	Green	Green	B	Blue	Blue	Blue	H.HV	GND/unused	HV sync	H-sync	V	GND/unused	GND /unused	V-sync
Connector	Sync on Green	Composite sync	Separate sync																							
R	Red	Red	Red																							
G/SYNC	Sync on Green	Green	Green																							
B	Blue	Blue	Blue																							
H.HV	GND/unused	HV sync	H-sync																							
V	GND/unused	GND /unused	V-sync																							
16	Termination Switches	Selects the appropriate line impedance for the video (75Ω or infinity) and sync (1 KΩ or 4.7 KΩ) signals. If the monitor is used in a loop-through operation, set the switches to the left (infinity for video, 4.7 K for sync). For single unit operation, or when the monitor is the last unit in a loop-through string, set the switches to the right (75 for video, 1 K for sync). The switches of unused BNC connectors should also be set to the right.																								

5.5.2 IDEK Model MF-8221 Monitor

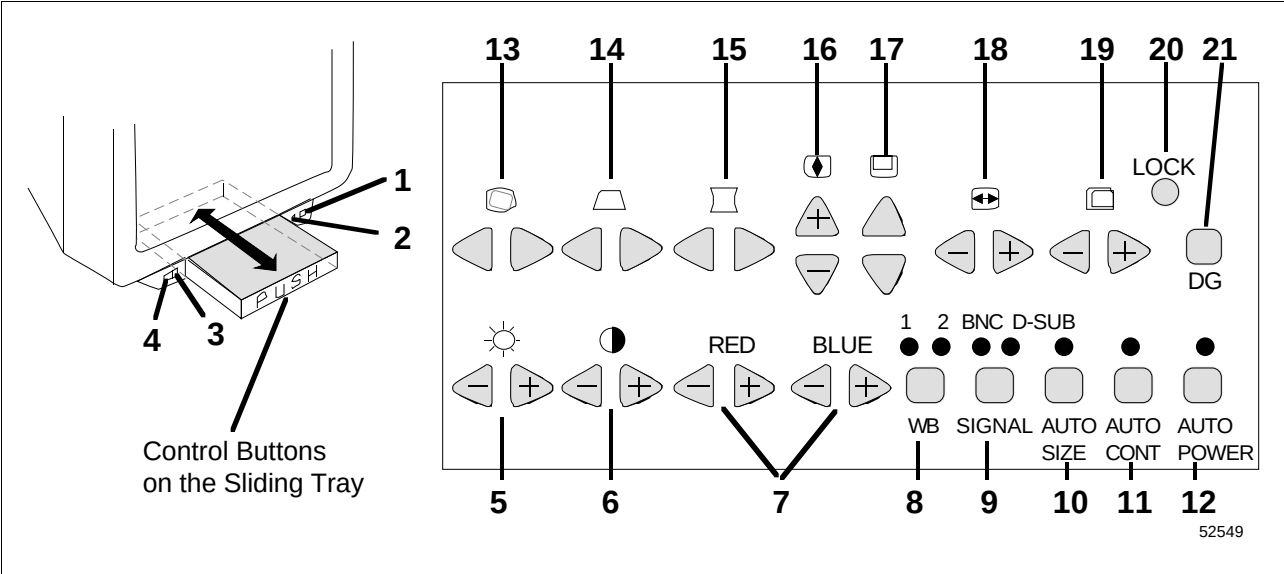
Introduction

This subsection describes the operational controls of the newer IDEK model MF-8221 21-inch Vision Master monitor. The numbered description in Table 5-2 for the digital, pushbutton screen controls, switches, and connectors are a reference to the identification numbers that are illustrated in Figures 5-3 and 5-4.

Monitor front controls

Figure 5-3 illustrates the controls and switches at the front of the IDEK model MF-8221 monitor.

Figure 5-3 Model MF-8221 Monitor Front Controls and Switches



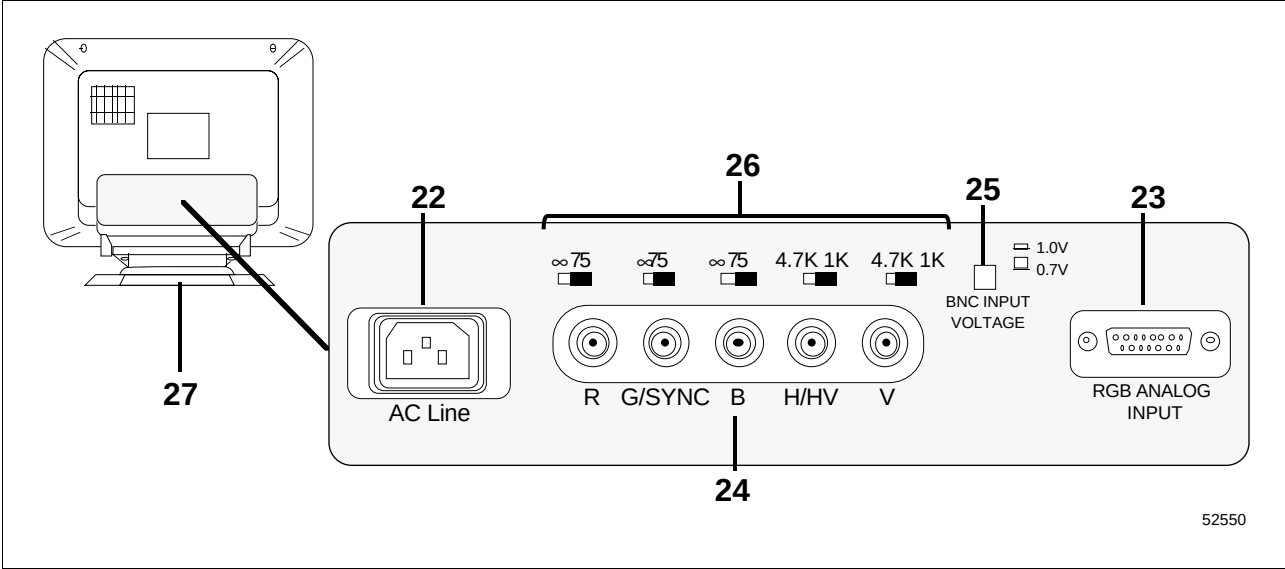
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5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor rear switches and connectors

Figure 5-4 illustrates the switches and connectors at the rear of the IDEK model MF-8221 monitor.

Figure 5-4 IDEK Model MF-8221 Monitor Rear Switches and Connectors



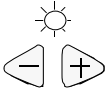
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5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors

Table 5-2 provides the function of the IDEK model MF-8221 monitor's controls, switches, and connectors.

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions

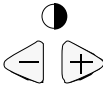
Reference Number	Control	Function
1	Power Switch (ON/OFF)	Pressing the switch applies power to or removes power from the monitor.
2	Power/Standby Indicator	Indicates the power-on/off or standby state of the monitor. If power is on, the green indicator is lit. When the green indicator is off, power is off. When the AUTO POWER-OFF sensor is activated and the monitor switches to the standby state, the green indicator becomes red.
3	AUTO POWER-OFF Sensor	To save energy, the sensor detects the presence of an Operator and powers down the monitor to the standby state automatically if no one is in front of the monitor for over an hour. To reapply power while in the standby state, two methods are available. • Press the AUTO POWER-OFF switch. • Press the POWER switch and be sure the red indicator extinguishes. Then, press the switch again and the green indicator will light. Do not place anything in front of the sensor. It will not operate properly if it is dirty or covered.
4	AUTO CONTRAST Sensor	To avoid eye strain and reduce energy consumption, the sensor adjusts the screen contrast automatically as the ambient light of the room changes. Do not place anything in front of the sensor. It will not operate properly if it is dirty or covered.
5	Brightness Control Buttons	 Adjusts the picture brightness. When you press both Brightness Control buttons at the same time for over 3 seconds, you will hear a beep, and the existing brightness adjustment will be canceled and returned to the factory "center" setting.

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

6	Contrast Control Buttons	 Adjusts the picture contrast. You will hear short beeps when the contrast reaches its maximum setting.
7	Tone Control Buttons (RED/BLUE)	RED  BLUE  Other than the WB1/WB2 tones that are selectable by the White Balance buttons, use the RED and BLUE buttons to adjust to the preferred white tone. When the + button is held, the color tone becomes vivid gradually. When the - button is held, the color tone fades gradually. The user adjusted tone is automatically stored. It is recalled and can be readjusted by momentarily pressing either one of the Tone Control buttons when using the WB1 or WB2 buttons. You will hear a beep and the user adjusted tone will be canceled and set back to the factory "WB1" setting when you press the White Balance (WB) button for over 3 seconds.
8	White Balance (WB) Button (WB1/WB2)	1 2  WB Use the position of the WB button to select the white balance (WB1 = 9300°K: bluish white, WB2 = 6500°K: reddish white). The white balance function alternates between the two white balances with each press of the button, and the appropriate indicator lights (1 or 2).

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

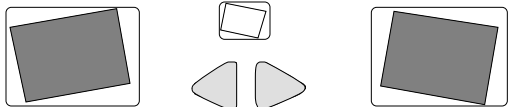
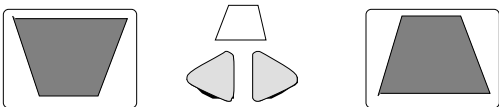
9	SIGNAL Button (BNC/D-SUB)	BNC D-SUB  SIGNAL Use the button to select the input signal from either the BNC connectors or the 15-pin D-SUB connector. Pressing the button alternates the connector selection, and the illuminated indicator identifies the selection.
10	AUTO SIZE Button (On/Off)	 AUTO SIZE Use the button to enable or disable the automatic horizontal and vertical picture sizing and centering function. The function recognizes the picture size and position based on the signal applied to the monitor and adjusts automatically. The indicator will be lit when the automatic function is on. While the automatic function is selected, it functions every time power is applied, or the input signal changes. It is better not to select the function when it is not necessary. When you select the function, you will need full-size picture information and maximum brightness video input such as a Windows screen. It may not operate properly with special video information such as a prompt C> on the screen. When the picture that is adjusted by the function needs to be changed, use the SIZE or POSITION button to make the change.
11	AUTO CONT (Contrast) Button (On/Off)	 AUTO CONT Use the button to enable or disable the automatic contrast sensor function. The indicator will be lit when the sensor is enabled. When the function is enabled, it will automatically adjust the CRT contrast as the ambient light of the room changes.

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

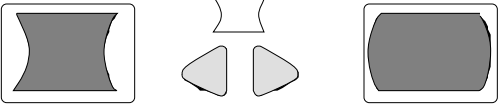
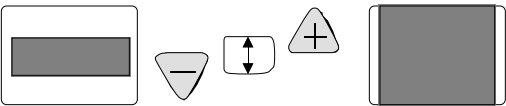
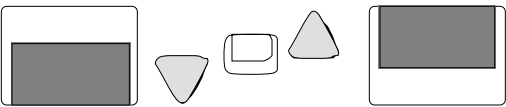
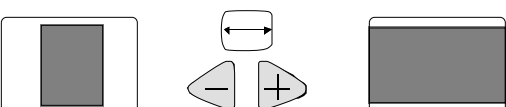
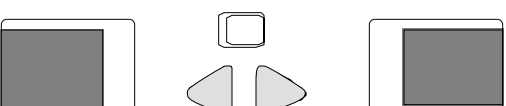
12	AUTO POWER Button (On/Off)	 AUTO POWER Use the button to enable or disable the automatic power-off sensor function. The indicator will be lit when the sensor is enabled. When the function is enabled, the presence of an Operator in front of the monitor will be sensed and automatically power down the monitor when an Operator is absent for more than an hour. The power indicator will light green during normal operation and red when in the standby state. To restore normal operation, either cycle power to the monitor or press the AUTO POWER button (disables the sensor).
13	RRC Control Buttons	 Use the buttons to correct raster rotation or poor purity that is caused by magnetism when the monitor is swiveled while in operation. When you press both RRC Control buttons at the same time for over 3 seconds, you will hear a beep, and the adjusted condition will be canceled and reset to the factory "center" setting. The RRC Control buttons must be used after degaussing the CRT.
14	Trapezoid Control Buttons	 Use the buttons to correct trapezoid distortion and to produce parallel vertical edges.

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

15	PCC Control Buttons	 Use the buttons to correct pincushion or barrel distortion and to produce straight vertical edges.
16	Vertical Size Control Buttons	 Use the buttons to adjust the vertical size of the picture.
17	Vertical Position Control Buttons	 Use the buttons to adjust the vertical position of the picture.
18	Horizontal Size Control Buttons	 Use the buttons to adjust the horizontal size of the picture.
19	Horizontal Position Control Buttons	 Use the buttons to adjust the horizontal position of the picture.

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

20	LOCK Button	LOCK  Use the button to protect the picture information from unintentional operation of the control buttons. When the LOCK button is pressed, you will hear a beep and the following control buttons will be disabled: Tone, AUTO SIZE, RRC, Trapezoid, PCC, Vertical Size, Vertical Position, Horizontal Size, Horizontal Position. If you try to use the locked control buttons, you will hear short beeps as an alarm. To disable the locked condition, press the LOCK button again, and you will hear a beep and the control buttons will be usable.
21	DEGAUSS (DG) Button	 DG Use the button to cancel magnetism when swiveling or moving the monitor causes poor purity. The function operates for approximately 10 seconds and goes off automatically. Degauss occurs automatically when powering on the monitor, and it is recommended that 30 minutes elapses between each degauss operation.
22	AC LINE Connector	AC power cord connection. The power cord connects to an ac outlet. Before connecting the power cord, be sure the POWER switch is in the OFF (O) position.
23	RGB ANALOG INPUT Connector	A 15-pin D-Sub input connector for equipment that provides an analog RGB output.
24	BNC Connectors	Five BNC input connectors that accept inputs from the EPDGP I/O or EPDGC I/O board.

Continued on next page

5.5.2 IDEK Model MF-8221 Monitor, Continued

Monitor controls and connectors, continued

Table 5-2 IDEK Model MF-8221 Monitor Control and Connector Descriptions, Continued

25	BNC INPUT VOLTAGE Switch	The switch is used to select the appropriate BNC video input voltage from the EPDGP I/O or EPDGC I/O board. Select the 0.7 V position for 0.74 volt peak-to-peak voltage or the 1.0 V position for 1.0 volt peak-to-peak voltage.
26	BNC Impedance Switches (Video: $\infty/75\Omega$, Sync: $4.7K\Omega/1K\Omega$)	Select the 75Ω position for line impedance. If the monitor is used in a loop-through operation, set the switches to the left (∞ for video, $4.7K\Omega$ for sync). For single unit operation, or when the monitor is the last unit in a loop-through series, set the switches to the right (75Ω for video, $1K\Omega$ for sync). The switches for the unused BNC connectors should also be set to the right.
27	Tilt/Swivel Stand	The stand allows the screen angle to be adjusted by 90 degrees horizontally, and 15 degrees up and 5 degrees down vertically.

5.5.3 Hitachi Model HM-4821-D Monitor

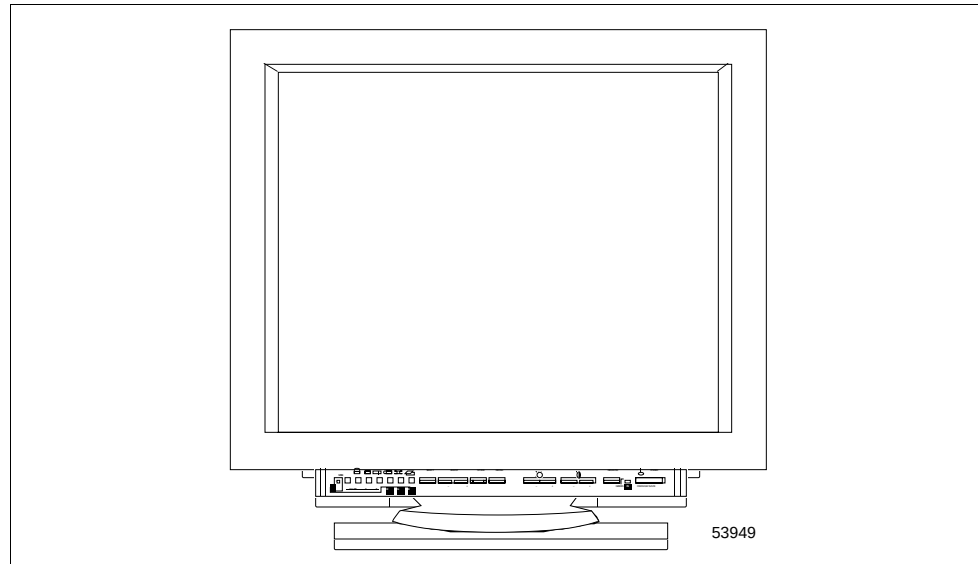
Introduction

This subsection describes the CE Compliant Hitachi model HM-4821-D 21-inch monitor's operational controls, selection switches, and connectors. The Hitachi monitor is now included with all current production UWSs.

Front of monitor

Figure 5-5 illustrates the front of the Hitachi model HM-4821-D monitor.

Figure 5-5 Hitachi Model HM-4821-D Monitor



Continued on next page

5.5.3 Hitachi Model HM-4821-D Monitor, Continued

Front controls, indicators, and switches

Figures 5-6 and 5-7 illustrate the location and function of the controls, indicators, and switches at the front of the Hitachi model HM-4821-D monitor.

Figure 5-6 Hitachi Model HM-4821-D Monitor Front Controls – Left Side

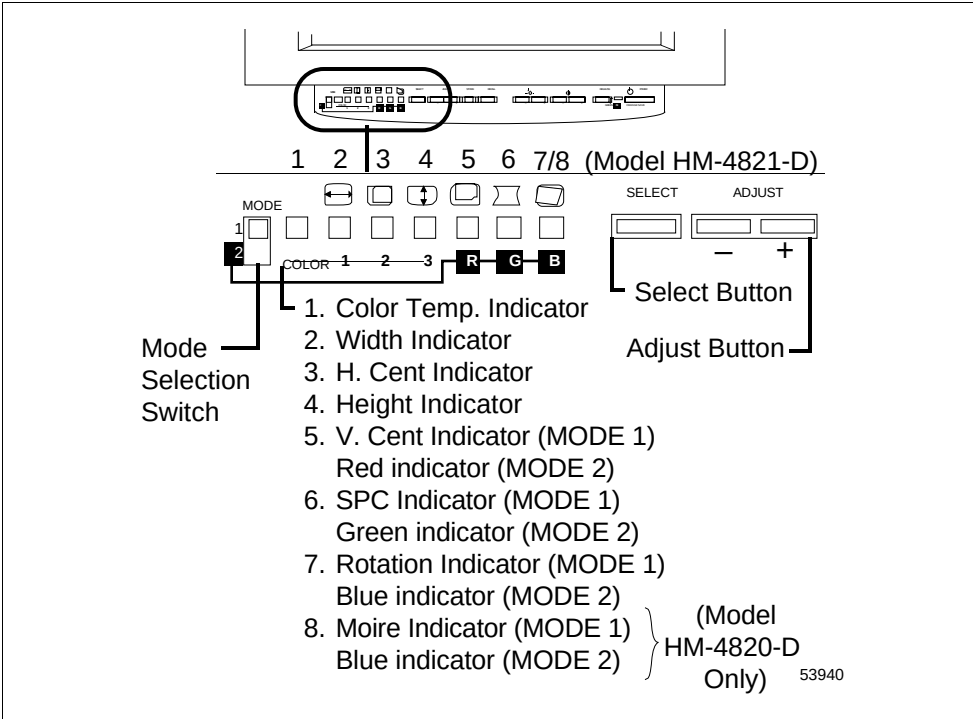
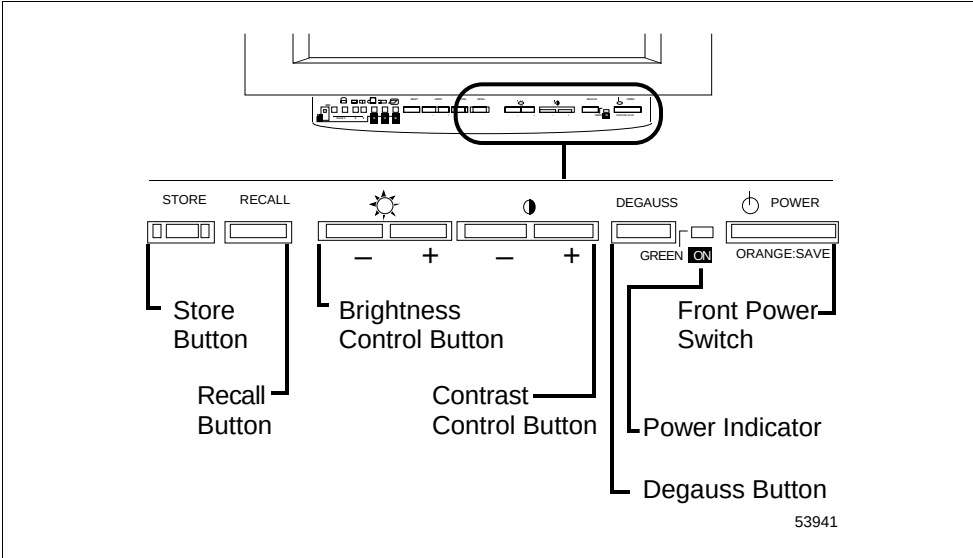


Figure 5-7 Hitachi Model HM-4821-D Monitor Front Controls – Right Side



5.5.3.1 Monitor Interface

CAUTION

CAUTION—Incorrect cable connections to the monitor may damage the monitor. Be sure the monitor and the UWS power switches are in the off position before connecting the monitor's interface cables.

Video signal cables

The five video signal cables from the EPDGP I/O or EPDGC I/O board connect to the BNC input connectors at the rear of the monitor. The monitor's connectors are identified in Figure 5-8.

Sync selection

Synchronization with the input is possible for composite sync on the green (G) signal, HV mixed external sync signal, and HV separate external sync signal (Auto selection).

Remote control interface port

The REMOTE port is preset in the factory and not user accessible.

AC input voltage

The ac input voltage is selected automatically.

Continued on next page

5.5.3.1 Monitor Interface, Continued

Rear switches and connectors

Figures 5-8 and 5-9 illustrate the location and function of the switches and connectors at the rear of the Hitachi model HM-4821-D monitor.

Figure 5-8 Hitachi Model HM-4821-D Monitor Rear Panel

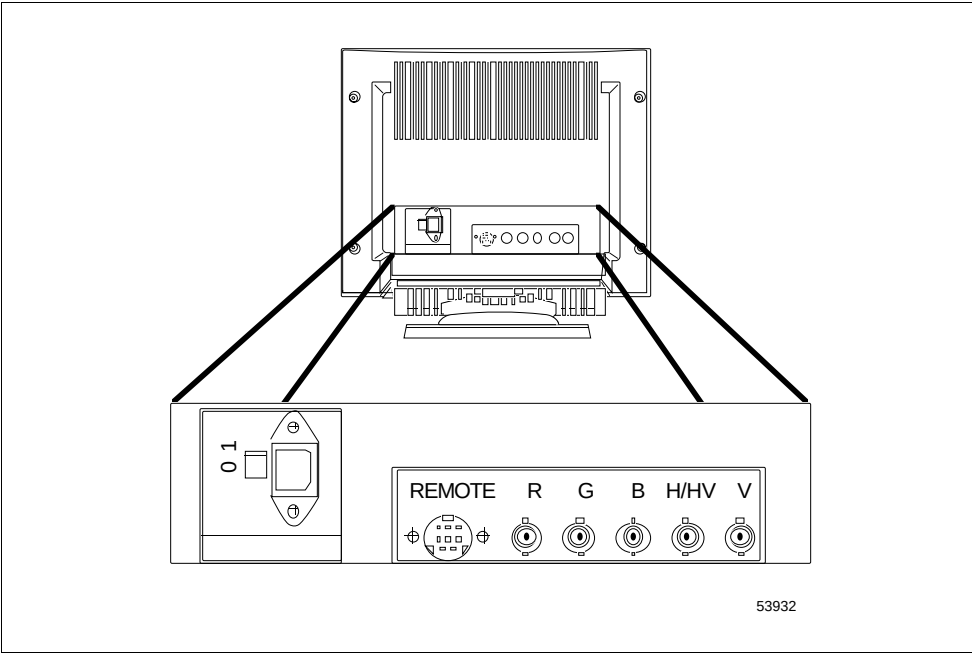
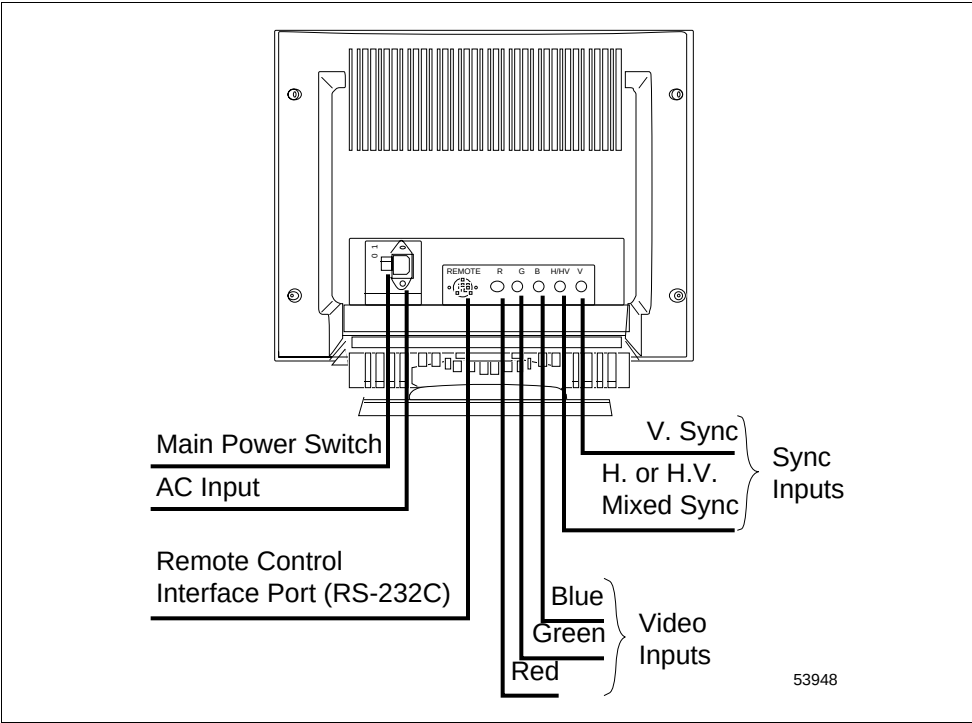


Figure 5-9 Hitachi Model HM-4821-D Monitor Rear Panel Functions



5.5.3.2 Monitor Operator Controls

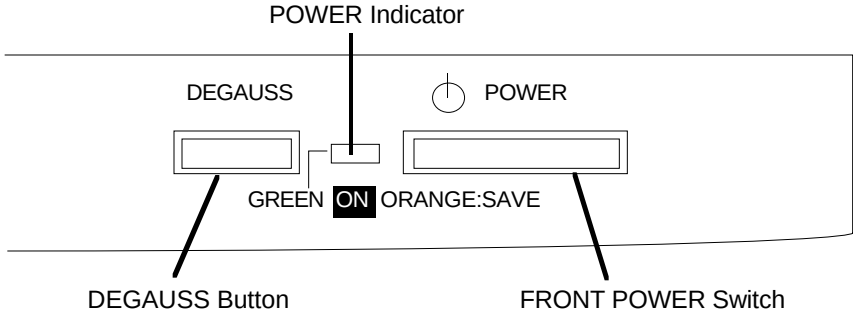
Introduction

The monitor usually operates in the Operation mode, and the screen is controlled by data in an internal memory for each video signal input.

Control and switch functions

Table 5-3 lists the monitor's control and switch functions.

Table 5-3 Hitachi Model HM-4821-D Monitor Functions

Control/Switch/Indicator	Function
Main POWER Switch	Controls all ac power entering the monitor. See Figure 5-8.
Front POWER Switch	Controls the ac power to the monitor controls. See Figure 5-7.
POWER Indicator—Green	Indicates that the monitor is powered up. See Figure 5-7.
POWER Indicator—Yellow (Power-saver function)	In the absence of any interface signals, the power-saver function is activated, and the POWER indicator is yellow. When the signals are restored, the power is activated and the POWER indicator is green.
DEGAUSS Button	 The degauss function is automatically initiated when power is applied to the monitor. By pressing the DEGAUSS button, the monitor will be manually degaussed. This is highly recommended following any tilt or swivel changes.
Contrast Controls	Refer to subsection 5.5.3.3, Display Adjustments.
Brightness Controls	Refer to subsection 5.5.3.3, Display Adjustments.

Continued on next page

5.3.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

Control/Switch/Indicator	Function
RECALL Button	You can recall the factory adjusted data or initial default data (not user stored data) by pressing and holding the RECALL button for more than 2 seconds. After this, if you press the STORE button, the displayed data is stored in the memory. If you want to recall the stored user data, press the front POWER switch. If you press the RECALL button and then place the main power switch in the off (0) position, placing the switch in the on (1) position before pressing the STORE button will display the original stored data on the screen. When the input signal is not a preset timing signal, if you press the STORE button, all the indicators light which indicates an error.
STORE Button	After making screen adjustments, pressing the STORE button stores the adjusted data in memory, and the monitor exits the Adjustment mode.
ADJUST + Control	When you press the ADJUST + button in Adjustment MODE 1 or MODE 2, the screen image changes as follows. MODE 1 WIDTH: Horizontal display width becomes wider. H. CENT: Display center shifts to the right. HEIGHT: Vertical display height becomes taller. V. CENT: Display center shifts upward. SPC (side pinchusion): Vertical edges barrel out. ROTATION: Display rotates clockwise. MOIRE: Moire pattern is changed. COLOR Temperature: Color temperature changes to Operator adjusted color temperature. MODE 2 R: Red of Color temperature No. 3 becomes strong. G: Green of Color temperature No. 3 becomes strong. B: Blue of Color temperature No. 3 becomes strong.

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5.3.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

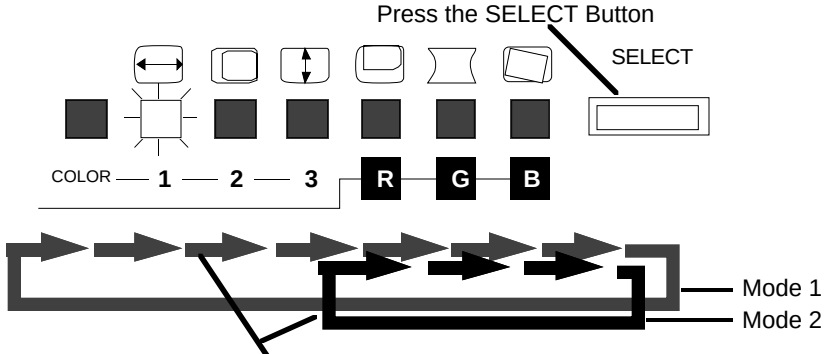
Control/Switch/Indicator	Function
ADJUST - Control	When you press the ADJUST - button in the Adjustment mode, the screen image changes as follows. MODE 1 - WIDTH: Horizontal display width becomes narrower. H. CENT: Display center shifts to the left. - HEIGHT: Vertical display height becomes shorter. V. CENT: Display center shifts downward. - SPC (side pinchushion): Vertical edges barrel in. - ROTATION: Display rotates counterclockwise. - MOIRE: Moire pattern is changed. - COLOR Temperature: Color temperature changes to color temperature No. 1. MODE 2 - R: Red of Color temperature No. 3 becomes weak. - G: Green of Color temperature No. 3 becomes weak. - B: Blue of Color temperature No. 3 becomes weak. When you adjust MOIRE, the moire pattern may be reduced. It is dependent upon the preset timing. If you compensate MOIRE excessively, horizontal jitter may appear on the screen. It is dependent upon the preset timing. When you adjust MOIRE, adjust it so that horizontal jitter does not appear on the screen.

Continued on next page

5.5.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

Control/Switch/Indicator	Function
SELECT Button	When the SELECT button is pressed, the monitor enters one of two Adjustment modes that is determined by the MODE switch. MODE 1 The WIDTH indicator lights, and you can adjust the width. MODE 2 The R indicator lights, and you can adjust red for Color Temperature No. 3. Pressing the SELECT button causes the indicator light to shift right and enable the adjustment function for the indicator that is lit. If you press the SELECT button in mode 1, while the Rotation (R) indicator is lit, the COLOR indicator lights. If you press the SELECT button in mode 2, while the B indicator is lit, the R indicator lights.  Indicator light shifts to the right by pressing the SELECT button.

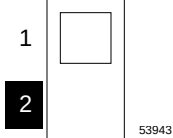
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5.5.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

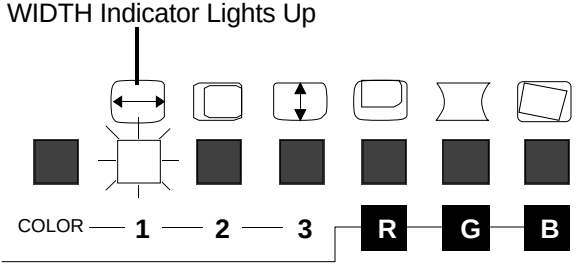
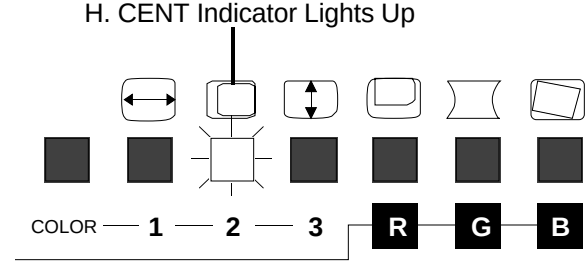
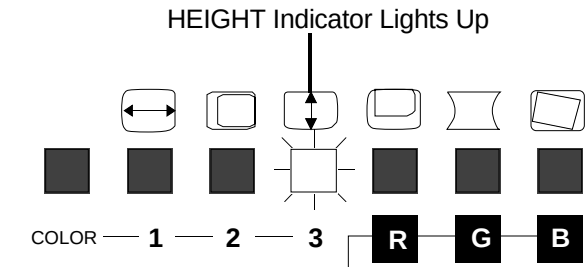
Control/Switch/Indicator	Function
MODE Switch (MODE 1/MODE 2), continued	MODE  You can select the Adjustment Mode 1 or Mode 2 with the MODE switch. When you select Mode 1, you can adjust (select) the following settings. • Horizontal width • Horizontal centering • Vertical height • Vertical centering • Moire • Color Temperature • SPC • Rotation When you select Mode 2, you can adjust Color Temperature number 3. • Red of Color Temperature number 3 • Green of Color Temperature number 3 • Blue of Color Temperature number 3 See subsection 5.5.3.3, Display Setting, for Color Temperature indicator information.

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5.5.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

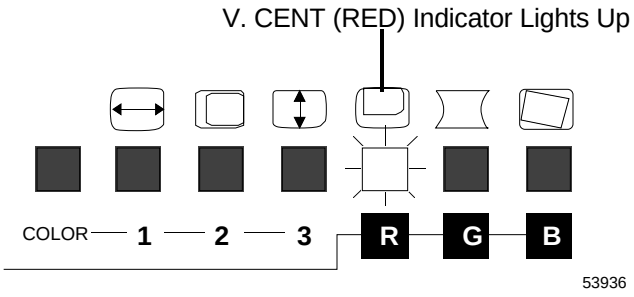
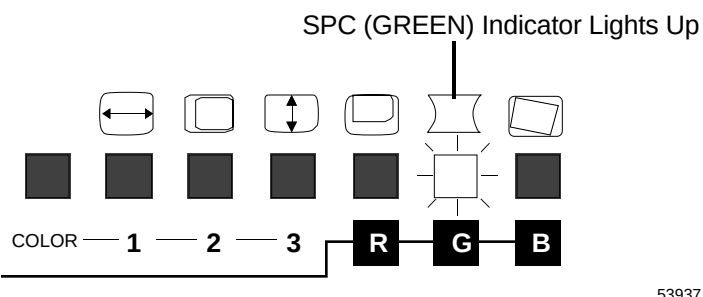
Control/Switch/Indicator	Function
MODE Switch (MODE 1/MODE 2), continued	The Width indicator lights when you select Width in Mode 1. WIDTH Indicator Lights Up  53933
	The Horizontal Centering indicator lights when you select Horizontal Centering in Mode 1. H. CENT Indicator Lights Up  53934
	The Height indicator lights when you select Height in Mode 1. HEIGHT Indicator Lights Up  53935

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5.5.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

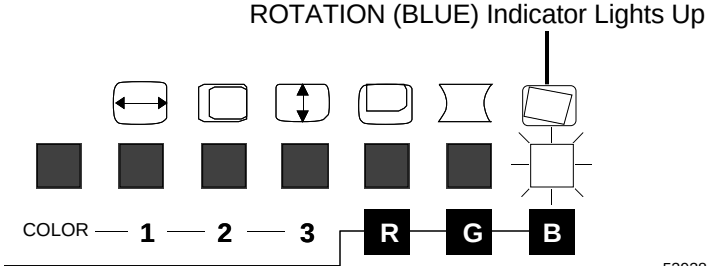
Control/Switch/Indicator	Function
MODE Switch (MODE 1/MODE 2), continued	The Vertical Centering indicator lights when you select Vertical Centering in Mode 1. The Red (R) indicator lights when you select Red in Mode 2. V. CENT (RED) Indicator Lights Up  53936 The SPC indicator lights when you select SPC in Mode 1. The Green (G) indicator lights when you select Green in Mode 2. SPC (GREEN) Indicator Lights Up  53937

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5.5.3.2 Monitor Operator Controls, Continued

Control and switch functions, continued

Table 5-3 Hitachi Model HM-4821-D Monitor Functions, Continued

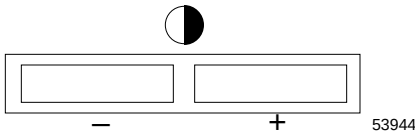
Control/Switch/Indicator	Function
MODE Switch (MODE 1/MODE 2), continued	The Rotation indicator lights when you select Rotation in Mode 1. The Blue (B) indicator lights when you select Blue in Mode 2. ROTATION (BLUE) Indicator Lights Up  COLOR — 1 — 2 — 3 — R — G — B 53938 The Moire indicator lights when you select Moire in Mode 1. The Blue indicator lights when you select Blue in Mode 2.

5.5.3.3 Display Setting Adjustment Procedures

Introduction	You can adjust the following screen settings by using the control buttons on the front panel: • Contrast • Brightness • Horizontal width • Horizontal centering • Vertical height • Vertical centering • SPC (side pin cushion) • Rotation • Moire • Color temperature • RGB color balance
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Contrast setting adjustment procedure	Use the procedure in Table 5-4 to adjust the screen’s contrast setting. The contrast control adjusts the contrast of the screen. If you want to retain the adjusted contrast setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.
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Table 5-4 Contrast Setting Adjustment Procedure

Step	Action
1	 To increase the screen contrast from light to dark, press the “+” button.
2	To decrease the screen contrast from dark to light, press the “-” button.
3	When you press “+” and “-” buttons at the same time, the contrast increases to maximum.

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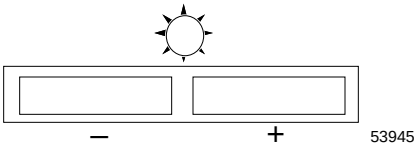
5.5.3.3 Display Setting Adjustment Procedures, Continued

Brightness setting adjustment procedure

Use the procedure in Table 5-5 to adjust the screen’s contrast setting. The contrast control adjusts the contrast of the screen.

If you want to retain the adjusted brightness setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-5 Brightness Setting Adjustment Procedure

Step	Action
1	 To increase the screen brightness, press the “+” button.
2	To decrease the screen brightness, press the “-” button.
3	When you press “+” and “-” buttons at the same time, the brightness increases to maximum.

Continued on next page

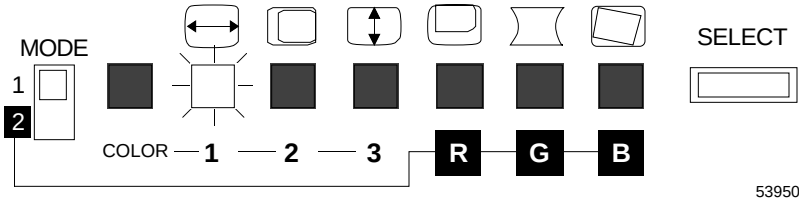
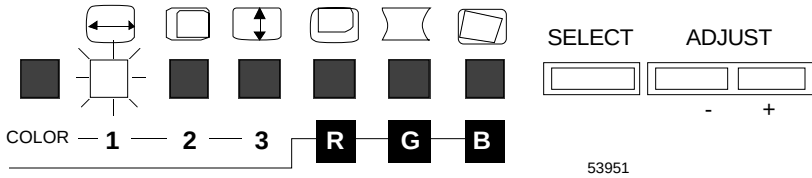
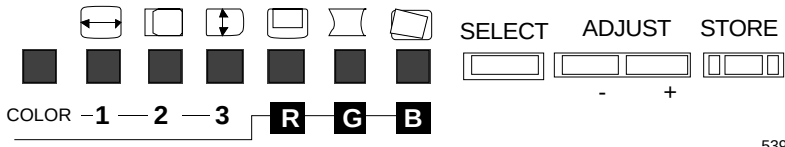
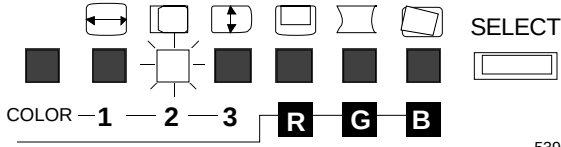
5.5.3.3 Display Setting Adjustment Procedures, Continued

Horizontal width setting adjustment procedure

Use the procedure in Table 5-6 to adjust the screen's horizontal width setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-6 Horizontal Width Setting Adjustment Procedure

Step	Action
1	 53950 Select Mode 1 with the MODE switch and press and hold the SELECT button until the Horizontal Width indicator lights.
2	 53951 Adjust the horizontal width by pressing an ADJUST button, "+" or "-."
3	 53952 After adjusting the horizontal width, to store the adjusted data, press the STORE button.
4	 53953 To proceed to the next adjustment mode, Horizontal Centering, press the SELECT button.

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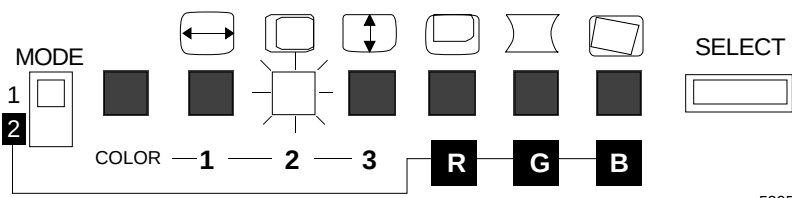
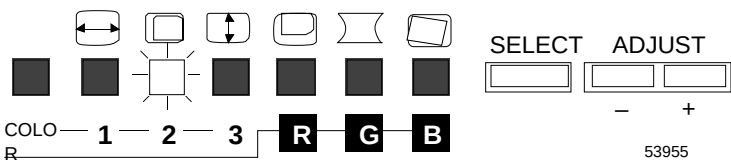
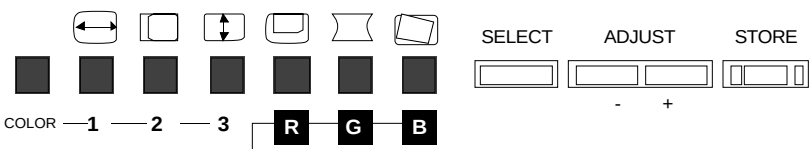
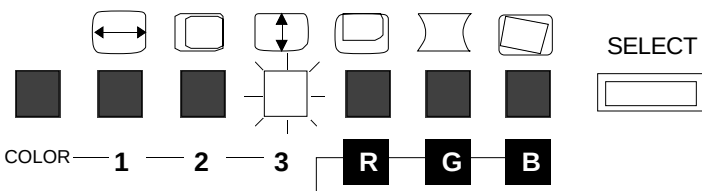
5.5.3.3 Display Setting Adjustment Procedures, Continued

Horizontal centering setting adjustment procedure

Use the procedure in Table 5-7 to adjust the screen’s horizontal centering setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-7 Horizontal Centering Setting Adjustment Procedure

Step	Action
1	 Select Mode 1 with the MODE switch and press and hold the SELECT button until the Horizontal Centering indicator lights.
2	 Adjust the horizontal centering by pressing an ADJUST button, “+” or “-.”
3	 After adjusting the horizontal centering, to store the adjusted data, press the STORE button.
4	 To proceed to the next adjustment mode, Vertical Height, press the SELECT button.

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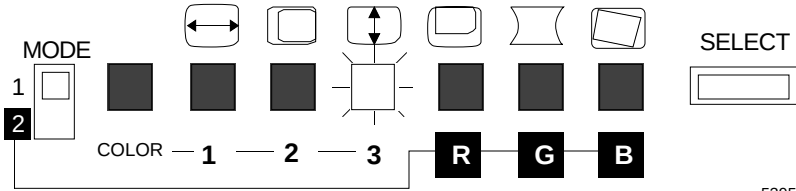
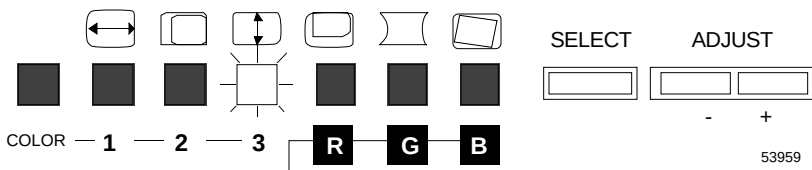
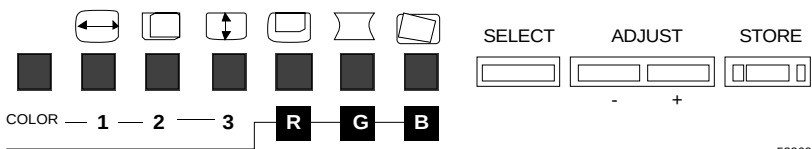
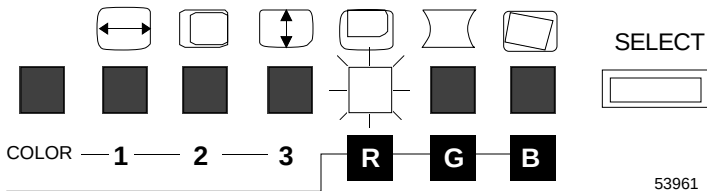
5.5.3.3 Display Setting Adjustment Procedures, Continued

Vertical height setting adjustment procedure

Use the procedure in Table 5-8 to adjust the screen's vertical height setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-8 Vertical Height Setting Adjustment Procedure

Step	Action
1	 Select Mode 1 with the MODE switch and press and hold the SELECT button until the Vertical Height indicator lights.
2	 Adjust the vertical height by pressing an ADJUST button, "+" or "-."
3	 After adjusting the vertical height, to store the adjusted data, press the STORE button.
4	 To proceed to the next adjustment mode, Vertical Centering, press the SELECT button.

Continued on next page

5.5.3.3 Display Setting Adjustment Procedures, Continued

Vertical centering
setting adjustment
procedure

Use the procedure in Table 5-9 to adjust the screen’s vertical centering setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-9 Vertical Centering Setting Adjustment Procedure

Step	Action
1	Select Mode 1 with the MODE switch and press and hold the SELECT button until the Vertical Centering indicator lights.
2	Adjust the vertical centering by pressing an ADJUST button, “+” or “-.”
3	After adjusting the vertical centering, to store the adjusted data, press the STORE button.
4	To proceed to the next adjustment mode, Side Pin Cushion (SPC), press the SELECT button.

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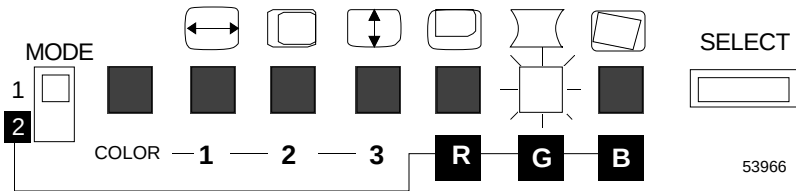
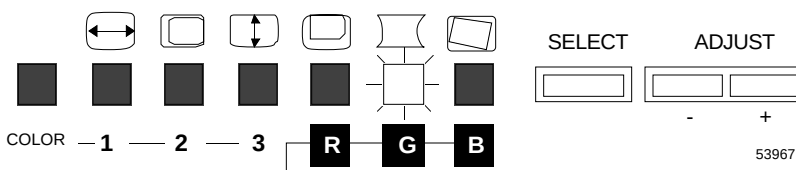
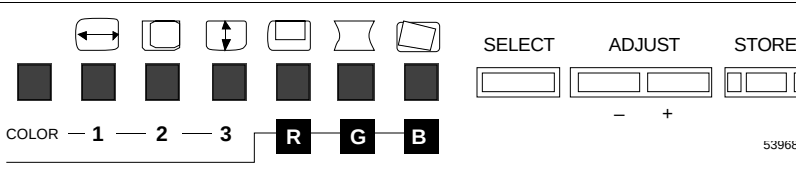
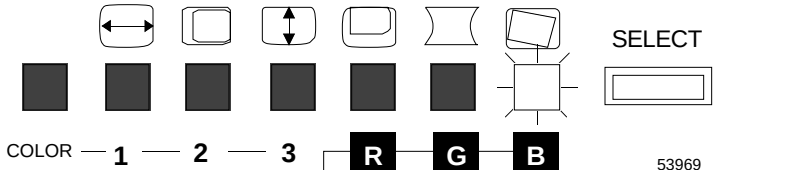
5.5.3.3 Display Setting Adjustment Procedures, Continued

SPC setting adjustment procedure

Use the procedure in Table 5-10 to adjust the screen's side pin cushion (SPC) setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-10 Side Pin Cushion (SPC) Setting Adjustment Procedure

Step	Action
1	 53966 Select Mode 1 with the MODE switch and press and hold the SELECT button until the SPC indicator lights.
2	 53967 Adjust the side pin cushion by pressing an ADJUST button, "+" or "-."
3	 53968 After adjusting the side pin cushion, to store the adjusted data, press the STORE button.
4	 53969 To proceed to the next adjustment mode, Rotation, press the SELECT button.

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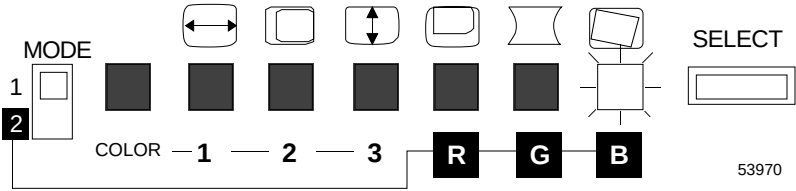
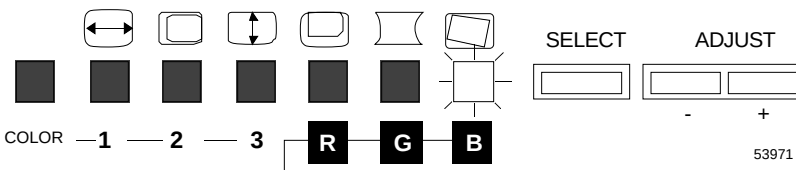
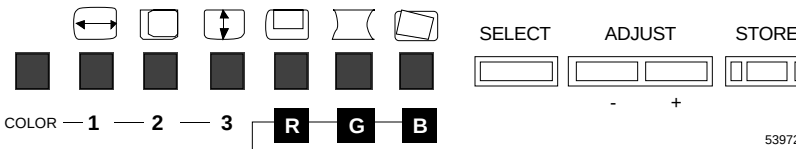
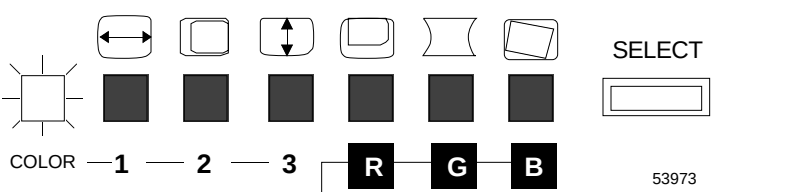
5.5.3.3 Display Setting Adjustment Procedures, Continued

Rotation setting adjustment procedure

Use the procedure in Table 5-11 to adjust the screen’s Rotation setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-11 Rotation Setting Adjustment Procedure

Step	Action
1	 Select Mode 1 with the MODE switch and press and hold the SELECT button until the Rotation indicator lights.
2	 Adjust the rotation by pressing an ADJUST button, “+” or “-.”
3	 After adjusting the rotation, to store the adjusted data, press the STORE button.
4	 To proceed to the next adjustment mode, Moire, press the SELECT button.

Continued on next page

5.5.3.3 Display Setting Adjustment Procedures, Continued

Moire setting adjustment procedure

Use the procedure in Table 5-12 to adjust the screen's moire setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-12 Moire Setting Adjustment Procedure

Step	Action
1	Select Mode 1 with the MODE switch and press and hold the SELECT button until the Moire indicator lights.
2	Adjust the moire by pressing an ADJUST button "+" or "-."
3	After adjusting the moire, to store the adjusted data, press the STORE button.
4	To proceed to the next adjustment mode, Color Temperature (COLOR), press the SELECT button.

Continued on next page

5.5.3.3 Display Setting Adjustment Procedures, Continued

Color temperature setting adjustment

Three types of color temperature have been stored prior to shipping the monitor, and they are selectable by using the front panel controls.

- 1 655K + 7MPCD
- 2 9300K + 8MPCD
- 3 User adjustable (default 9300K + 8MPCD)

Color temperature is adjustable in Mode 2 in RGB color balance. Refer to the display setting color balance adjustments.

ATTENTION

If you want a color temperature that complies with ergonomic regulation ZH1/618, you must select default color temperature (9300K + 8MPCD).

Color temperature setting adjustment procedure

Use the procedure in Table 5-13 to adjust the screen's color temperature setting.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-13 Color Temperature Setting Adjustment Procedure

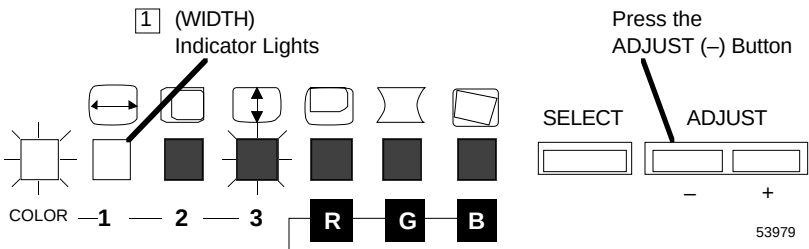
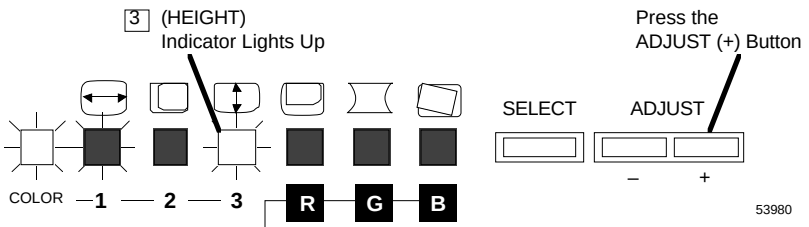
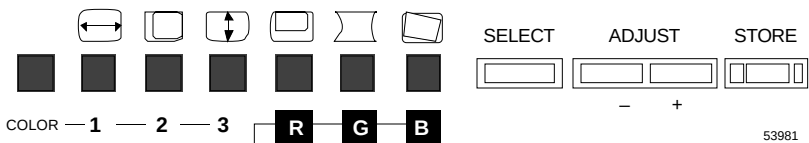
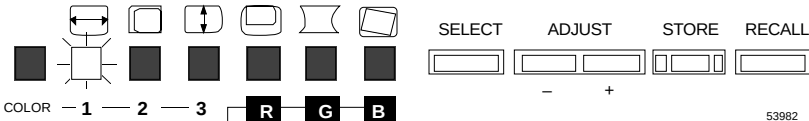
Step	Action
1	Diagram illustrating the control panel setup for Step 1. The MODE switch is set to position 2. The H.CENT Indicator Lights are shown. The SELECT button is highlighted. The color sequence is 1, 2, 3, R, G, B.

Continued on next page

5.5.3.3 Display Setting Adjustment Procedures, Continued

Color temperature
setting adjustment
procedure, continued

Table 5-13 Color Temperature Setting Adjustment Procedure, Continued

2	 1 (WIDTH) Indicator Lights Press the ADJUST (-) Button SELECT ADJUST - + COLOR — 1 — 2 — 3 R G B 53979 To change the color temperature to 6550K + 8MPCD, press the ADJUST “-” button, and the “1” indicator will light.
3	 3 (HEIGHT) Indicator Lights Up Press the ADJUST (+) Button SELECT ADJUST - + COLOR — 1 — 2 — 3 R G B 53980 To change the color temperature to a user-adjustable temperature, press the ADJUST “+” button and the “3” indicator will light.
4	 SELECT ADJUST STORE - + COLOR — 1 — 2 — 3 R G B 53981 After selecting the color temperature, to store the adjustment data, press the STORE button.
5	 SELECT ADJUST STORE RECALL - + COLOR — 1 — 2 — 3 R G B 53982 When the COLOR indicator is lit, one of the three indicators (“1,” “2,” or “3”) will be lit. If you press the ADJUST “+” button, the lit indicator shifts to the right. If you press the ADJUST “-” button, the lit indicator shifts to the left. To proceed to the next adjustment mode, Red Color Balance, press the SELECT button.

Continued on next page

5.5.3.3 Display Setting Adjustment Procedures, Continued

Red color balance
setting adjustment
procedure

Use the procedure in Table 5-14 to adjust the screen’s red color balance.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-14 Red Color Balance Setting Adjustment Procedure

Step	Action
1	MODE 1 2 COLOR — 1 — 2 — 3 R G B 53983 Select Mode 2 with the MODE switch and press and hold the SELECT button until the “R” indicator lights.
2	COLOR — 1 — 2 — 3 R G B 53984 Adjust the red color balance by pressing an ADJUST button “+” or “-.”
3	COLOR — 1 — 2 — 3 R G B 53985 After adjusting the red color balance, to store the adjusted data, press the STORE button.
4	COLOR — 1 — 2 — 3 R G B 53986 To proceed to the next adjustment mode, Green Color Balance, press the SELECT button.

Continued on next page

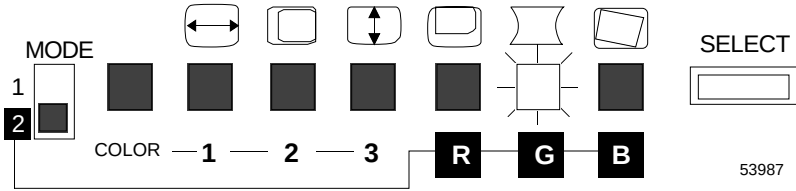
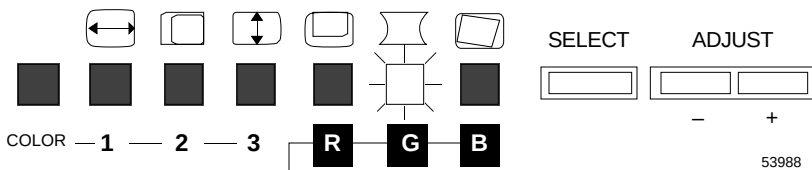
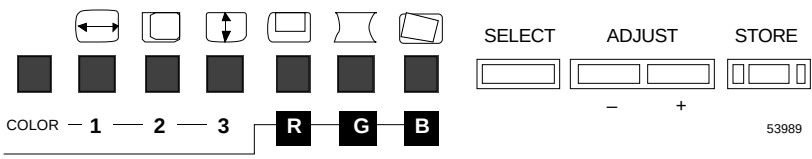
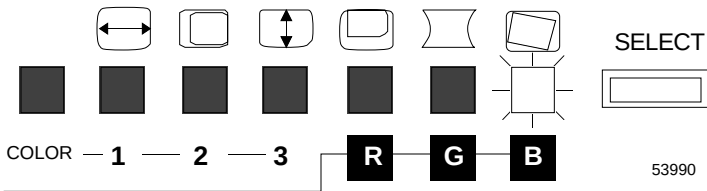
5.5.3.3 Display Setting Adjustment Procedures, Continued

Green color balance setting adjustment procedure

Use the procedure in Table 5-15 to adjust the screen's green color balance.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-15 Green Color Balance Setting Adjustment Procedure

Step	Action
1	 Select Mode 2 with the MODE switch and press and hold the SELECT button until the "G" indicator lights.
2	 Adjust the green color balance by pressing an ADJUST button "+" or "-."
3	 After adjusting the green color balance, to store the adjusted data, press the STORE button.
4	 To proceed to the next adjustment mode, Blue Color Balance, press the SELECT button.

Continued on next page

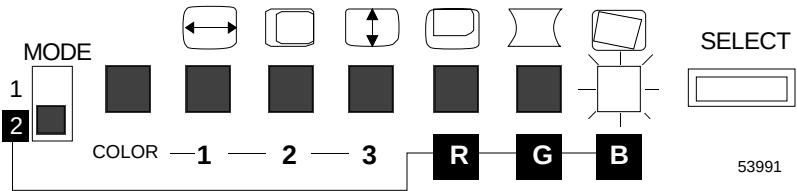
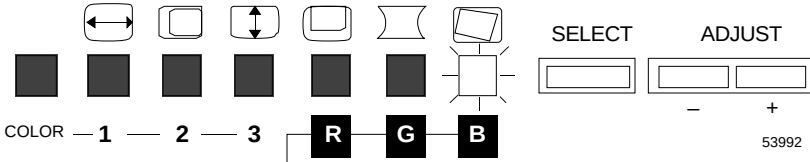
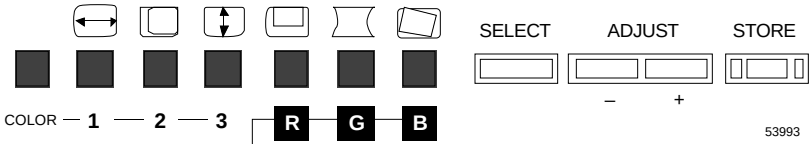
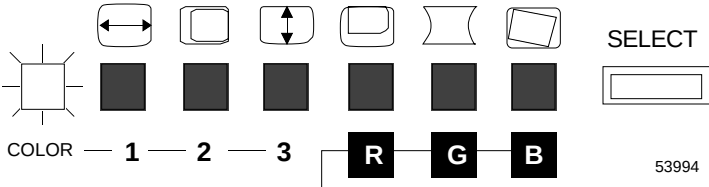
5.5.3.3 Display Setting Adjustment Procedures, Continued

Blue color balance setting adjustment procedure

Use the procedure in Table 5-16 to adjust the screen’s blue color balance.

If you want to retain the setting until the next time you apply power to the monitor, you can store the adjusted data by pressing the STORE button before removing power from the monitor.

Table 5-16 Blue Color Balance Setting Adjustment Procedure

Step	Action
1	 Select Mode 2 with the MODE switch and press and hold the SELECT button until the “B” indicator lights.
2	 Adjust the blue color balance by pressing an ADJUST button “+” or “-.”
3	 After adjusting the blue color balance, to store the adjusted data, press the STORE button.
4	 To return to the COLOR adjustment mode, press the SELECT button.

ATTENTION

ATTENTION—RGB color balance is factory preset to a maximum value. When balance is adjusted for the first time, or after pressing the RECALL button, if you press the “+” button, the color balance does not change.

Section 6 – Integrated Keyboard

6.1 Overview

Section contents

These are the topics covered in this section:

	Topic	See Page
6.1	Overview	89
6.2	Functional Partition of Integrated Keyboard	90
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6.4	Relegendable Function Key Panel.....	92
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6.13	Operator Control Panel Replacement.....	104
6.14	Integrated Keyboard Replacement Parts	105

Introduction

The Integrated Keyboard (IKB) is an Universal Work Station option that provides Operator functions as well as a QWERTY full-travel keyboard. The IKB can have an optional trackball.

The desktop IKB rests on any flat work surface, and the keyboard interface cable(s) connects to the EPDGC or EPDGP I/O board.

Continued on next page

6.1 Overview, Continued

IKB illustration

Figure 6-1 is an illustration of the Integrated Keyboard.

Figure 6-1 Integrated Keyboard Illustration



6.2 Functional Partition of Integrated Keyboard

Description

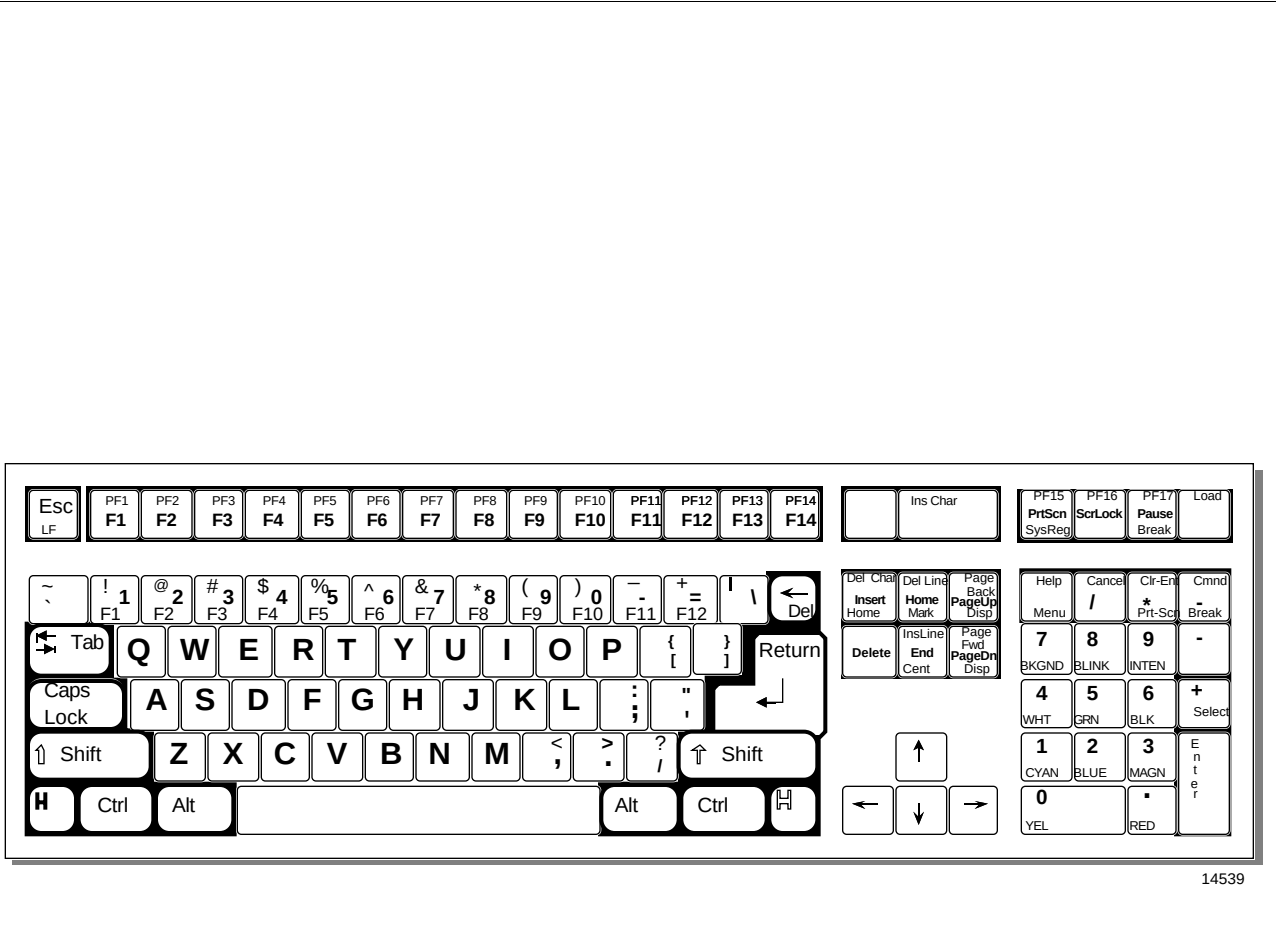
The Integrated Keyboard integrates the Operator and Engineer functions into a single keyboard and has the following functional parts.

- Full-travel keyboard keys
- Relegendable Function Key panel (RFKP)
- Operator Control panel
- Optional cursor control unit (trackball)

6.3 Full-Travel Keyboard

Description The QWERTY keyboard functions as both an Operator’s and Engineer’s keyboard with multifunctional keys with black, red, and blue lettering, similar to the U^XS Auxiliary keyboard. Figure 6-2 illustrates the layout of the full-travel QWERTY keyboard.

Figure 6-2 Full-Travel QWERTY Keyboard Layout



6.4 Relegendable Function Key Panel

Description

Relegendable function keys (buttons) can be defined or configured by the user to perform certain functions, such as invoking certain critical or frequently used displays or cross-screen displays within a console, or to activate a particular Control Language program.

Refer to *Button Configuration Form Instructions* and *Button Configuration Data Entry* manuals for button configuration information. Refer to Sections 7 and 8 of the *Process Operation Manual* for alarm annunciation information.

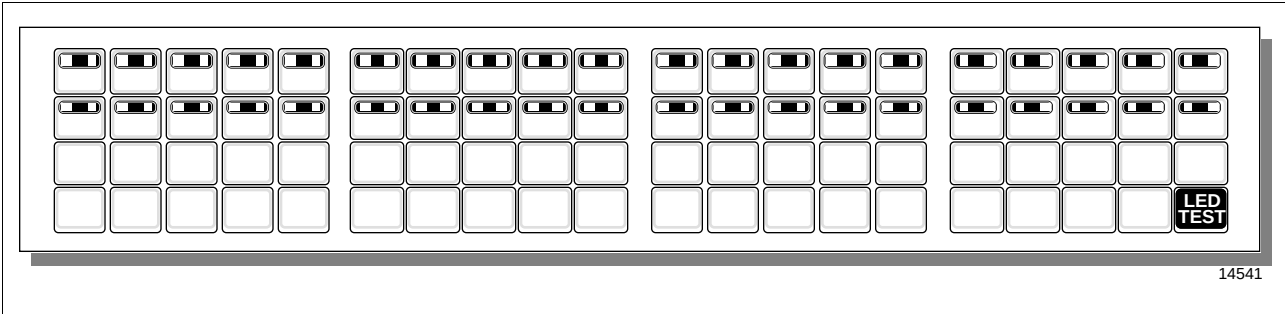
ATTENTION

ATTENTION—The paper strip that can be labeled for the key legends can be removed and reinserted behind the panel surface.

Relegendable Function Key panel

Figure 6-3 illustrates the layout of the Relegendable Function Key panel (RFBK).

Figure 6-3 Relegendable Function Key Panel (RFBK)



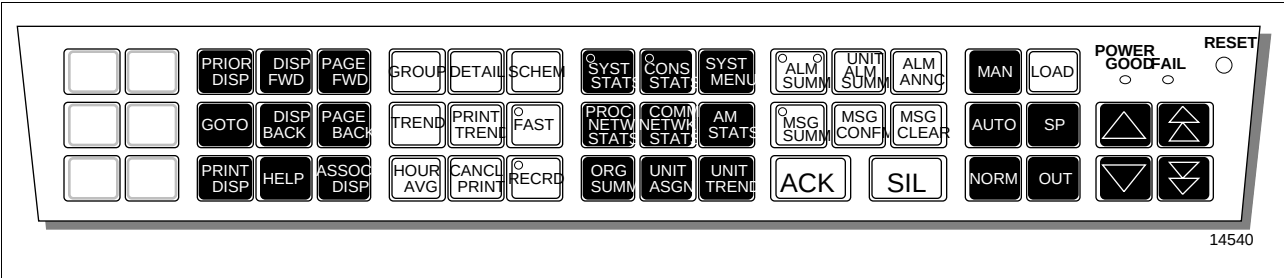
6.5 Operator Control Panel

Description

The Operator Control panel includes the **RESET** switch and power **POWER GOOD/FAIL** LEDs, as well as the Operator function keys.

Figure 6-4 illustrates the layout of the Operator Control panel.

Figure 6-4 Operator Control Panel Layout



RESET button

The **RESET** button resets the Universal Work Station hardware. It provides the same function as the white **RESET** button on the Five-Slot Module's power supply.

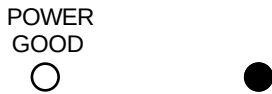
POWER GOOD indicator

The **POWER GOOD** indicator is lit when the keyboard has power applied and is functioning properly.



FAIL indicator

The **FAIL** indicator lights when a hardware failure in the keyboard is detected.



Releagendable keys

There are six releagendable keys that are located on the far left side of the Operator Control panel. To label the keys, raise the keyboard to access the paper insert, pull the paper insert out, label it, and replace the paper insert.

6.6 IKB Trackball

Introduction

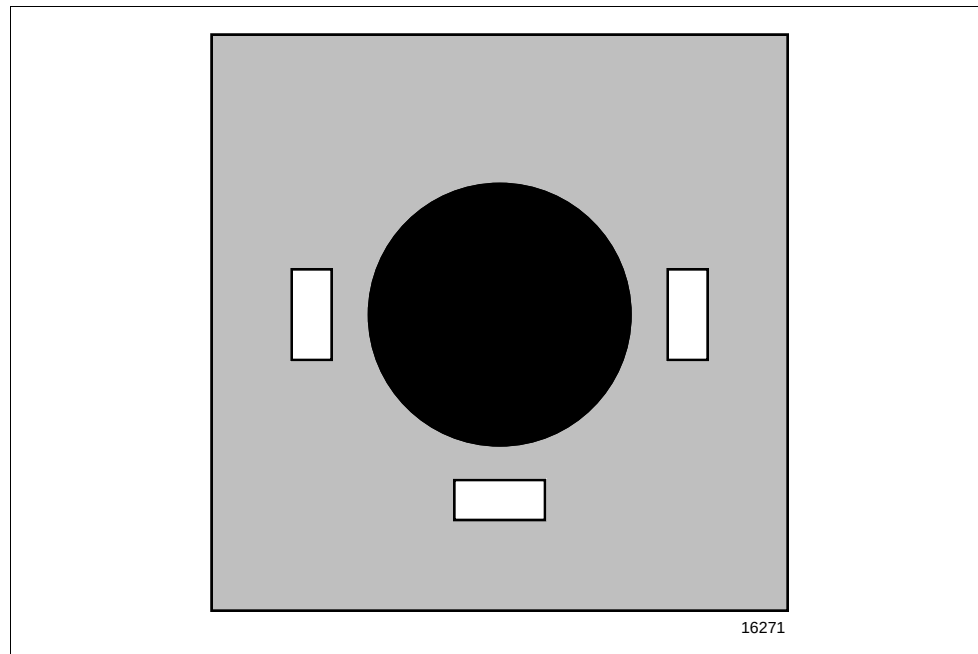
The trackball is an IKB option. It controls the movement of the cursor on the monitor screen.

Operation

The trackball button functions depend upon the application window that the cursor is in when the buttons are used.

Figure 6-5 is an illustration of the trackball.

Figure 6-5 Integrated Keyboard Trackball



6.7 Cleaning the Integrated Keyboard

Cleaning the keyboards Both keyboards can be cleaned with a damp (not saturated) soft cloth that is moistened with a light dishwashing soap solution.

Use the procedure in Table 6-1 to clean the keyboards.

Table 6-1 Keyboard Cleaning Procedure

Step	Action
1	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
2	Using a soft cloth that is slightly dampened with a dishwashing soap in water, clean the surface of the keyboard while ensuring that no excess solution (there should not be any excess solution) runs into the keyboard assembly.
3	Polish the surfaces of the keyboards with a soft cloth to remove any remaining soap residue.
4	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.

Cleaning the trackball The trackball can be cleaned with a damp (not saturated) soft cloth that is moistened with a light dishwashing soap solution.

Use the procedure in Table 6-2 to clean the trackball.

Table 6-2 Trackball Cleaning Procedure

Step	Action
1	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
2	Using a soft cloth that is slightly dampened with a dishwashing soap in water, clean the surface of the trackball while ensuring that no excess solution (there should not be any excess solution) runs into the trackball assembly.
3	Polish the surfaces of the trackball with a soft cloth to remove any remaining soap residue.
4	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.

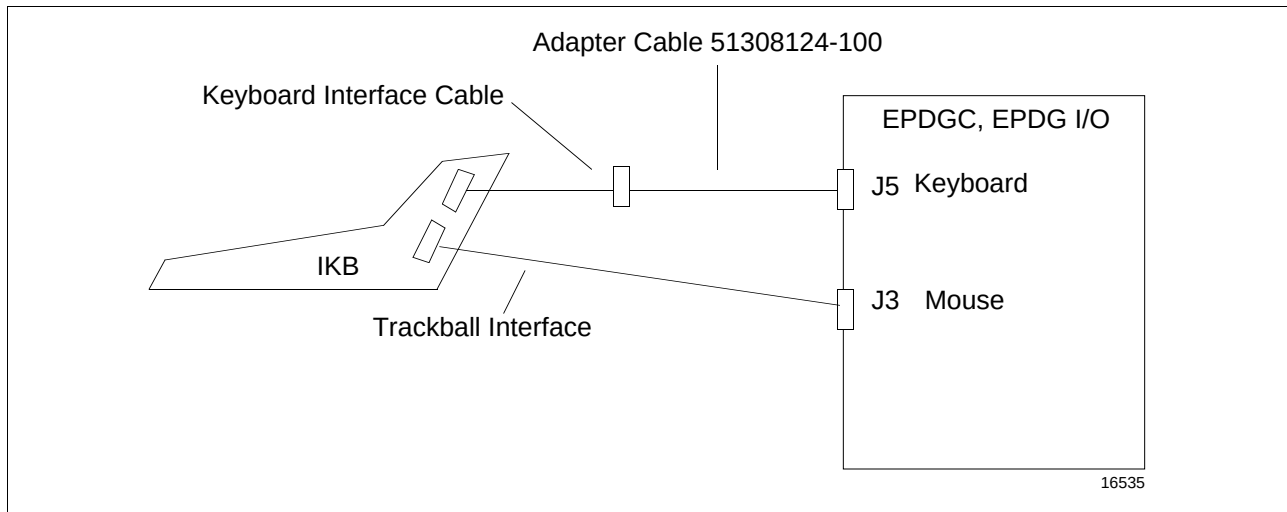
6.8 Integrated Keyboard Troubleshooting

Introduction

All four functional parts of the Integrated Keyboard (IKB) are integrated into one Optimal Replaceable Unit (ORU). However, the individual parts can be special ordered and replaced. Replacement procedures are included in this section.

The cabling diagram, Figure 6-6, illustrates the keyboard interface and optional cursor (trackball) interface cable connections.

Figure 6-6 Keyboard Cabling



Continued on next page

6.8 Troubleshooting the Integrated Keyboard, Continued

Keyboard troubleshooting procedure

Before replacing the keyboard, use the procedure in Table 6-3 to troubleshoot the keyboard.

Table 6-3 Keyboard Troubleshooting Procedure

Step	Action
1	Check the POWER GOOD LED on the Operator Control panel. If it indicates a good condition, try more than one functions on each of the keyboard panels. If just the trackball fails, the problem could be the cursor interface cable. If either the QWERTY keyboard or the button panel fails, the problem could be the keyboard interface cable.
2	Try to reset the keyboard by pressing the RESET button with a pointed, but not sharp, object.
3	Check the connections to the EPDGC or EPDGP I/O board.
4	If the problem still is not identified, reseal the EPDG2, EPDGC, or EPDGP I/O boards in the Five-Slot Module.
5	If the problem persists, replace the circuit boards one at a time.
6	If the problem is still is not identified, replace the IKB.

6.9 Integrated Keyboard Replacement

Overview

The complete Integrated Keyboard assembly can be replaced by disconnecting the cables from the EPDGP I/O or EPDGC I/O board and removing the keyboard. However, the individual IKB components can be replaced separately.

6.10 Relegendable Function Key Panel Replacement

Function Key Panel replacement procedure

Use the procedure in Table 6-4 to replace the Relegendable Function Key panel (RFKP).

Table 6-4 Relegendable Function Key Panel Replacement Procedure

Step	Action
1	Shut down the UWS node and go to another station and do a SHUTDOWN of the node before removing the ac power at the UWS. This prevents the LCN from isolating the station.
2	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
3	Turn the keyboard upside down and remove the bottom cover plate.
4	Disconnect the RFKP ribbon cable from the center connector at the edge of the IKB Controller board as illustrated in Figure 6-7.
5	Carefully move a screw driver through the cable slot behind the RFKP assembly until the panel is released from the adhesive that holds it in place. Remove the RFKP assembly.
6	Pass the replacement RFKP assembly through the slot in the IKB bezel and connect it to the RFKP connector on the IKB Controller board.
7	Spread new adhesive under the RFKP assembly and press it in place.
8	Reinstall the bottom cover plate.
9	Rotate the IKB to its upright position.
10	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.
11	Restart the UWS node.

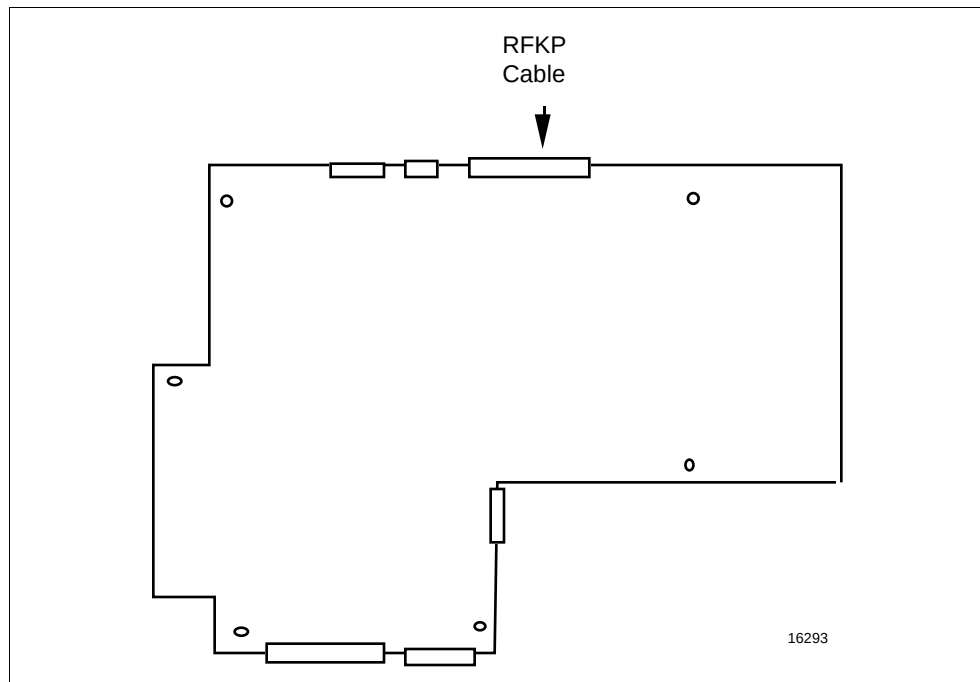
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6.10 Relegendable Function Key Panel Replacement, Continued

RFKP interface cable connection

Figure 6-7 illustrates the location of the Relegendable Function Key panel (RFKP) interface cable connection.

Figure 6-7 Relegendable Function Key Panel Interface Cable Connection Location



6.11 IKB Controller Board Replacement

Replacing the IKB Controller board

The IKB Controller board can be replaced after the Integrated Keyboard has been disconnected from the UWS.

Use the procedure in Table 6-5 to replace the IKB Controller board.

Table 6-5 IKB Controller Board Replacement Procedure

Step	Action
1	Shut down the UWS node and go to another station and do a SHUTDOWN of the node before removing the ac power at the UWS. This prevents the LCN from isolating the station.
2	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
3	Turn the keyboard upside down and remove the bottom cover plate.
4	Disconnect the cables that are illustrated in Figure 6-8 from the IKB Controller board.
5	Remove the six screws that secure the IKB Controller board in place.
6	Remove the board from the IKB assembly.
7	Install the replacement IKB Controller board and secure it with the six screws that were previously removed.
8	Reconnect the cables that were previously removed as illustrated in Figure 6-8.
9	Reinstall the bottom cover plate.
10	Rotate the IKB to its upright position.
11	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.
12	Restart the UWS node.

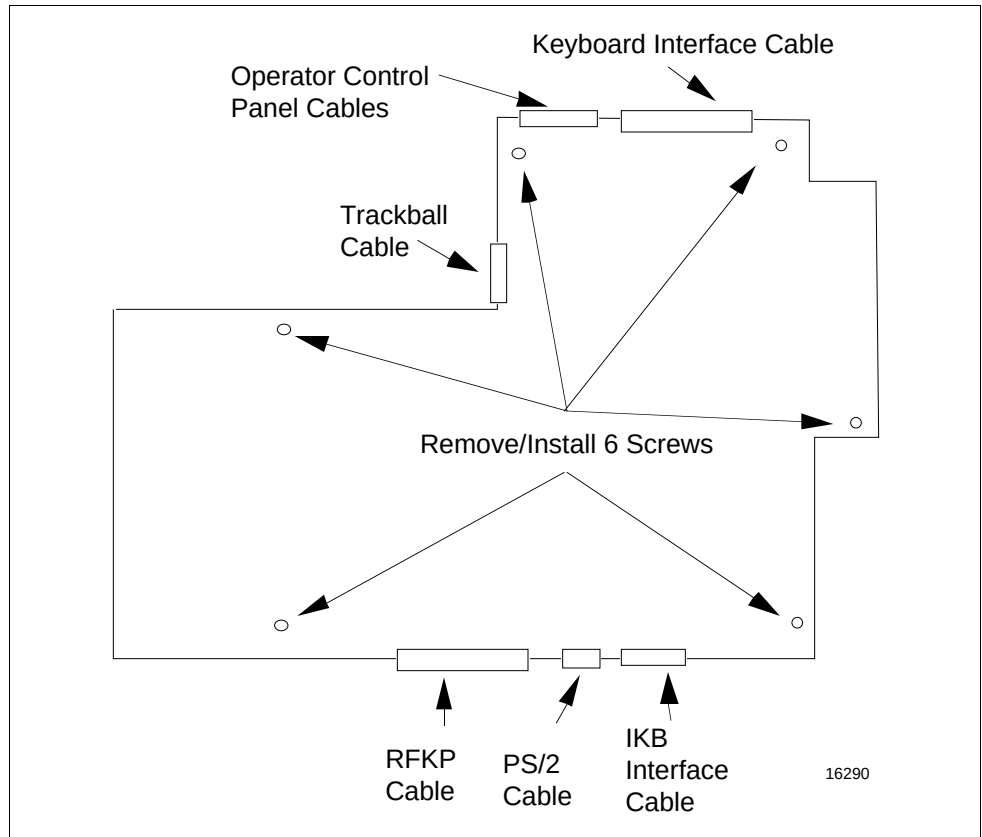
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6.11 IKB Controller Board Replacement, Continued

IKB Controller board cabling

Figure 6-8 illustrates the cabling to the IKB Controller board.

Figure 6-8 IKB Controller Board Cabling



6.12 Full-Travel Keyboard Replacement

Replacing the existing Full-travel keyboard

The full-travel keyboard can be replaced after the Integrated Keyboard assembly is removed from the UWS.

Use the procedure in Table 6-6 to replace the full-travel keyboard.

Table 6-6 Full-Travel Keyboard Replacement Procedure

Step	Action
1	Shut down the UWS node and go to another station and do a SHUTDOWN of the node before removing the ac power at the UWS. This prevents the LCN from isolating the station.
2	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
3	Turn the keyboard upside down and remove the bottom cover plate.
4	Locate and remove the screws in both ground/retaining straps that attach to the full-travel keyboard. The straps on each side of the full-travel keyboard prevent the keyboard from tilting upward more than 4 inches.
5	Remove the IKB Controller board. Refer to Table 6-5.
6	Slide the full-travel keyboard interface cable through the slot in the IKB bezel that is illustrated in Figure 6-9.
7	Lift the keyboard carefully away from the keyboard bezel by sliding the ground straps through the openings in the bezel.
8	Place the IKB assembly on its left side with the front edge of the assembly nearest you.
9	With the replacement keyboard held flat in front of you as if you were using it, grasp the rear edge and rotate it counterclockwise 90° to the vertical position ready to place inside the IKB bezel. Hold it in place next to the keyboard bezel.
10	Insert the keyboard interface cable through the slot in the keyboard bezel as shown in Figure 6-8.
11	Feed the ground straps through the two slots in the bezel and fit the two pivot tabs, located on the front edge of the keyboard, into the pivot slots in the bezel.
12	Install the two screws in the ground straps and tighten them so that the straps hold the keyboard securely in the bezel.
13	Press the keyboard into the bezel.
14	Replace the IKB Controller board and reconnect the cables. Refer to Table 6-5.

Continued on next page

6.12 Full-Travel Keyboard Replacement, Continued

Replacing the existing full-travel keyboard, continued

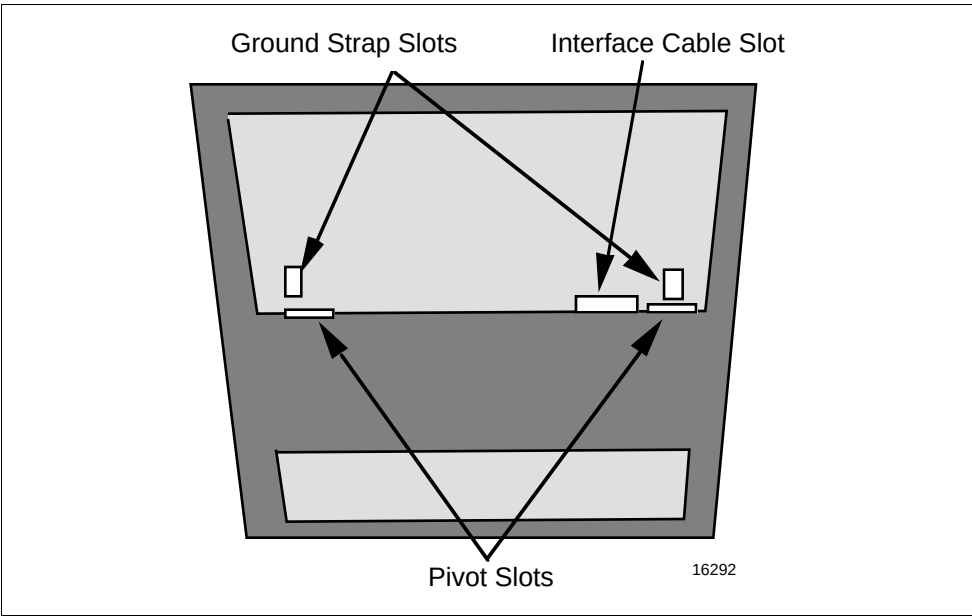
Table 6-6 Full-Travel Keyboard Replacement Procedure, Continued

Step	Action
15	Reinstall the IKB bottom cover plate.
16	Rotate the IKB to its upright position.
17	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.
18	Restart the UWS node.

IKB bezel cable and bezel slots

Figure 6-9 illustrates the locations of the cable and ground strap slots in the IKB bezel.

Figure 6-9 IKB Bezel Cable and Ground Strap Slots



6.13 Operator Control Panel Replacement

Operator Control panel replacement

The Operator Control panel can be replaced after the Integrated Keyboard assembly is removed from the UWS.

Use the procedure in Table 6-7 to replace the Operator Control panel.

Table 6-7 Operator Control Panel Replacement Procedure

Step	Action
1	Shut down the UWS node and go to another station and do a SHUTDOWN of the node before removing the ac power at the UWS. This prevents the LCN from isolating the station.
2	Remove power from the UWS by placing the power switch at the front panel in the OFF (0) position.
3	Turn the keyboard upside down and remove the bottom cover plate.
4	Remove the IKB Controller board. Refer to Table 6-6.
5	Loosen the four nut and washer sets that are located near the front edge of the IKB assembly that secure the Operator Control panel.
6	Remove the three screws and one retaining bracket that secure the other edge of the Operator Control panel.
7	Lift the free edge of the board slightly to clear any obstructions and pull forward until the board is free.
8	Transfer the relegendable insert to the replacement Operator Control panel.
9	Transfer the four screw, washer, and nut sets from the edge of the old board to the replacement panel.
10	Insert the replacement panel into the IKB bezel, leaving the four nuts loose.
11	Replace the three screws and one bracket that secure the free edge of the board.
12	Firmly tighten the four nuts and three screws.
13	Replace the IKB Controller board. Refer to Table 6-6.
14	Reinstall the IKB bottom cover plate.
15	Rotate the IKB to its upright position.
16	Reapply power to the UWS by placing the power switch at the front panel in the ON (1) position.
17	Restart the UWS node.

6.14 Integrated Keyboard Replacement Parts

Parts List

The IKB can be replaced as a complete assembly or by replacement of individual functional parts. Table 6-8 is a list of replaceable parts.

Table 6-8 Integrated Keyboard Replacement Parts

Description	Part Number
Integrated Keyboard Assembly without Trackball (complete)	*51196694-100
IKBI2 Integrated Keyboard Interface – Two Boards	*51305378-100
IKB Keyboard Cable—72-Inch Length	51196694-913
IKB Controller board – PS/2 Trackball Protocol Firmware	51196694-903
Full-Travel Keyboard	51196694-904
Operator Control Panel	51196694-907
Relegendable Function Key Panel	51196694-909
PS/2 KB Cable (Connects IKBI2 to mother board's KB connector)	51305381-100
IKB Relegendable Insert Set (three inserts)	51196694-191

* = Optimum Replaceable Unit

Section 7 – Service

7.1 Overview

Section contents

These are the topics covered in this section:

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7.3	Service Support/Troubleshooting.....	113
7.4	Cleaning.....	113
7.5	Troubleshooting.....	113
7.6	Spare Parts.....	114

Introduction

The UWS Five-Slot Module is the same assembly that is used in all other LCN Five-Slot Modules. Also, the media storage devices (Zip drives, Bernoulli cartridge drives, and floppy disk drives) and their power supplies, when present, are the same or similar to those used in the Universal Station (US) except for their physical mounting hardware. For drive service information, refer to the *Universal Station Service* manual.

The current production UWS has a higher capacity enhanced power supply in the Five-Slot Module. This enhancement has eliminated the need for a separate power supply to provide power to the Zip or Bernoulli cartridge drives. The Winchester power supply has been replaced by a cable harness that provides power from the Five-Slot Module.

UWS requirements

This section provides service information for UWSs that have been upgraded to R5xx performance level. Some requirements are:

- Software Release 5xx will not accommodate MCPU or EMPU processor boards.
- A UWS that uses the Operator/Supervisor personality requires 4 megawords of memory.
- A UWS that uses the Universal personality requires 5 megawords of memory.

ATTENTION

ATTENTION—The current UWS production hardware is CE Compliant. CE Compliance requires a unique set of I/O boards, cables, and a ZIP or Bernoulli cartridge drive assembly that encloses the drives to prevent EMI emissions.

The spare parts for previous, non-CE Compliant production UWSs, are listed in this section.

Service information for the CE Compliant UWS is supported in Section 7.

7.2 UWS Board Configurations

Board complements

Tables 7-1 through 7-6 lists the acceptable non-CE Compliant UWS Five-Slot Module circuit board complements that will support R520 software for the Operator and Supervisor personalities. In each case, the minimum memory requirement is given.

UWS Operator personality

Table 7-1 lists the board complements for a UWS Operator personality when an HMPU or HPK2 processor board is installed in the Five-Slot Module. The minimum memory requirement is 4 megawords.

Table 7-1 UWS Operator Personality (HMPU or HPK2 Board)

Slot	Board Type	I/O Board Type
5	Do not use	Do not use
4	EPDG or EPDG2	EPDGP I/O
3		
2	EMEM or QMEM -2 ¹	
1	HMPU, HPK2-2, HPK2-3	LCN I/O

Note 1: HMPU + QMEM-2 = 4 Mw

HPK2-2 + QMEM-2 = 4 Mw

HPK2-3 + EMEM = 4 Mw

UWS Universal personality

Table 7-2 lists the board complements for a UWS Universal personality when an HMPU or HPK2 processor board is installed in the Five-Slot Module. The minimum memory requirement is 5 megawords.

Table 7-2 UWS Universal Personality (HMPU or HPK2 Board)

Slot	Board Type	I/O Board Type
5	Do not use	Do not use
4	EPDG or EPDG2	EPDGP I/O
3		
2	EMEM or QMEM -2 ²	
1	HMPU, HPK2-2, HPK2-3	LCN I/O

Note 2: HMPU + QMEM-2 + EMEM = 5 Mw

HMPU + QMEM-3 = 5 Mw

HPK2-2 + QMEM-2 + EMEM = 5 Mw

HPK2-3 + QMEM-3 = 5 Mw

Continued on next page

7.2 UWS Board Configurations, Continued

UWS Operator personality

Table 7-3 lists the board complement for a UWS Operator personality when a K2LCN processor board is installed in the Five-Slot Module. The minimum memory requirement is 4 megawords.

Table 7-3 UWS Operator Personality (K2LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG or EPDG2	EPDGP I/O
3		
2		
1	K2LCN-4 ³	LCN I/O

Note 3: Current production K2LCN boards have 2, 4, or 8 Mw of memory.

UWS Universal personality

Table 7-4 lists the board complement for a UWS Universal personality when a K2LCN processor board is installed in the Five-Slot Module. The minimum memory requirement is 5 megawords.

Table 7-4 UWS Universal Personality (K2LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG or EPDG2	EPDGP I/O
3		
2		
1	K2LCN-8 ³	LCN I/O

Note 3: Current production K2LCN boards have 2, 4, or 8 Mw of memory.

Continued on next page

7.2 UWS Board Configurations, Continued

UWS Operator personality

Table 7-5 lists the board complement for a UWS Operator personality when a K4LCN processor board is installed in the Five-Slot Module. The minimum memory requirement is 4 megawords.

Table 7-5 UWS Operator Personality (K4LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG or EPDG2	EPDGP I/O
3		
2		
1	K4LCN-4 ⁴	LCN I/O

Note 4: Current production K4LCN boards have 4, 8, or 16 Mw of memory.

UWS Universal personality

Table 7-6 lists the board complement for a UWS Universal personality when a K4LCN processor board is installed in the Five-Slot Module. The minimum memory requirement is 5 megawords.

Table 7-6 UWS Universal Personality (K4LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG or EPDG2	EPDGP I/O
3		
2		
1	K2LCN-8 ⁴	LCN I/O

Note 4: Current production K4LCN boards have 4, 8, or 16 Mw of memory.

Continued on next page

7.2 UWS Board Configurations, Continued

Board complements

Tables 7-7 through 7-10 list the acceptable CE Compliant UWS Five-Slot Module board complements that will support R520 software for the Operator and Supervisor personalities. In each case, the minimum memory requirement is given.

UWS Operator personality

Table 7-7 lists the board complement for a UWS Operator personality when a K2LCN processor board is installed in the CE Compliant Five-Slot Module. The minimum memory requirement is 4 megawords.

Table 7-7 CE Compliant UWS Operator Personality (K2LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG2	EPDGC I/O
3		
2		
1	K2LCN-4 ³	CLCN A/B

Note 3: Current production K2LCN boards have 2, 4, or 8 Mw of memory.

UWS Universal personality

Table 7-8 lists the board complement for a UWS Universal personality when a K2LCN processor board is installed in the CE Compliant Five-Slot Module. The minimum memory requirement is 5 megawords.

Table 7-8 CE Compliant UWS Universal Personality (K2LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG2	EPDGC I/O
3		
2		
1	K2LCN-8 ³	CLCN A/B

Note 3: Current production K2LCN boards have 2, 4, or 8 Mw of memory.

Continued on next page

7.2 UWS Board Configurations, Continued

UWS Operator personality

Table 7-9 lists the board complement for a UWS Operator personality when a K4LCN processor board is installed in the CE Compliant Five-Slot Module. The minimum memory requirement is 4 megawords.

Table 7-9 CE Compliant UWS Operator Personality (K4LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG2	EPDGC I/O
3		
2		
1	K4LCN-8 ⁴	CLCN A/B

Note 4: Current production K4LCN boards have 4, 8, or 16 Mw of memory.

UWS Universal personality

Table 7-10 lists the board complements for a UWS Universal personality when a K4LCN processor board is installed in the CE Compliant Five-Slot Module. The minimum memory requirement is 5 megawords.

Table 7-10 CE Compliant UWS Universal Personality (K4LCN Board)

Slot	Board Type	I/O Board Type
5	Do Not Use	Do Not Use
4	EPDG or EPDG2	EPDGC I/O
3		
2		
1	K4LCN-8 ⁴	CLCN A/B

Note 4: Current production K4LCN boards have 4, 8, or 16 Mw of memory.

7.3 Service Support/Troubleshooting

Introduction

Service support for the UWS hardware can be found in the TPS publications that are listed in Table 7-11. Troubleshooting information is not detailed in this manual, but is referenced to the more general publications in order to maintain up-to-date information.

Table 7-11 Service Publication References

Description of Service	Publication
Five-Slot Module	<i>Five/Ten-Slot Service</i>
Bernoulli Cartridge Drive	<i>Universal Station Service</i>
Zip Disk Drive	<i>Universal Station Service</i>
Power Supply	<i>Five/Ten-Slot Module Service</i>
Keyboard	This publication
Monitor	See adjustments in this publication.
Printer	<i>Universal Station Service</i>

7.4 Cleaning

Reference

The monitor screen and Bernoulli cartridge drive heads must be cleaned periodically. For cleaning instructions and the cleaning interval, refer to the *Universal Station Service* manual.

7.5 Troubleshooting

CAUTION

CAUTION—DO NOT REMOVE OR REPLACE THE CIRCUIT BOARDS WHILE POWER IS ON.

Function board removal

After the front panel is removed, the function boards can be removed from the front of the Five-Slot Module. To remove the front panel, grasp both sides of the panel near the bottom and pull forward.

I/O board removal

After the rear panel is removed, the I/O circuit boards can be removed from the Five-Slot Module's rear card cage. To remove the rear panel, remove the six screws and star washers that secure the panel in place.

7.6 Spare Parts

Introduction

The parts listed in Tables 7-12 and 7-13 represent the Optimum Replaceable Units (ORUs) when the part number is prefixed by an asterisk (*). Other parts may be required, but are not standard replaceable parts and must be special ordered.

The replacement parts for non-CE Compliant hardware is listed in Table 7-12, while the CE Compliant hardware is listed in Table 7-13.

Non-CE Compliant UWS parts list

Table 7-12 lists the replaceable parts for a non-CE Compliant UWS.

Table 7-12 Non-CE Compliant UWS Parts List

Description	Part Number
14-Inch Color Monitor	*51108311-400
21-Inch Multisync FST Color Monitor (IDEK 5221)	*51196075-100
Touchscreen for 51196075-100 Monitor	*51197007-100
21-Inch Multisync FST Color Monitor (IDEK 8221)	*51196363-100
Touchscreen for 51196363-100 Monitor	*51197019-100
Touchscreen Controller Assembly	*51197005-100
Enhanced Operator's Keyboard	*51401437-100
Engineer's Keyboard	*51190975-100
Supervisor's Keyboard	*51400990-100
Mouse with Attached Cable	*51195792-100
LCN I/O Board	*51107403-100
EPDG Board	*51401286-100
EPDGP I/O Board	*51304584-300
K2LCN Board with 4 Megawords of Memory	*51402615-400
K2LCN Board with 8 Megawords of Memory	*51402615-800
K4LCN Board without Daughter (Mezzanine) Memory Board	*51401946-100
4-Megaword Daughter (Mezzanine) Memory Board	*51201645-400
8-Megaword Daughter (Mezzanine) Memory Board	*51201645-800

Continued on next page

7.6 Spare Parts, Continued

Non-CE Compliant UWS parts list, continued

Table 7-12 Non-CE Compliant UWS Parts List, Continued

K4LCN Board without DIMM Memory Board Memory	*51402755-100
4-Megaword DIMM Memory Board	*51201745-400
8-Megaword DIMM Memory Board	*51201745-800
Floppy Disk Drive	*51109701-100
20 MB Bernoulli Cartridge Drive	*51195156-200
150 MB Bernoulli Cartridge Drive	*51196483-100
100 MB Zip Disk Drive	*51196929-100
Floppy Drive Power Supply	*51109818-100
Winchester Power Supply	*51107595-100
Video Copier to EPDG Cable	51195904-XXX
SCSI Bus Interface Cable	51304199-100
Drive DC Power Cable	51308055-100
AC Power Cord for Five Slot-Slot Module	51303508-100
AC Power Cord for Drive Power Supply	51303654-092
AC Power Cord for Five-Slot Module	51304037-183

CE Compliant UWS parts list

Table 7-13 lists the replaceable parts for a CE Compliant UWS.

Table 7-13 CE Compliant UWS Parts List

Description	Part Number
21-Inch Multisync FST Color Monitor	*51197021-100
Touchscreen for 51197021-100 Monitor	*51197022-200
Touchscreen Controller Assembly	*51197005-100
Enhanced Operator's/Supervisor's Keyboard	51402497-200
Engineer's Keyboard	51196712-200
EPDG2 Board	*51402089-100
EPDGC I/O Board	*51402447-100

Continued on next page

7.6 Spare Parts, Continued

CE Compliant UWS parts list, continued

Table 7-13 CE Compliant UWS Parts List, Continued

K2LCN Board with 4 Megawords of Memory	*51402615-400
K2LCN Board with 8 Megawords of Memory	*51402615-800
K4LCN Board without Daughter (Mezzanine) Memory Board	*51401946-100
4-Megaword Daughter (Mezzanine) Memory Board	*51201645-400
8-Megaword Daughter (Mezzanine) Memory Board	*51201645-800
K4LCN Board without DIMM Memory Board Memory	*51402755-100
4-Megaword DIMM Memory Board	*51201745-400
8-Megaword DIMM Memory Board	*51201745-800
CLCN A/B Board	*51305072-100
Media Drive Assembly	51402693-100
100 MB Zip Disk Drive	*51196929-100
150 MB Bernoulli Cartridge Drive	*51196483-100
Interface Cable with Ferrite Beads (for 51197021-100 Monitor)	51192126-400
Touchscreen Interface Cable (EPDGC to Touchscreen Controller Assembly)	51308097-100
Touchscreen Interface Cable (Touchscreen to Touchscreen Controller Assembly)	51308097-100
Mouse Interface Cable (EPDGC Board to Mouse/Trackball)	51308099-100
Keyboard Interface Cable (EPDGC Board to Keyboard)	51308105-200
SCSI Bus Cable	51192054-101
DC Power Cable	51308110-100
AC Power Cord, 120 Vac	51190921-110
AC Power Cord, 240 Vac	51190921-210

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